

**FIFTH FIVE-YEAR REVIEW REPORT FOR
KYSOR INDUSTRIAL CORP. SUPERFUND SITE
WEXFORD COUNTY, MICHIGAN**



Prepared by

**U.S. Environmental Protection Agency
Region 5
Chicago, Illinois**

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LIST OF ABBREVIATIONS & ACRONYMS

ARAR	Applicable or Relevant and Appropriate Requirement
bgs	Below ground surface
CERCLA	Comprehensive Environmental Response, Compensation, and Liability Act
CFR	Code of Federal Regulations
CIC	Community Involvement Coordinator
CIP	Cadillac Industrial Park
COC	Contaminant of Concern
DCE	Dichloroethene
EGLE	Michigan Department of Environment, Great Lakes and Energy
EPA	United States Environmental Protection Agency
ESD	Explanation of Significant Differences
EW	Extraction Well
FYR	Five-Year Review
GAC	Granular Activated Carbon
IC	Institutional Control
ICIAP	Institutional Controls Implementation and Assurance Plan
LDFA	Local Development Finance Authority
LTS	Long-Term Stewardship
MCL	Maximum Contaminant Level
MDEQ	Michigan Department of Environmental Quality
MDNR	Michigan Department of Natural Resources
MW	Monitoring Well
NCP	National Oil and Hazardous Substances Pollution Contingency Plan
ND	Non-detect
NPDES	National Pollution Discharge Elimination System
NPL	National Priorities List
O&M	Operation and Maintenance
OU	Operable Unit
PCE	Tetrachloroethylene or perchloroethylene
P&T	Pump-and-Treat
PFAS	Per- and polyfluoroalkyl substances
PFOA	Perfluorooctanoic acid
PFOS	Perfluorooctanesulfonic acid
PMP	Performance Monitoring Plan

ppt	Parts-Per-Trillion
PQL	Practical Quantitation Limit
PRP	Potentially Responsible Party
PTAS	Packed Tower Air Stripper
RA	Remedial Action
RAO	Remedial Action Objective
RD	Remedial Design
RI/FS	Remedial Investigation and Feasibility Study
ROD	Record of Decision
RPM	Remedial Project Manager
SDWA	Safe Drinking Water Act
SVE	Soil Vapor Extraction
SWRAU	Site-Wide Ready for Anticipated Use
TCA	Trichloroethane
TCE	Trichloroethylene
TCL	Target Cleanup Level
UAO	Unilateral Administrative Order
µg/L	Micrograms per liter or parts-per-billion
UU/UE	Unlimited Use and Unrestricted Exposure
VI	Vapor Intrusion
VOC	Volatile Organic Compound

I. INTRODUCTION

The purpose of a Five-Year Review (FYR) is to evaluate the implementation and performance of a remedy in order to determine if the remedy is and will continue to be protective of human health and the environment. The methods, findings, and conclusions of reviews are documented in FYR reports such as this one. In addition, FYR reports identify issues found during the review, if any, and document recommendations to address them.

The United States Environmental Protection Agency (EPA) is preparing this FYR report pursuant to the Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA) Section 121, consistent with the National Oil and Hazardous Substances Pollution Contingency Plan (NCP)(40 CFR Section 300.430(f)(4)(ii)) and considering EPA policy.

This is the fifth FYR for the Kysor Industrial Corp. Superfund Site (the “Kysor Site” or “Site”). The triggering action for this statutory review is the completion date of the previous FYR on August 31, 2015. This FYR report has been prepared due to the fact that hazardous substances, pollutants, or contaminants remain at the Site above levels that allow for unlimited use and unrestricted exposure (UU/UE).

The Site consists of a single operable unit (OU) addressing Site-wide Remedial Actions (RAs) which will be addressed in this FYR report.

The Kysor Site FYR was led by EPA Remedial Project Manager (RPM) Sheila Sullivan. Participants included the Michigan Department of Environment, Great Lakes, and Energy (EGLE)¹ Site Manager Nicholas Swiger. Participants also included the EPA Community Involvement Coordinator, Heriberto Leon. The relevant entities such as the potentially responsible parties (PRPs) and EGLE were notified of the initiation of the FYR. The review began on February 10, 2020.

Site Background

The Kysor Site is located about one-half mile northwest of and within the limits of the City of Cadillac (“City”), Michigan (Figure 1, Appendix C). The Site is defined by the former Kysor facility plume within the Cadillac Industrial Park (CIP) and areas in Haring Township to which the plume has migrated.

Kysor Corporation began operating in 1955 at its location at 1100 Wright Street. The company manufactured truck parts and temperature control systems for the automotive industry. Processes at the plant included stamping, machining metal parts, and painting. A variety of solvents, degreasers and paints were used at the 30-acre facility. The Kysor manufacturing facility still operates and is now owned by Borg-Warner. The immediate surrounding area is largely industrial but also includes a 23-acre municipal well field, a number of residences and a trailer park. Haring Township (“Township”) is largely forest/recreational and rural.

¹ In 2019, the Michigan Department of Environmental Quality (MDEQ) was renamed the Michigan Department of Environment, Great Lakes, and Energy (EGLE) and is the state regulatory agency for the Superfund program. These terms are used throughout this FYR in historical context.

The municipal well field (known as the Eighth Street field) is about 2,500 feet north of the Kysor facility within the CIP (Figure 2 Appendix C). The City well fields are the sole community drinking water supply for its nearly 10,500 residents. Four of the seven wells in this well field are no longer used. In 2021, the City will begin operating a new well field southwest of the City in a different aquifer. The City will subsequently close the Eighth Street well field and sell or redevelop the property.

Studies have shown that there are three distinct aquifers at, or near, the Site, referred to as shallow, intermediate, and well field (or deep). The shallow and intermediate aquifers consist of sand with some silty clay and gravel. Clay layers separating the three aquifers from one another are predominantly seen in the southwest part of the CIP. The shallow and intermediate aquifers are hydraulically connected northeast of the Kysor Corporation, near the North Park subdivision. The clay layer appears to be absent near the City well field. The City wells are screened at 240-340 feet below ground surface (bgs) and draw from a deep, predominantly sand aquifer.

There are no environmentally sensitive ecosystems in the Site area. Figure 4 (Appendix C) provides the City land use.

The Haring Township water system draws from two groundwater wells. The well field is located two miles northeast of the North Park area and supplies about 3,400 residents. Figure 5 in Appendix C is a map of the current Township distribution system. A number of residents use private wells for potable water in the areas not supplied by public water. The Township source water assessment found the wells to be moderately susceptible to potential of contamination. The Township also delineated a Well Head Protection Area.

Kysor and several other facilities contributed to six commingled groundwater contaminant plumes containing volatile organic compounds (VOCs) found under the Site.

The Kysor Site has long been closely associated with the adjacent Northernair Plating Superfund Site (“Northernair”) because the Northernair facility released hexavalent chromium [Cr(VI)] to the groundwater, which commingled with the other plumes at the Kysor Site. Northernair is a former electroplating facility which operated from a prefabricated metal building on 12.75 acres of land at the corner of Sixth Street and Eighth Avenue. Groundwater contamination emanating from Northernair consisted of two small Cr(VI) plumes in the shallow and intermediate aquifers, respectively. Figure 6, Appendix C depicts the size and relative locations of the plumes described above.

FIVE-YEAR REVIEW SUMMARY FORM

SITE IDENTIFICATION		
Site Name:	Kysor Industrial Corp. Site	
EPA ID:	MID043681840	
Region: 5	State: MI	City/County: Cadillac/Wexford
SITE STATUS		
NPL Status: Final		

Multiple OUs? No	Has the site achieved construction completion? Yes
REVIEW STATUS	
Lead agency: EPA <i>[If "Other Federal Agency", enter Agency name]:</i>	
Author name (Federal or State Project Manager): Sheila Sullivan	
Author affiliation: EPA	
Review period: 2/10/2020 - 7/29/2020	
Date of site inspection: 7/29/2020	
Type of review: Statutory	
Review number: 5	
Triggering action date: 8/31/2019	
Due date (five years after triggering action date): 8/31/2020	

II. RESPONSE ACTION SUMMARY

Basis for Taking Action

The Kysor facility and several other facilities contributed to six commingled groundwater contaminant plumes containing VOCs beneath the Site. Figure 6, Appendix C depicts the size and relative locations of these plumes. The following Contaminants of Concern (COCs) were identified in groundwater during the baseline risk assessment.

Noncarcinogens

Toluene
1,1,1-Trichloroethane (1,1,1-TCA)
Trans-1,2-dichloroethene (*trans*-1,2-DCE)
Xylenes
Hexavalent Chromium [Cr(VI)]

Carcinogens

1,1-Dichloroethene (1,1-DCE)
1,2-Dichloroethane (1,2-DCA)
Methylene Chloride
Tetrachloroethane (PCE)
Trichloroethane (TCE)

The investigations showed the approximate extent of the plume to be 4,000 feet long (north to south and beginning at the Kysor property) and, at its widest (east to west), 2,500 feet wide. The eastern boundary of the plume extends to the west edge of the Cadillac municipal well field; the western boundary extends approximately to Leeson Avenue, although trace amounts of COCs have been found west of Leeson.

The exposure pathway of concern at the Site is groundwater ingestion. Trichloroethylene was detected in production well CW-7 above its Safe Drinking Water Act (SDWA) Maximum Contaminant Level (MCL) of five micrograms per liter (µg/L) a number of times since 1993. The municipal wells draw water from about 340 feet bgs, while the deepest of the Kysor Site plume is estimated at slightly over 200 feet bgs, though a clay confining layer extends over part of the plume between the intermediate aquifer and the aquifer from which the municipal wells draw. Several residents in the North Park subdivision, where the northern extent of the plume has migrated, potentially use their private wells for

drinking water or for other domestic purposes. Groundwater in the shallow aquifer where carcinogens were detected incurred Excess Lifetime Cancer Risks (ELCR)² ranging from 4×10^{-4} to 5×10^{-2} , while in the intermediate aquifer the ELCR ranged from 1×10^{-5} to 4×10^{-3} . Non-carcinogenic hazard indices (HI) ranged from less than 1 to 11 in the shallow aquifer. The range in the intermediate aquifer was from below 1 to 66.³

Groundwater on-Site in the shallow and intermediate aquifers was not used as a drinking water source and the contaminants would be removed as part of the remedy. Residents in the area were connected to a municipal water supply; however, many residents who have been connected to the municipal water supply have retained their old wells for non-consumptive uses.

Future risks posed from ingestion of contaminated groundwater were modeled to theoretical receptor locations. Assuming no remediation was performed at the sites, residents would be exposed to ELCRs ranging from 1×10^{-5} to 1×10^{-3} for the shallow aquifer and 6×10^{-5} to 5×10^{-3} for the intermediate aquifer for at least another 10 to 15 years. Noncarcinogenic HIs resulting from future consumption of the groundwater would range from below 1 to 1.29 in the shallow aquifer and in the intermediate aquifer the range is from below 1 to 17.14, indicating there would still be significant future risks if no remediation was done.

Response Actions

Initial Response/Pre-ROD activities

On July 22-23, 1981, Kysor voluntarily excavated and disposed of approximately 700 cubic yards of contaminated soil from the two disposal areas (including the Ingraham property) behind its facility. The average depth of the excavation was six feet, and the average depth to the water table is 25 feet. Drums, buckets, rags, and paint sludge were removed along with the soil. The Michigan Department of Natural Resources (MDNR) noted strong odors remaining in the completed excavation pit. Samples taken by MDNR indicated greater than 100 parts per million (ppm) of TCE, TCA, PCE, ethylbenzene, toluene, and xylenes remained in the soils. The excavation was backfilled with clean material.

Other facilities that contributed to the commingled VOC plumes and the actions taken include the following, also depicted in Figure 6, Appendix C:

- Paulstra/CRC Corporation, formerly known as Cadillac Molded Rubber, Inc.-- located northeast of Kysor. After an above-ground tank containing TCE ruptured in 1984, Paulstra installed

² Excess lifetime cancer risks are probabilities that are generally expressed in scientific notation (e.g., 1×10^{-6} or $1 \text{E-}6$). An excess lifetime cancer risk of 1×10^{-6} indicates that, as a plausible upper bound, an individual has a one-in-one-million chance of developing cancer as a result of site-related exposure to a carcinogen over a 70-year lifetime under the specific exposure conditions at a site.

³ An HI greater than 1.0 indicates an unacceptable systemic or noncarcinogenic risk. The risk assessment assumed that a 70-kg adult would drink two liters of groundwater per day over a 70-year lifetime, and a 35-kg child would drink one liter of groundwater per day over a 10-year span.

recovery wells that operated until 1986. An additional extraction well (EW) was installed in 2000 and is still operating.

- Mitchell-Bentlev Corporation-- located 1,400 feet east of and side-gradient to Kysor began operating in late 1960s. A 1991 investigation showed a PCE spill near a former underground tank. An EW system was installed to capture PCE plume. After the company declared bankruptcy in 2006, MDEQ assumed operation of the extraction system until 2011.
- Rexair, Inc.-- located east-northeast of and side-gradient to Kysor. A groundwater investigation in late 1980s revealed a TCE plume emanating from the facility. After a State of Michigan Consent Judgment in 1991, Rexair installed a number of EWs and monitoring wells (MWs). A MDEQ report maintains that the Rexair EW is not adequately capturing its plume, allowing it to commingle with the Kysor plume. This resulted in the Kysor remediation system capturing and treating part of the contamination arising from Rexair since 1997.

Due to the number of groundwater plumes near the Kysor Site, more than a dozen different parties have installed MWs and EWs. A total of 340 wells are present in the area. Some are seen in Figure 7 (Appendix C).

Decision Documents

EPA issued a Record of Decision (ROD) in September 1989 to address:

- A VOC plume in the shallow aquifer containing up to 115 milligrams per liter (mg/L) of VOCs
- A less concentrated VOC plume (up to 12 mg/L) in the intermediate aquifer
- A Cr(VI) plume in both the intermediate and shallow aquifers
- An area of soil contaminated with VOCs at the Kysor Site

This remedy was the first and final ROD for the Kysor Site, and the second and final ROD for the Northernnaire Site.⁴ The sites were treated as one in the 1989 ROD because the designs for the treatment systems were developed jointly. Further, the plume from Northernnaire was contained within the Kysor Site plume, so designing separated groundwater extraction systems would not have been feasible.

The Remedial Action Objectives (RAOs) at the Kysor Site were:

- to eliminate any human exposure to residual hazardous waste disposed of or contaminated materials at the Site, and;
- to address all potential risks to human health and/or impacts to the environment.

The selected remedy addressed all Site concerns by a combination of treatment, and Site use restrictions. In addition, the ROD stated that corrective action measures would be taken should monitoring indicate a failure of any component of the remedy. The estimated time to achieve the goals was 64 years. The major components of the selected remedy are listed below. A more detailed description of the components is included in the following section.

⁴ The Northernnaire Plating Site was initially addressed via a ROD in 1985 (OU1), in which the soil was addressed. The groundwater for the Northernnaire site was addressed through the 1989 ROD and identified as OU2 for Northernnaire. The 1989 ROD was also the only OU for the Kysor Site. The sites were often referred to as the Northernnaire/Kysor Site.

- Install groundwater extraction and treatment (pump-and-treat or P&T) system to remove groundwater contamination from area surrounding the Site.
- Install a soil vacuum extraction (SVE) system to remove VOC contamination from the soils.
- Conduct long term groundwater monitoring to assess quality of area groundwater.
- Impose access and use restrictions.

The Target Cleanup Levels (TCLs) were calculated for each of the groundwater COCs as site-specific risk-based values that were not to exceed their respective MCLs under the SDWA. Further, the additive risk of all contaminants must fall within EPA's target ECLR range of 1×10^{-4} to 1×10^{-6} .

<u>COCs</u>	<u>TCLs (µg/L)</u>
1,1,1-Trichloroethane (1,1,1-TCA)	200
Trans-1,2-dichloroethylene (trans-1,2-DCE)	70
1,1-Dichloroethylene (1,1-DCE)	5
1,2-Dichloroethane (1,2-DCA)	5
Methylene chloride	5
Tetrachloroethylene (PCE)	1
Trichloroethylene (TCE)	5
Hexavalent Chromium (Cr(VI))	50
Xylenes	440
Toluene	40

Soil clean up levels were calculated so that residual contamination in the soil would not result in a continuing source of contamination to the groundwater above groundwater TCLs. The TCLs used to determine protectiveness for soil remediation are as follows:

<u>COCs</u>	<u>TCLs (mg/kg)</u>
TCE	0.07
Xylenes	141.00
1,1,1-TCA	7.60
Toluene	724.00

In 1994, Michigan prepared a Substantive Requirements Document (SRD) to establish requirements and limits for discharging treated groundwater to the Clam River. The limits were to meet the substantive requirements of a National Pollution Elimination Discharge System (NPDES) permit and the Michigan Wastewater Discharge Permit Rules. The surface water discharge limits for the Clam River that were established for VOCs have not changed and appear as below in the most recent NPDES (2017) permit. The 2017 discharge limit for Cr(VI) is listed below, however Cr(VI) monitoring has not been required as of 2019 since no Cr(VI) has been detected in the treatment plant influent.

<u>COCs</u>	<u>Weekly Maximum Effluent Limit (µg/L)</u>
1,1,1-TCA	5
1,2-DCA	5
PCE	5
TCE	5

In 1994, EPA issued an Explanation of Significant Differences (ESD) to formally expand the targeted area of the selected remedy to include a potentially impacted subdivision (North Park Subdivision) in Haring Township. This area is beyond the border of the City of Cadillac and not covered by the Cadillac groundwater restriction ordinance.

Status of Implementation

The selected remedy in the 1989 ROD and modified by the ESD included the following specific components:

- A groundwater extraction system consisting of 18 pumping wells
- Pipelines to convey VOC- and VOC/Cr(VI)-contaminated groundwater from the wells to the treatment system
- Discharge pipe to convey treated groundwater to the Clam River
- Packed tower air stripper (PTAS) system to remove VOCs from groundwater
- Granular activated carbon (GAC) contactors to remove Cr(VI) from groundwater
- Treatment building to house the treatment system components
- Expansion of a pilot SVE and treatment system to treat VOC-contaminated source soils at the Kysor facility, as discussed below
- Conduct long term groundwater monitoring to assess quality of area groundwater
- Impose access and use restrictions.

In 1995, EPA issued several Unilateral Administrative Orders (UAOs) to the PRPs to implement the 1989 ROD remedy, as modified by the 1994 ESD. On-Site construction of the RA began in June 1995. A public/private sector partnership was formed to construct the remedy and conduct operation and maintenance (O&M). While the City of Cadillac was not identified as a PRP and is not a Respondent to the UAO, it has taken the lead in constructing and operating the remedy. The City formed a Local Development Finance Authority (LDFA) in the early 1990s to facilitate construction of the remedy and assist with the capital expenditures of building the treatment system. The City established a Special Assessment District to finance the O&M of the treatment plant, monitoring and other costs. By 1996, the groundwater P&T and SVE systems at the Site were built and operating.

The SVE system consisted of 23 extraction/induction wells at a depth of approximately 25 feet (about five feet above the water table). A mounted blower created a negative pressure in the system. Collected vapors were blown through two vapor phase carbon units for VOC removal and then discharged through a stack. The SVE system operated from the spring through the fall each year from 1996 to 2010. From 1996 to 2000, 669 pounds of VOCs were removed. From 2000 to 2010, no VOCs were detected at the influent sampling ports, meaning no VOCs from the soil entered the system, and a change-out of the activated carbon had not been necessary since April 1999. The findings indicated that soil TCLs had been achieved. Completion of the soil remediation was correlated with the groundwater COC concentrations in the two closest wells downgradient of the Site. By December 1999, most VOC contaminants in these wells were below the groundwater TCLs and have remained so. With the approval of EPA and MDEQ, the LDFA decommissioned the SVE system in 2010 and dismantled it in fall 2016. The soil remediation has addressed potential health risks due to direct contact and/or ingestion of VOC-contaminated soil and eliminated significant contaminant source areas.

The groundwater P&T system continues to operate. Groundwater captured by the EW network is conveyed to the treatment facility. The VOC treatment system consists of two PTAS units, centrifugal fans, transfer pumps, and associated valves and piping. One EW (I-5) is used to capture Cr(VI) contamination in groundwater. The Cr(VI) treatment system consists of two GAC contactors operated in series, a pH control system, and associated valving and piping. The discharge from the GAC system is combined with the influent groundwater from the remaining EWs to the VOC treatment system.

Historical monitoring data indicate a sustained reduction of Cr(VI) to concentrations below its TCL, indicating that the majority of Cr(VI) has been effectively removed by the extraction system(s). Because the TCL is met at the influent monitoring point, the chromium testing requirement was removed from the Michigan discharge permit. The LDFA continues to perform chromium analyses and treatment via the GAC system, however EPA and EGLE provided confirmation (August 3, 2020) that operation of the Cr(VI) treatment system can be discontinued as documented in the 2010 FYR for the Northern Site.

Institutional Controls

Table 1: Summary of Planned and/or Implemented Institutional Controls (ICs)

Media, engineered controls, and areas that do not support UU/UE based on current conditions	ICs Needed	ICs Called for in the Decision Documents	Impacted Parcel(s)	IC Objective	Title of IC Instrument Implemented and Date (or planned)
On-Site and Off-Site Groundwater Areas with groundwater contamination do not support UU/UE. The current area that exceeds groundwater cleanup standards is within the CIP and areas north of Thirteenth Street in Haring Township (Figures 16 and 17 of Appendix C).	Yes	Yes	City of Cadillac	(1) Restrict groundwater use for drinking purposes in the shallow and intermediate aquifer. (2) Ensure that residences with contaminated wells are not using these wells for consumption purposes. (3) Take appropriate actions to ensure that current or future landowners do not use the contaminated groundwater as a drinking water source. (4) Ensure that no wells are installed that may interfere with the O&M of treatment and monitoring systems required by the UAO.	The City of Cadillac ordinance (Ord. No. 97-10, Sect. 42-164) ⁵ prohibits the installation of drinking water wells on-Site; the installation of any wells that might interfere with the O&M of the groundwater extraction and treatment system, except with written consent by EPA; and, tampering with or removing the containment or monitoring systems at the Site. (Implemented Nov. 3, 1997) The MDEQ Water Well Construction and Pump Installation Code (Part 127, Act 368, PA 1978, as amended) requires well drillers and pump installers to notify and obtain a permit from the local health department before the well is installed. (Implemented Feb. 14, 1967, revised April 21, 1994). Cadillac Ordinance No. 2018-10: <i>Ordinance Amending Chapter 42, Article II, Division 6 of the Cadillac City Code Regarding Institutional Controls for the Kysor Industrial</i>

⁵ Previously Ordinance No. 88-07, Sec. 2.300, Ch. 24

			Haring Township	Restrict groundwater use for drinking purposes in the shallow and intermediate aquifer by requiring all residents to connect to the Township water supply system. Reinforces connection to public supply and establishes required connection fees. Require the abandonment of all private wells in Haring Township where connections to the Township system are required. The resident must use a professional well driller. The driller must provide paper certification to the Department of Public Works of the abandonment.	<i>Corporation/ Northernaire Plating Company</i> (Implemented Sept 24, 2018) Haring Township Water Ordinance No. 86-01, Sect.2.1) requires Township homeowners and property owners located within 200 feet of a water distribution line to connect to the system within 180 days after the line becomes available; restricts cross connections between the Township system and any other source; requires all new structures or uses to hook up to the public system. (Implemented, April 14, 1986). Haring Township Water Ordinance No. 12-1016 (Amendment to Ordinance No. 86-01, Sect. 2.2). Establishes and requires all users to pay direct and/or indirect connection fees to the Township water supply. (Implemented October 30, 2012) Haring Township Water Ordinance No. 2013-81 (amendment to Ordinance No. 86-01, as amended). Prohibits use of private wells in the Township water service area; imposes a connection deadline; provides enforcement measures; prevents the installation of private wells and requires the abandonment of all pre-existing private wells where a connection has been made, or upon connection. (Implemented June 10, 2013). Haring Township Water Ordinance No. 2017-107. The ordinance consolidated Ordinance No. 86-01 and all subsequent ordinances. It also allows residents with private wells to maintain them and postpone connecting to the Township system until the well fails, at which time the owner is required to connect to the system. (Implemented June 28, 2017)
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A map showing the areas in which the ICs apply is included in Figures 16 and 17 of Appendix C.

Status of Access Restrictions and ICs:

City of Cadillac

The 1989 ROD required that access and use restrictions be imposed on groundwater use for drinking water purposes in the affected shallow and intermediate aquifers. The PRPs were required to install and maintain a fence to limit access to portions of the Site containing RA components. A six-foot high perimeter chain link fence topped by three-strand barbed wire surrounds the SVE and groundwater treatment facilities. Based on discussions with the City of Cadillac, the fence is well-maintained, and no trespassing has been evident.

The ICs were also to prevent interference with remedial activities and were intended to be deed restrictions unless the City or other entity adopts and maintains an enforceable ordinance. As such, the City adopted Ordinance No. 97-10 in November 1997. The 2015 FYR found that the ordinance was not sufficiently robust because it did not require property owners to abandon their private wells; however, new structures were required to connect to the municipal water supply. No penalties were associated with violating the ordinance. Outside of the restricted area covered by the ordinance, residents could install wells but were required to connect to water lines when the private well fails or as municipal lines become available. The District #10 Health Department, which includes Wexford County, requires well installation permits and well location approval. MDEQ provides drilling contractor and pump installer registrations.

As a result of the 2015 FYR recommendations, the City strengthened Ordinance 97-10, now identified as Ordinance No. 2018-10 (*Ordinance Amending Chapter 42, Article II, Division 6 of the Cadillac City Code Regarding Institutional Controls for the Kysor Industrial Corporation/Northern Plating Company*) and is provided in Attachment 1 (Appendix D). The amended ordinance, effective September 24, 2018, provides comprehensive requirements for the abandonment of existing wells, prohibits the installation and use of wells in the restricted zone, and prohibits groundwater use from and within the restricted zone. The ordinance provides exceptions for wells within the restricted zone under certain circumstances, such as if the well is needed for groundwater monitoring under a response activity, construction dewatering, non-contact processing operations (i.e., heating or cooling), or public emergencies. The ordinance also specifies enforcement procedures and penalties. Figure 16 (Appendix C) shows a map of the area covered by the ordinance.

Haring Township

The previous FYR report discussed and provided the April 1986 Haring Township Water Ordinance No. 86-01 and both of its 2012 and 2013 amendments. The Ordinance required property owners to connect to the Township water system if they were within 200 feet of a water line. Residents were still allowed to keep and use their private wells. In areas where public water lines are available, new users are required to hook up to the public system.

On October 30, 2012, Haring Township amended Ordinance No. 86-01 (Ordinance No. 12-1016) to require property owners to pay connection fees to the Township water system and to use water meters. Haring Township amended the Ordinance a second time on June 10, 2013 (Ordinance No. 2013-81) to require mandatory connection of all structures to the distribution line at the owner's expense if the water line is within 200 feet from the nearest point of the structure. The ordinance also prohibits the installation and use of private wells on premises already connected to the water system. If a property is required to connect to the system, then private wells must be abandoned, except for those within a

closed geothermal heating system or those required for groundwater monitoring. The ordinance established a timeline and procedures for achieving compliance, and the assessment of penalties.

On June 28, 2017, Haring Township adopted Ordinance 2017-107 (Attachment 2, Appendix D), which is a comprehensive ordinance integrating the original Ordinance 86-01 and all amendments thereto. The revision was to accommodate residents located along the 39 Road loop, where a new water transmission main was installed in 2017. The provision allowed residents with private wells to maintain the well and postpone connecting to the Township system until the well fails, or if there is a change of ownership or change of use, at which time the owner is required to connect. This was a one-time allowance for specific residents to provide consistency between commercial and residential requirements as well as between the sewer and water ordinance with respect to mandatory connection distances and the establishment of connection charges. All other residential and commercial users must connect to the water system if they are located within 200 feet, as stated previously. Figure 17 (Appendix C) shows the area covered by the ordinance.

It should be noted that the Haring Township water supply includes those areas, such as the North Park subdivision, that were previously and/or potentially affected by the Kysor Site groundwater contamination plumes. Residences located outside the Township water supply use private wells; however, these wells are not located in the plume area.

Current Compliance:

The FYR Site inspection (July 29, 2020) as well as daily, weekly and monthly O&M activities and Site visits by local entities, such as the City of Cadillac and Haring Township utilities departments provide the opportunity to assess and ensure that no inconsistent uses occur on or near the Site. Further, based on recent discussions in June and July 2020 with the City, Haring Township officials and consultants, the Wexford County Health Department and EGLE, EPA is not aware of Site or media uses that are inconsistent with the stated objectives of the ICs provided in Table 1. The ICs have been effective to date, and no compliance issues are apparent.

IC Follow-up Actions Needed:

Long-term protectiveness at the Site requires compliance with use restrictions to assure the remedy continues to function as intended. Since compliance with ICs is necessary to assure the long-term protectiveness of the remedy, planning for long-term stewardship (LTS) is required. LTS will ensure that effective ICs are maintained, monitored and enforced, and that the remedy continues to function as intended with regard to ICs. To assure proper maintenance and monitoring of effective ICs, LTS procedures will be included in the Institutional Control Implementation and Assurance Plan (ICIAP) being developed by the LDFA. The LTS procedures include regular inspection of ICs and annual certification to EPA and EGLE that ICs are in place and effective.

Systems Operations/Operation & Maintenance

The City of Cadillac water utilities on behalf of the LDFA performs routine O&M activities for remedial system components, including regular groundwater monitoring in accordance with the O&M manual and has maintained the existing institutional and Site access controls.

The groundwater treatment plant is checked daily. If a problem is detected, such as low EW withdrawal rate, then maintenance/cleaning of the well is performed. Each EW vault and panel are opened and

checked thoroughly three-to four times annually. All equipment, such as the blowers, well pumps, both air stripping towers are checked three times per week. The towers are generally cleaned out every three years. Maintenance performed on the remediation system in 2019 included:

- Replacing the pump and motor for EW S-6;
- Replacing the pump and motor for EW I-1, and;
- Replacing the reversing contactor on Tower 1 level control valve.

Remediation Facility Flow

The effluent from the discharge sump of the groundwater remediation facility, which includes flow from the Cr(VI) treatment system⁶, is monitored at the discharge weir by an ultrasonic type level indicator. The effluent flow is continually recorded on a circular chart recorder located at the main control panel. Approximately 3,254 million gallons were treated from January 2016 to December 2019.

Air Stripping System Performance

The PTAS system is sampled at the influent, midpoint, and effluent of the system. Grab samples for VOCs analyses are collected in accordance with the Performance Monitoring Plan (PMP).⁷ Analytical methodologies are in accordance with the PMP and the Quality Assurance Project Plan (QAPP). Weekly samples of the influent and effluent, along with monthly samples of the midpoint are collected and submitted to the analytical laboratory.

As shown in Table 2, about 1,414 pounds of VOCs were removed from groundwater from January 2015 to December 2019. During this reporting period, the VOC concentration of extracted groundwater delivered to the treatment system ranged from 25 to 92 µg/L. A summary of influent VOC concentration ranges, total system flow and annual VOC removal are provided in the table below. Since 2008, influent concentrations have declined with a corresponding decline in VOC mass removed by the system.

Table 2: Mass Removal of VOCs from Groundwater from 2015-2019

Year	Minimum Influent Total VOC Concentration (µg/L)	Maximum Influent Total VOC Concentration (µg/L)	Total System Flow (million gallons)	Estimated Mass of VOCs Removed (lbs)
2015	31	70	826.9	369
2016	29.2	75.1	867.4	350.1
2017	33.4	92.3	844.0	299.2
2018	25.0	42.8	782.7	210.8

⁶ The dedicated chromium EW (1-5) is still being pumped, however, because there is no longer chromium in the groundwater, the influent is now piped to bypass the carbon treatment before mixing with the influent from the VOC EWs. The combined streams flow into the PTAS. The effluent streams are discharged to the Clam River.

⁷ Under the UAO issued by EPA in May 1990 for the RD, a PMP was prepared to define the data collection requirements for evaluating the effectiveness of the RA. An annual report is required by the PMP to summarize the data and to verify that RAOs are met.

2019	17.3	37.3	760.0	184.4
2015-2019 Totals			4,081	1,413.6

In order to track the comingled VOC contaminant plumes, the City on behalf of the LDFA monitors 21 shallow and 21 intermediate MWs and EWs each year for VOCs. The shallow wells are screened between 35 and 70 feet bgs and intermediate wells are screened between 120 to 200 feet bgs. Although the cleanup level for Cr(VI) has been met, the City continues to monitor for chromium. Data confirm that chromium is either not detected or remains below the TCLs at all monitored locations. Figure 7 (Appendix C) provides the locations of the MWs. These wells are sampled on quarterly and annual cycles. The breakdown of wells sampled is as follows:

Quarterly:

- 13 chromium MWs
- 3 VOC MWs (fringe wells F-6, CMS-5W, SLW-34)

Annually in September:

- 15 chromium MWs
- 15 shallow VOC MWs
- 10 intermediate VOC MWs
- 17 EWs
 - 6 shallow wells (S-1 through S-6)
 - 11 intermediate wells (I-1 through I-11)

Other measurements collected include:

- Annual static water level measurements at about 75 wells.
- Monthly sampling at the treatment plant of the influent, midpoint and effluent flows
- Weekly sampling of effluent for compliance with Discharge Monitoring Requirements

As part of O&M, the City monitors its municipal wells annually for regulated contaminants under the SDWA. No VOCs or other regulated contaminants have been detected in these wells over the past five years.

Haring Township

The Township water is supplied by two production wells, each over 260 feet deep. The Kysor Site COC concentrations are monitored every three years and no contaminants have exceeded detection levels. The most recent monitoring was conducted in 2017. As required by EGLE, the Township annually monitors contaminants regulated under the SDWA.

III. PROGRESS SINCE THE LAST REVIEW

This section includes the protectiveness determinations and statements from the last FYR as well as the recommendations from the last FYR and the current status of those recommendations.

Table 3: Protectiveness Determinations/Statements from the 2015 FYR

OU #	Protectiveness Determination	Protectiveness Statement
Sitewide	Short-term Protective	The remedy at the Kysor Industrial Site is currently protective of human health and the environment because the implemented RAs are functioning as intended and exposure pathways that could result in unacceptable risks are being controlled. ICs are in place in the form of local ordinances for the City of Cadillac and Haring Township that prohibit groundwater use. The existing Site use is consistent with the RAOs set forth in the decision documents. However, in order for the remedy to be protective in the long term, the following actions need to be taken to ensure protectiveness: the City of Cadillac should strengthen its ordinance by requiring all current and future residents in the City to abandon private wells and connect to the municipal system; procedures should be implemented to monitor and track compliance with current and future ICs; and the reasons for the recent increases in TCE concentrations in EW 1-9 should be explored. Long-term protectiveness requires compliance with effective groundwater use restrictions. This will be accomplished by ensuring that effective ICs are in place and planning for LTS, which includes maintaining, monitoring and enforcing effective ICs.

Table 4: Status of Recommendations from the 2015 FYR

OU #	Issue	Recommendations	Current Status	Current Implementation Status Description*	Completion Date (if applicable)
Site-wide	ICs to restrict groundwater use should be strengthened.	EPA should again encourage the City of Cadillac to strengthen its ordinance by requiring all current and future residents in the City to abandon private wells and connect to the municipal system.	Completed	The City strengthened Ordinance 97-10, now identified as Ordinance No. 2018-10 and is provided in Attachment 1. The public notice of Proposed Ordinance and Public Hearing, as well as the Public Notice of Ordinance Adoption were published in the <i>Cadillac News</i> on September 7, 2018. The amended ordinance, effective September 24, 2018, provides comprehensive requirements for the abandonment of existing wells, and prohibits the installation and use of wells in the restricted zone, as well as a general prohibition of groundwater use from and within the restricted zone. The ordinance provides for exceptions to the use of wells within the restricted zone under certain circumstances, such as groundwater monitoring under a response activity, construction dewatering, non-contact processing operations (i.e., heating or cooling), or emergencies. Enforcement procedures and penalties are also specified.	9/24/2018

Site-wide	An LTS plan needs to be put in place.	The City of Cadillac and Haring Township should implement an LTS plan to include procedures for monitoring and tracking compliance with the ICs, communications procedures, and annual certification to EPA that ICs remain in place and are effective.	Ongoing	Although the City and Township monitor compliance with ICs, annual certification to EPA that ICs are in place and effective has not been performed consistently. The City is in the process of developing an ICIAP that will include the required LTS elements. The ICIAP is expected to be completed by 3/31/2021.	N/A
Site-wide	TCE concentrations at EW 1-9 have been increasing.	Explore the reasons for the recent increases in TCE concentrations in EW 1-9.	Completed	From 2012 through 2014, TCE levels increased from below 3.1 µg/L in 2012 to 21 µg/L in 2014. Since then, levels have declined and appear to have stabilized at about 17.5 µg/L for the last four years. The reason for the increase is not known but may be attributable to a pocket of TCE that was previously outside of the influence of I-9 and was eventually drawn in and is now being detected at higher levels. The TCE trend in this well will continue to be tracked as part of the groundwater monitoring program.	5/1/2020

IV. FIVE-YEAR REVIEW PROCESS

Community Notification, Involvement & Site Interviews

A public notice was made available by a newspaper posting in the *Cadillac News* on June 20-21, 2020, stating that EPA was beginning a FYR and inviting the public to submit any comments to EPA (Attachment 3, Appendix D). The same announcement was published on the Site's profile page, www.epa.gov/superfund/kysor-industrial. No comments were received. The results of the review and the report will be made available at the Site information repository located at the Cadillac-Wexford Library, 411 S. Lake St., Cadillac, Michigan and at www.epa.gov/superfund/kysor-industrial.

During the FYR process, the RPM conducted interviews with the following representatives: 1) Jeffrey Dietlin, Director of Utilities, City of Cadillac; 2) Robert Scarborough, Supervisor, Haring Township and Building Official, Wexford County; 3) Bob Polanic, Infrastructure Alternatives, Inc. O&M contractor, Haring Township utilities; 4) Nicholas Swiger, Project Manager, EGLE; 5) Tom Reichard, Environmental Health Director, Wexford County Health Department, District #10.

Mr. Dietlin indicated that the remedy has performed well and there have been no unusual O&M issues or costs over the past five years. There are no major changes to land use in the Cadillac area. The City currently has three industrial parks, including the CIP; however no notable industrial expansion has occurred. The water distribution system was expanded due to the addition of 250 acres to the City near Clam Lake Township. This acreage is undeveloped and currently supports a gas station and coffee shop.

The Cadillac Ordinance of September 2018 has been effective, and all residents are connected to the municipal water supply. Records indicate that all private wells have been abandoned. Mr. Dietlin indicated that occasionally they receive inquiries from residents living near the City borders regarding approval to install private wells. The City does not regularly check residents' water supply sources, but he can provide annual certification as he has in the past.

Also discussed were the other VOC plumes beneath the CIC released by three companies (Rexair, Mitchell-Bentley, and AAR) that contributed to the Site. These companies were not part of EPA enforcement actions but were being addressed by EGLE. Mr. Dietlin indicated that the Rexair plume is still likely being captured and treated by the LDFA system. Rexair appears to be maintaining its P&T system and shares quarterly monitoring data with the City and EGLE. EGLE reviews the data as per the consent judgement and work done under its agreement with Rexair. Rexair is in the process of shutting down their EWs as most of the VOC concentrations are below MCLs.

Mr. Swiger indicated that AAR has met the shutdown criteria for its EW but still uses it for non-contact cooling water. Once the operations change, they will begin post operation sampling. Mitchell-Bentley closed in 2014.

Mr. Scarborough discussed the Township status, indicating that the population has not changed significantly. The Township has not changed any land zoning, and, except for a new FedEx distribution center, no commercial or industrial entities have moved to the area. New private well permits are issued if the requestor provides a letter from the local water authority, and regulated by the Wexford County Health Department, District #10.

Mr. Polanic indicated that the Township expanded its water distribution system by 8,000 feet to accommodate the FedEx facility and additional fire suppression. The Township water supply includes those areas, such as the North Park subdivision, previously and potentially affected by the Kysor Site groundwater contamination plumes. Residents within the Township supply area have abandoned their wells, except those used for irrigation. A well driller must certify the well abandonment and submit logs. Residences located outside the Township supply use private wells; however, these wells are not located in the plume area. The private wells are not monitored by the Township or District #10 for the COCs. Mr. Reichard added that private well sample results (mainly bacteriological and nitrates) are tracked on GIS. Private wells are rarely screened for VOCs and very few local laboratories are able to perform the analyses.

Data Review

The following wells were sampled for VOCs between 2016 and January 2020 to evaluate performance of the remediation system as summarized below:

Shallow Aquifer Wells		Intermediate Aquifer Wells	
Extraction		Extraction	
S1	S4	I-1	I-6
S2	S5	I-2	I-7
S3	S6	I-3	I-8
		I-4	I-9

			I-5	I-10
Monitoring				
DNR-11	K-6-S			
F-1-S	K-11			
F-5	K-13			
FWMW-9	K-22			
FWMW-16	SLW-9			
GW-9S	SLW-25			
GW-11	SLW-31			
GW-14S				
		MW-10		

The primary COCs in groundwater are chlorinated compounds, specifically PCE, TCE, and 1,1,1-TCA. Monitoring results are discussed for the shallow and intermediate aquifers in the following sections. The evaluations are based on annual sampling of the compliance MWs and EWs and quarterly sampling of select MWs. All wells sampled during the reporting period from 2016 to 2019 are shown in Figure 7 (Appendix C).

Shallow Aquifer VOC Status

Two COCs, PCE and TCE, were detected in the shallow aquifer at concentrations greater than the TCLs of 1 µg/L and 5 µg/L, respectively. TCE consistently exhibits the highest concentrations. As shown in Table 5 below, the following wells had at least one detection of PCE or TCE during the reporting period that exceeded the TCL: MWs K-6S, GW-14S, SLW-9, SLW-31, and EWs S-2, S-3, S-4 and S-5.

Monitoring Wells

The maximum concentrations detected during the reporting period in the shallow MWs was 4.6 µg/L for PCE at SLW-9 and 50 µg/L for TCE at K-6S. Both of these results were from samples collected in 2016. The maximum detection of 1,1,1-TCA in the shallow aquifer was 15 µg/L at K-6S in 2016, which is well below the TCL of 200 µg/L. The most recent data (January 2020) for the shallow MWs indicates that none exhibited concentrations above the TCLs for PCE and TCE during the reporting period.

Extraction Wells

The maximum concentrations detected during the reporting period in the shallow EWs was 4.7 µg/L for PCE and 30 µg/L for TCE at well S-3 from 2016. The January 2020 data for the shallow extraction wells showed that well S-2 has PCE and TCE at concentrations of 1.4 and 6.5 µg/L, respectively (slightly above the TCLs), and well S-4 has PCE and TCE at concentrations of 1.3 and 18.4 µg/L, respectively. The maximum detection of 1,1,1-TCA in the shallow EWs during the monitoring period was 19.4 µg/L at S-2 in 2017, which is well below the TCL of 200 µg/L.

The extents of groundwater impacts for years from 1998 to 2020 in the shallow aquifer are illustrated in Figure 8 for PCE, Figure 9 for TCE, and Figure 10 for 1,1,1-TCA. The concentration contours represent the TCL for each COC and demonstrate a sustained decrease in the horizontal extent of impacts.

As of January 2020, PCE in the shallow aquifer is limited to the area between EWs S-2 and S-4 (Figure 8, Appendix C). This area contains PCE concentrations slightly above the TCL of 1 µg/L and represents a significant decrease of PCE concentrations and affected area within the CIP. The extent of TCE is also limited to the same area and is a significant improvement in groundwater quality relative to TCE in the

area. Detections of 1,1,1-TCA greater than 200 µg/L have not occurred in this aquifer since 2003 (Figure 10, Appendix C).

The current extent of shallow aquifer groundwater impacts is within the historically estimated capture zone⁸ depicted in Figures 8-10 (Appendix C). During this reporting period, there were no detections of PCE and TCE outside the capture zone that exceeded TCLs, with the exception of a TCE detection (12.5 µg/L) in GW-14S in 2017, on the western edge of the capture zone. Recent results (2018 and 2020) at this well are below the laboratory Practical Quantitation Limits (PQLs) for all VOCs.

Table 5: Shallow Wells with TCL Exceedances (2015-2020)

Well	COC	Highest Concentration (µg/L)	TCL (µg/L)	Year
F-5	PCE	1.1	1	2015
	TCE	5.2	5	
GW-14S	TCE	12.5	5	2017
SLW-31	PCE	3.9	1	2016
	TCE	25	5	
S-4	PCE	1.6	1	2016
	TCE	27.1	5	2017
S-5	PCE	1.6	1	2016
SLW-9	PCE	4.6	1	2016
	TCE	30	5	
K-6S	TCE	50	5	2016
	PCE	2.4	1	
S2	PCE	2.1	1	2016
	TCE	7.9	5	2017/2018
S3	PCE	4.7	1	2016
	TCE	30	5	

Shallow Aquifer VOC Trends

Analytical results for PCE, TCE, and 1,1,1-TCA for select years from 2012 through 2020 for key locations in the shallow aquifer are shown on the charts in Figure 11 (Appendix C). The contaminant concentrations at each location is plotted along the y-axis between 0 and 40 µg/L. Although *cis*-1,2-dichloroethene (*cis*-1,2,-DCE) is not a COC in the ROD, it is a daughter product in the dechlorination of PCE and TCE and is included in trend charts.

Generally, the 2020 VOCs concentrations in the shallow aquifer have decreased relative to the previous FYR period. None of the groundwater MWs sampled in January 2020 contain PCE or TCE above the TCLs. For EW S-2, PCE and TCE concentrations are stable and slightly above their respective TCLs. Extraction well S-4 concentrations of PCE are also stable at a concentration slightly above the TCL. The TCE concentrations in S-4 trend downward but are currently about three times higher than the TCL. A limited regression analysis was performed on S-2 and S-4 using TCE data as it is the predominant COC in most wells and typically is an indicator of increases and decreases of other contaminants for a given location. Graphs illustrating TCE concentration trends for shallow EWs S-2 and S-4 are presented in

⁸ The capture zone was estimated by analytical methods developed by Todd (1980) and Kasenow (1994) and was found to be generally consistent with the numerical modeling presented in the April 2007 *Hydraulic Capture Zone Evaluation*, prepared by S.S. Papadopoulos & Associates, Inc.

Appendix D⁹. A regression analysis was not performed for S-6 because TCE concentrations have been below the laboratory PQL since 2016.

Each graph includes a trendline that estimates the decline of VOCs over time. The analysis suggests that S-4 has the longest estimated time until TCE detections are consistently below the TCL, which is nine years (after March 2029). This is generally consistent with the projected date in the ROD of 2025. The analysis suggests that S-2 will reach the cleanup objectives in early 2022.

Intermediate Aquifer VOC Status

As shown in Table 6, both PCE and TCE were detected in the intermediate aquifer during this reporting period at concentrations exceeding the TCLs of 1 µg/L and 5 µg/L, respectively. TCE consistently exhibits the highest concentrations. The following intermediate aquifer wells had at least one detection of PCE or TCE that exceeded the TCL during the past five years: MWs DNR-14D, F-6, K-6D, K-16D, MW-10, MW-11, MW-18, GW-17, and EWs I-1 through I-6, I-9, and I-10.

Monitoring Wells

The maximum concentrations detected during the reporting period in the intermediate MWs were 2.5 µg/L for PCE at DNR-14D and 66.3 µg/L for TCE at MW-11. These results were from samples collected during the fourth quarter of 2017 and 2018, respectively. The January 2020 data for the intermediate MWs indicate that PCE concentrations range from ND to 2.5 µg/L, while TCE ranges from ND to 52.8 µg/L. The maximum detection of 1,1,1-TCA was 17.8 µg/L at DNR-14D in 2017, which is well below the TCL of 200 µg/L.

Extraction Wells

The maximum concentrations detected during the reporting period in the intermediate EWs were 4.4 µg/L for PCE at well I-2 and 130 µg/L for TCE at well I-10. These results were from samples collected during the fourth quarter of 2018 and 2016, respectively. The January 2020 data for the intermediate EWs indicate that PCE ranges from ND to 3.2 µg/L and TCE ranges from ND to 93.5 µg/L. The maximum detection of 1,1,1-TCA in the intermediate EWs during the reporting period was 13.7 µg/L at I-1 in 2020, well below the TCL of 200 µg/L.

The extent of groundwater impacts for select years from 1998 to 2020 for the intermediate aquifer are illustrated in Figure 12 for PCE, Figure 13 for TCE, and Figure 14 for 1,1,1-TCA (Appendix C). The boundary of the areas represents the TCL for each respective contaminant. The extent of impacts in the intermediate aquifer is limited to an area downgradient of the Ingraham Site, extending from I-9 and I-10 at the Ingraham Site to I-3 and I-11 north of 13th Street. This is within the historically estimated capture zone.

⁹ Monitoring data typically ranges from 1996 to 2020. The data used to generate trendline equations were based on the data distribution and the ability of the trendline equation to estimate present-day concentrations when compared to actual data. The trendlines were calibrated to the more recent data by excluding some of the early data generated during the first years after system startup.

Table 6: Intermediate Wells with TCL Exceedances (2015-2020)

Well	COC	Highest Concentration (µg/L)	TCL (µg/L)	Year
K-6D	<i>cis</i> -1,2-DCE*	68.3	-	2018
	PCE	1.6	1	
	TCE	20.6	5	
K-16D	<i>cis</i> -1,2-DCE*	180	-	2015
	PCE	2.2	1	2016
	TCE	28	5	2015
F-6	PCE	1.7	1	2017
	TCE	17.4	5	
DNR-14D	PCE	2.5	1	2017
	TCE	52.8	5	
GW-17	PCE	2.4	1	2017
	TCE	50.5	5	
MW-10	PCE	1.4	1	2018
MW-11	PCE	1.3	1	2019
	TCE	66.3	5	2018
MW-18	PCE	1.0	1	2018
	TCE	27.2	5	
I-1	1,1-DCE	5.5	5	2015
	PCE	3.7	1	2018
	TCE	86	5	2015
I-2	PCE	4.6	1	2015
	TCE	41	5	
I-3	PCE	1.1	1	2015
	TCE	33	5	
I-4	PCE	15	1	2015
	TCE	16	5	
I-5	PCE	2.5	1	2015
	TCE	8.4	5	
I-6	PCE	2.5	1	2018
	TCE	6.5	5	
I-9	TCE	19	5	2015/2016
I-10	PCE	1.6	1	2015
	TCE	150	5	
I-11	PCE	1.1	1	2015

**cis*-1,2-DCE is not a COC, but was included as a daughter product of TCE

Intermediate Aquifer VOC Trends

Analytical results for PCE, TCE, and 1,1,1-TCA for select years from 2012 through 2020 for key locations are shown on charts in Figure 15 (Appendix C). Generally, the recent concentrations of VOCs are trending downward or remaining stable when compared to the previous reporting period, as highlighted below:

- I-1: TCE detections further decreased from 86 µg/L in the 2015 report to 44 µg/L, continuing a decreasing trend dating to 1999 when the concentration was 670 µg/L.

- I-2: PCE, TCE and 1,1,1-TCA concentrations that were reported to increase by 11 µg/L or less during the previous FYR period are now declining.
- I-4: PCE results have trended to ND and TCE results have further declined.
- I-5: PCE and TCE detections that were reported to increase by less than 2 µg/L in the 2015 report are now declining.
- I-9: TCE concentrations peaked at 21 µg/L in 2014, but have since trended downward to the current concentration of 17.7 µg/L
- I-10: PCE detections declined from slightly above the TCL to ND. TCE remains elevated at 93.5 µg/L but is steadily declining.

The following wells showed exceptions to the downward trend:

- I-6: TCE detections have increased slightly from 5.4 µg/L in 2015 to the current result of 7.1 µg/L.
- K-16D: TCE detections have declined significantly from the middle to late 2000s but have recently increased from 9.4 µg/L in 2016 to 24.2 µg/L in 2020.

A regression analysis was performed for several EWs, except I-5 because TCE concentrations have been below the laboratory PQL since 2018. The *2015 Performance Monitoring Report* estimated a cleanup date of September 2016. Plots for contaminant concentration over time are included in Appendix E.

The trends seen in the intermediate aquifer wells suggest earlier cleanup times for the area along the eastern and northern portions of the affected area. Most locations are estimated to be below the TCL by 2031. Under present operating conditions, EWs I-1 and I-10 appear to have the longest estimated cleanup times until TCE detections are consistently below the TCL. Extraction well I-1 and I-10, near the former Ingraham Site, are estimated to be below the TCL in 2038 and 2048, respectively. The regression analysis for I-5 during the previous FYR estimated a cleanup date of December 2016 and results at this well for TCE have been below the TCL of 5 µg/L since 2018. The 1989 ROD estimated a cleanup time of 64 years for the intermediate aquifer. Based on the remediation startup date of 1996, the cleanup date in the ROD was estimated to be 2060. The maximum 2016-2019 Compendium Performance Monitoring LDFA cleanup date was estimated to be 2046, as per the April 2007 *Hydraulic Capture Zone Evaluation*, prepared by S.S. Papadopoulos & Associates, Inc.

Emerging contaminants

1,4-Dioxane

In October 2017, MDEQ promulgated the 1,4-Dioxane (1,4-D) rule that established Part 201 residential Drinking Water Criterion (DWC) of 7.2 µg/L. A federal drinking water standard for 1,4-D has not been established. EPA's current tap-water regional screening level is 0.46 µg/L based on a carcinogenic risk of 1×10^{-6} and EPA risk assessments indicate that the drinking water concentration representing a 1×10^{-6} ELCR level is 0.35 µg/L. 1,4-D is a likely contaminant at sites where 1,1,1-TCA is detected because of the widespread use of 1,4-D as a stabilizer. Since 1,1,1-TCA is a COC at the Site, 1,4-D should be added to the list of analytes for the City and Township during the next round of sampling.

Polyfluoroalkyl Substances (PFAS):

Per- and polyfluoroalkyl substances (PFAS) are a group of chemicals that are resistant to heat and water and have been used in many industrial applications and consumer products, such as carpeting, waterproofing, upholstery, fire-fighting foams, and metal plating. In 2018, the EGLE's Remediation and Redevelopment Division established cleanup criteria for groundwater used as drinking water of 70 parts-per-trillion (ppt) of perfluorooctanoic acid (PFOA) and perfluorooctanesulfonic acid (PFOS), individually or combined. Michigan recently promulgated new drinking water and groundwater cleanup standards of 8 ppt for PFOA and 16 ppt for PFOS, which became effective August 3, 2020.

In 2018 and 2019, the City sampled the Crosby Street and 44th Street wells and did not detect PFAS in any of the wells. The City's wastewater treatment plant showed low levels of PFAS. No Site MWs are currently tested for PFAS; however, since PFAS is a common contaminant at plating sites, a select few wells around the Northern Plating Site should be sampled for PFAS.

PFAS sampling for Haring Township was conducted last year under the State-wide sampling program. Township monitoring results showed PFAS at concentrations below the State cleanup standards.

Site Inspection

The FYR Site inspection was conducted on July 29, 2020 by EGLE Site Manager, Nick Swiger on behalf of EPA. In attendance were David Stearns, Assistant to the Director of Cadillac Utilities, and other staff from the Cadillac Utilities. The purpose of the inspection was to assess the protectiveness of the remedy. The inspection focused on the groundwater EWs, groundwater treatment systems, and effluent discharge. The overall Site was found to be in good condition. The treatment facility appeared to be well-maintained and all machinery, equipment and electrical panels are labeled. No problems or issues impacting the current or future protectiveness of the remedy were evident. The completed inspection checklist and photos are provided in Attachment 4.

V. TECHNICAL ASSESSMENT

QUESTION A: Is the remedy functioning as intended by the decision documents?

Question A Summary:

Yes. Based upon review of the relevant documents; RPM interviews with the respective utilities departments and local officials from the City of Cadillac and Haring Township, and the EGLE Site representative; groundwater data for the MWs and EWs, and the data from the groundwater remediation system; and O&M reports, the remedy continues to operate as designed and the plume appears to be contained. The PTAS system continues to operate at 99 percent efficiency. The average influent concentration exceeds the TCLs for PCE and TCE but the discharge from the system exhibits no detectable concentrations of COCs and is in compliance with the NPDES Permit.

The LDFA treatment system is meeting the RAOs in a reasonable time frame as per the ROD. The monitoring data suggests that the shallow aquifer will take about nine years until TCE concentrations are consistently below the TCL, which is generally consistent with the ROD projection of 2025. The

analysis suggests that S-2 will reach the cleanup objectives in early 2022, which is about one year longer than estimated in the *2015 Performance Monitoring Report*.

The 1989 ROD estimated a cleanup time of 64 years for the intermediate aquifer. Based on the remediation startup date of 1996, the cleanup date in the ROD was estimated to be 2060. The maximum Cadillac LDFA cleanup date was estimated to be 2046 in the April 2007 *Hydraulic Capture Zone Evaluation*, prepared by S.S. Papadopoulos & Associates, Inc. Analysis using concentrations measured in this FYR period indicate that TCLs could be met by 2048, which is consistent with that 2007 estimate.

The groundwater monitoring programs have been effective. The combined monitoring networks of the LDFA, Rexair and adjacent remedies provide an opportunity for wide spatial coverage of groundwater hydrogeology and water quality data for remedy evaluation. As stated in the previous FYR, the groundwater P&T system for the Kysor Site is capturing and treating contaminated water from the Rexair Site, which may increase the time needed to achieve groundwater TCLs.

Opportunities for Optimization

Influent VOC concentrations to the PTAS units are approaching very low levels, allowing for the system to be optimized to retain efficacy while reducing system O&M resources. The LDFA is exploring the possibility of using only one of the towers in the future based on the projected plume contraction. The LDFA plans to study the efficiency of one PTAS tower while the other tower is taken offline for cleaning. Other optimization opportunities include discontinued treatment of influent from the dedicated I-5 chromium EW prior to combining with VOC influent before entering the PTAS.

Operation and Maintenance

The groundwater treatment system continues to operate as designed. The current O&M procedures are effectively maintaining the remedy. Operation and maintenance costs have been stable over the past five years and no potential problems with the remedy are evident. There are no indicators of potential issues. Monitoring is conducted according to the O&M plan.

Implementation of ICs and Other Measures

Access controls in the form of perimeter fencing, locked gates, and signs are in place and effectively maintained. Since the 2015 FYR, ICs have been implemented via local ordinances in 2017 and 2018 for the City of Cadillac and Haring Township, respectively. Both ordinances are more robust and enforceable in prohibiting groundwater use; however, both the City and Township need to ensure LTS procedures are implemented to assure the protectiveness of the remedy. These procedures include tracking and certifying compliance with ICs to the agencies and identify the need for future ICs. This can be achieved by developing and implementing an ICIAP.

QUESTION B: Are the exposure assumptions, toxicity data, cleanup levels, and RAOs used at the time of the remedy selection still valid?

Question B Summary:

No. There have been no changes in the physical conditions at the Site or land use that have occurred that would affect the protectiveness of the remedy. Further, no changes in ARARs, standards, to-be-considereds (TBCs), TCLs, or risk assessment methodologies have been adopted that would affect or question the protectiveness of the remedy. There have been no changes in toxicity, other contaminant characteristics, and standardized risk assessment methodologies that could affect the protectiveness of

the remedy. As previously mentioned, Michigan promulgated new drinking water and groundwater cleanup standards of 8 ppt for PFOA and 16 ppt for PFOS, which became effective August 3, 2020.

No ecological routes of exposure or receptors have been newly identified or changed in a way that could affect the protectiveness of the remedy. Groundwater ingestion of COCs was the principle exposure pathway of concern at the Site, which has been mitigated under the 1989 ROD-selected remedy. The presence of COCs in groundwater beneath the CIP indicated that the VI pathway should be considered as a potential exposure pathway. Recent groundwater monitoring data from the shallow and intermediate wells were analyzed with respect to the VI screening levels and criteria. Potential receptors in the CIP were also identified. The current assessment indicates that VI is an unlikely exposure pathway of concern. This pathway will continue to be evaluated with respect to regular groundwater monitoring data.

As previously discussed, the potential presence of two emerging groundwater contaminants, 1,4-D and PFAS should be assessed to determine if they are present at elevated concentrations; however, the principle exposure pathway has been largely eliminated through the implementation of the P&T system, monitoring, and ICs. The remedy is progressing and is expected to meet all cleanup goals of the 1989 ROD within a reasonable time frame.

QUESTION C: Has any other information come to light that could call into question the protectiveness of the remedy?

No. There is no new information that has come to light that could impact the protectiveness of the remedy.

VI. ISSUES/RECOMMENDATIONS

Issues/Recommendations				
Issues and Recommendations Identified in the Five-Year Review:				
OU 1 and Sitewide	Issue Category: Institutional Controls			
	Issue: LTS procedures should be developed and implemented to ensure that the required ICs are effective and properly maintained, monitored, and enforced.			
	Recommendation: The City of Cadillac and Haring Township should develop an ICIAP which includes a plan for: 1) IC evaluation activity; 2) taking corrective measures to existing ICs, if needed; 3) placing additional ICs, if needed; and, 4) ensuring the LTS of the Site, which includes ongoing monitoring, maintenance, and enforcement of ICs.			
Affect Current Protectiveness	Affect Future Protectiveness	Party Responsible	Oversight Party	Milestone Date
No	Yes	Other	EPA/State	3/31/2021

OU 1 and Sitewide	Issue Category: Monitoring			
	Issue: No Site MWs are currently tested for PFAS; however, since PFAS is a common contaminant at plating sites, a select few wells around the Northernair Plating Site should be sampled for PFAS.			
	Recommendation: The LDFA should work with EGLE to identify and monitor appropriate wells for PFAS compounds. Confirmatory sampling should be conducted at locations where the groundwater concentrations exceed the newly promulgated Michigan criteria of 8 ppt for PFOA and 16 ppt for PFOS.			
Affect Current Protectiveness	Affect Future Protectiveness	Party Responsible	Oversight Party	Milestone Date
No	Yes	PRP	EPA/State	6/30/2021

OU 1 and Sitewide	Issue Category: Monitoring			
	Issue: 1,4-D is a likely contaminant at sites and may be present particularly when 1,1,1-TCA is detected because of the widespread use of 1,4-D as a stabilizer.			
	Recommendation: Since 1,1,1-TCA is a COC at the Site, 1,4-D should be added to the list of analytes for the City and Township during the next round of sampling. 1,4-D analysis should continue at locations where concentrations exceed Michigan and EPA residential drinking water screening levels.			
Affect Current Protectiveness	Affect Future Protectiveness	Party Responsible	Oversight Party	Milestone Date
No	Yes	PRP	EPA/State	12/31/2020

OU 1 and Sitewide	Issue Category: Other			
	Issue: Although it is unlikely that VI is a potential exposure pathway for an industrial worker within the Cadillac Industrial Park based on VI assessments performed to date, this should continue to be tracked.			
	Recommendation: The LDFA should continue to evaluate the regular groundwater monitoring data with respect to the nonresidential VI screening criteria in the Cadillac Industrial Park to ensure that VI is not a potential exposure pathway.			
Affect Current Protectiveness	Affect Future Protectiveness	Party Responsible	Oversight Party	Milestone Date
No	Yes	PRP	EPA/State	12/31/2020

VII. PROTECTIVENESS STATEMENT

OU 1 and Sitewide Protectiveness Statement	
<i>Protectiveness Determination:</i> Short-term Protective	
<i>Protectiveness Statement:</i> The remedy at the Kysor Site currently protects human health and the Environment. The implemented remedy is functioning as intended and no complete exposure pathways are present that could result in unacceptable risks to human health or the environment. The P&T system is reducing concentrations of COCs in both aquifers, and ICs are in place in the form of local ordinances for the City of Cadillac and Haring Township which prohibit groundwater use. The existing Site use is consistent with the RAOs set forth in the ROD. In order for the remedy to be protective in the long term, the following actions need to be taken to ensure protectiveness: 1. The City of Cadillac and Haring Township should develop an ICIAP which includes a plan for: IC evaluation activity; taking corrective measures to existing ICs, if needed; placing additional ICs, if needed; and, ensuring LTS of the Site, which includes ongoing monitoring, maintenance, and enforcement of ICs. 2. 1,4-Dioxane should be added to the list of analytes for the City and Township during the next round of groundwater sampling. Analyses should continue at locations where 1,4-D concentrations exceed the State of Michigan and EPA residential drinking water screening levels. 3. Since PFAS is a common contaminant at plating sites, a select few wells around the Northernnaire Plating Site should be sampled for PFAS. Confirmatory analyses should be conducted as necessary. 4. The LDFA should continue to evaluate the regular groundwater monitoring data with respect to the nonresidential VI screening criteria in the Cadillac Industrial Park to ensure that VI is not a potential exposure pathway.	

VIII. NEXT REVIEW

The next FYR report for the Kysor Industrial Corp. Superfund Site is required five years from the completion date of this review.

APPENDIX A – REFERENCE LIST

Charter Township of Haring, Wexford Co., 1986. Michigan Ordinance No. 86-01 Water Ordinance (Ord. No. 86-01, Sect. 2.1). SDMS ID# 494187

Charter Township of Haring, Wexford Co. October 30, 2012 Amendment to Michigan Ordinance No. 86-01 Water Ordinance (12-1016). See Fourth Five-Year Review, SDMS ID# 494193.

Charter Township of Haring, Wexford County, MI. June 10, 2013 Amendment to Michigan Water Ordinance (2013-81)—Amendment to Ordinance No. 86-01, As Amended. See Fourth Five-Year Review, SDMS ID# 494193.

Charter Township of Haring, Wexford County, MI. Ordinance No. 2017-107: *Water Ordinance*, SDMS ID# 549551.

City of Cadillac, 1997. Ordinance No. 97-10, ch.25, sect. 2.300), Title II, Chpt. 24, Water Supply and Wastewater Systems, Recertification in August 2005 and July 2015. See Fourth Five-Year Review, SDMS ID# 494193.

Cadillac Ordinance No. 2018-10: *Ordinance Amending Chapter 42, Article II, Division 6 of the Cadillac City Code Regarding Institutional Controls for the Kysor Industrial Corporation/Northernair Plating Company*. Sept. 4, 2018, SDMS ID# 943505.

Gosling Czubak Engineering Sciences, Inc., New Well Location Report for the Charter Township of Haring Wellhead Protection Program, September 2017, SDMS ID# 959011.

S.S. Papadopoulos and Associates, Inc., April 2007. Hydraulic Capture Zone Evaluation, Kysor Northernair Superfund Site, Cadillac, Michigan, prepared for U.S. EPA.

State of Michigan, Department of Environmental Quality, June 2017. Authorization to Discharge under the National Pollutant Discharge Elimination System to the Clam River by the Cadillac Financing Authority GWCU (Permit No. MI0054925), SDMS ID# 943506.

Tetra-Tech 2015 Annual Performance Monitoring Report, Cadillac Local Development Finance Authority, Northernair/Kysor Sites, Cadillac, Michigan, February 2017. SDMS ID# 549553.

Tetra-Tech 2016-2019 Compendium Performance Monitoring Report, Cadillac Local Development Finance Authority, Northernair/Kysor Sites, Cadillac, Michigan, May 2020. SDMS ID# 959012.

U.S. EPA, September 29, 1989. Record of Decision, Northernair/Kysor sites, OU2, SDMS ID# 143861.

U.S. EPA, March 3, 1994. Explanation of Significant Differences, Northernair/Kysor Sites, OU2, SDMS ID# 234897.

U.S. EPA, January 30, 1995. Unilateral Administrative Order for Remedial Action, RA-V-W-95-C-282. SDMS ID# 143919.

U.S. EPA, August 31, 2010. Third Five-Year Review Report, Kysor Industrial Corp. Site. Cadillac, Michigan, SDMS ID# 373482.

U.S. EPA, August 31, 2015. Fourth Five-Year Review Report, Kysor Industrial Corp. Site. Cadillac, Michigan, SDMS ID# 494193.

U.S. EPA, September 21, 2018. Site-Wide Ready for Anticipated Use (SWRAU), Kysor Industrial Corp. Site, Cadillac, Michigan, SDMS ID# 516376.

APPENDIX B – SITE CHRONOLOGY

EVENT	DATE
Kysor Corporation begins operations	1955
Disposal practices at Kysor include dumping barrels of spent solvent directly on the ground behind the plant	As early as 1955 (exact date unknown) to 1980s
Four Winns (Frisbie Street) begins operations	1975
A.H. Joynt begins operations	1977
Voluntary excavation of some soils from waste pits behind Kysor plant and from nearby property	July 22 and 23, 1981
Hydrogeological Study, Kysor of Cadillac, Inc.	March 1982 (Phase I) and August 1983 (Phase II)
MDNR inspection of Four Winns (Frisbie Street)	1986
Re-proposal of Kysor site to NPL	June 24, 1988 (initially proposed in September 1985)
General notice letters and 104(e) requests sent	May 20, 1988
Feasibility Study for Cadillac area groundwater completed	August 1988
ROD (OU2) for groundwater cleanup signed	September 29, 1989
Final NPL listing	October 4, 1989
UAO issued for Remedial Design of OU2	May 16, 1990
Remedial Design Additional Studies report	October 1992
TCE detected above its MCL in city drinking well	1993
ESD signed	March 3, 1994
Two separate UAOs for RA issued (Kysor Corporation and Four Winns/A.H. Joynt)	January 30, 1995
Third UAO for RA issued (Northernair Plating)	April 11, 1995
On-site construction for OU2 begins	June 29, 1995
Groundwater extraction and treatment begins	August 1996
Final inspection of RA construction by EPA	September 19, 1996
Construction Completion	January 24, 1997
First FYR completed	July 26, 2000
Second FYR completed	September 30, 2005
Approval to city of Cadillac to shut down SVE system	April 16, 2010
Third FYR completed	August 31, 2010
Fourth FYR completed	August 31, 2015
City amends Ordinance 97-10, now identified as Ordinance No. 2018-10	September 24, 2018
EPA issues SWRAU	September 21, 2018
Fifth FYR Site inspection conducted	July 29, 2020

APPENDIX C - FIGURES

Figure 1 – Site Location Map

Figure 2 – Cadillac Industrial Park Area

Figure 3 – Haring Township Current Zoning Map

Figure 4 – City of Cadillac Zoning Map

Figure 5 – Haring Township Water Distribution System

Figure 6 – Map of Commingled Plumes and Well Locations

Figure 7 – Map of Monitoring Wells

Figure 8 – Tetrachloroethylene in Shallow Wells

Figure 9 – Trichloroethylene in Shallow Wells

Figure 10 – 1,1,1-Trichloroethane in Shallow Wells

Figure 11 – VOC Trends for Shallow Wells

Figure 12 – Tetrachloroethylene in Intermediate Wells

Figure 13 – Trichloroethylene in Intermediate Wells

Figure 14 – 1,1,1-Trichloroethane in Intermediate Wells

Figure 15 – VOC Trends for Intermediate Wells

Figure 16 – Cadillac Area Covered by Groundwater Restriction Ordinance

Figure 17 – Haring Township Area Covered by Groundwater Restriction Ordinance

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Figure 1

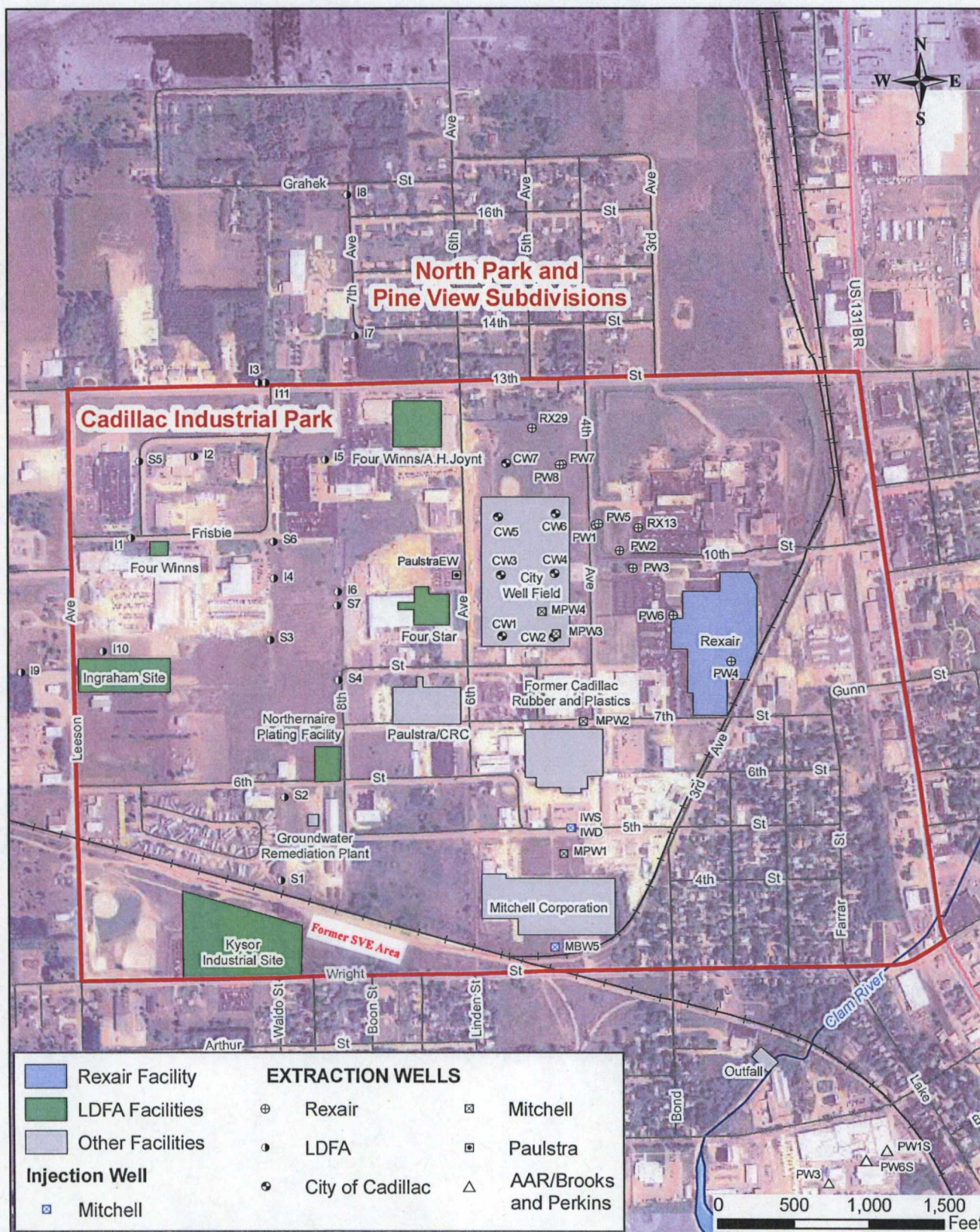


Figure 2 Cadillac Area Location Map

Figure 3: Haring Township Zoning Overlay

- Legend**
- Other Road
 - State Highway
 - Interstate Highway
 - Mile Marker
 - Railroad
 - Airport Runway
 - Hydrology
 - Township Boundary
 - Water Body
 - Parcel Boundary
 - AGRICULTURE
 - COMMERCIAL
 - FOREST/RECREATIONAL
 - INDUSTRIAL
 - LIGHT INDUSTRIAL
 - OFFICE RESIDENTIAL
 - RESIDENTIAL
 - RESORT/LAKE
 - PUD Overlay Zone



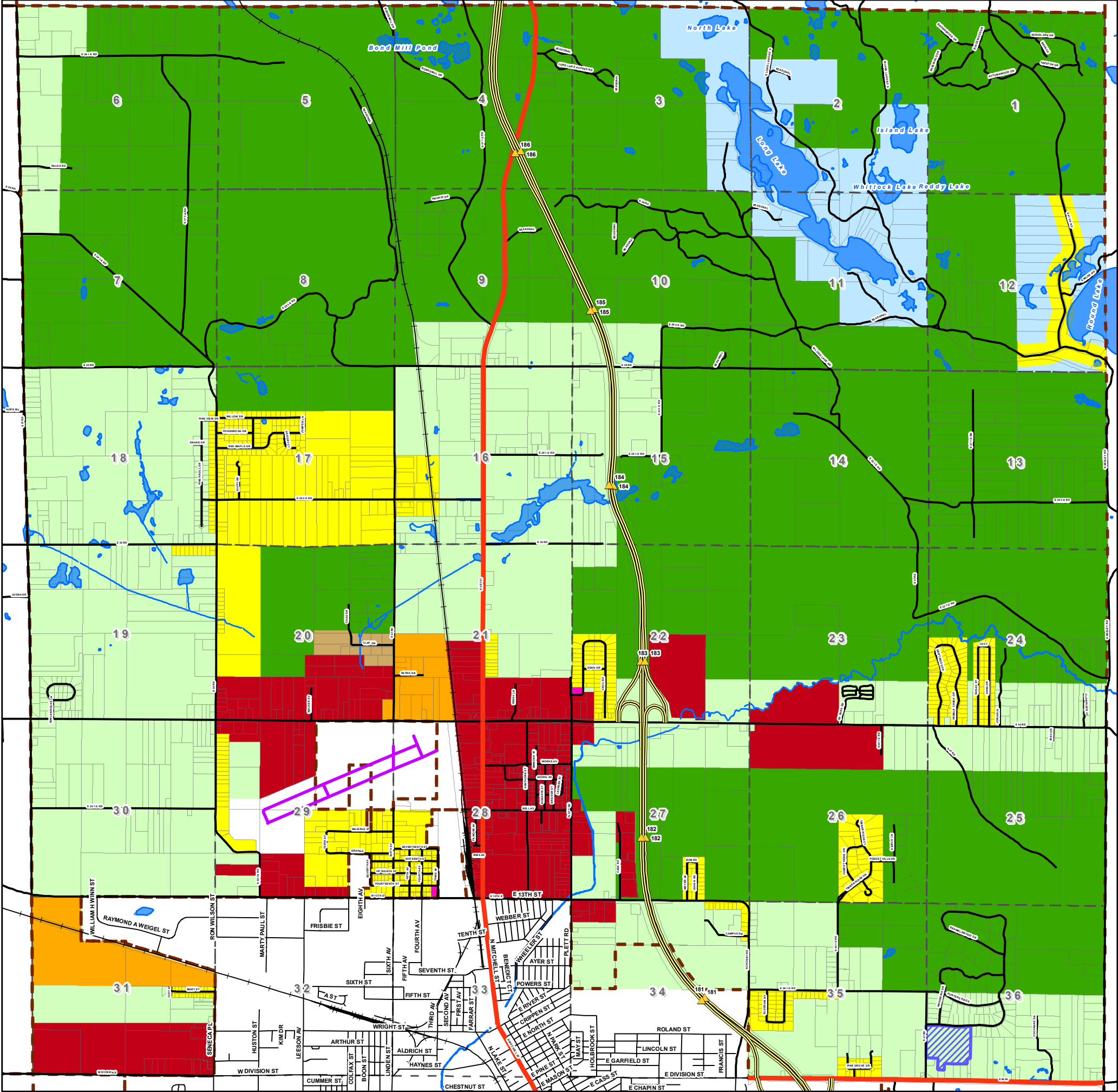
Adopted By the Haring Township Board of Supervisors.
Updated: July 15, 2016

0 3,075 6,150 Feet

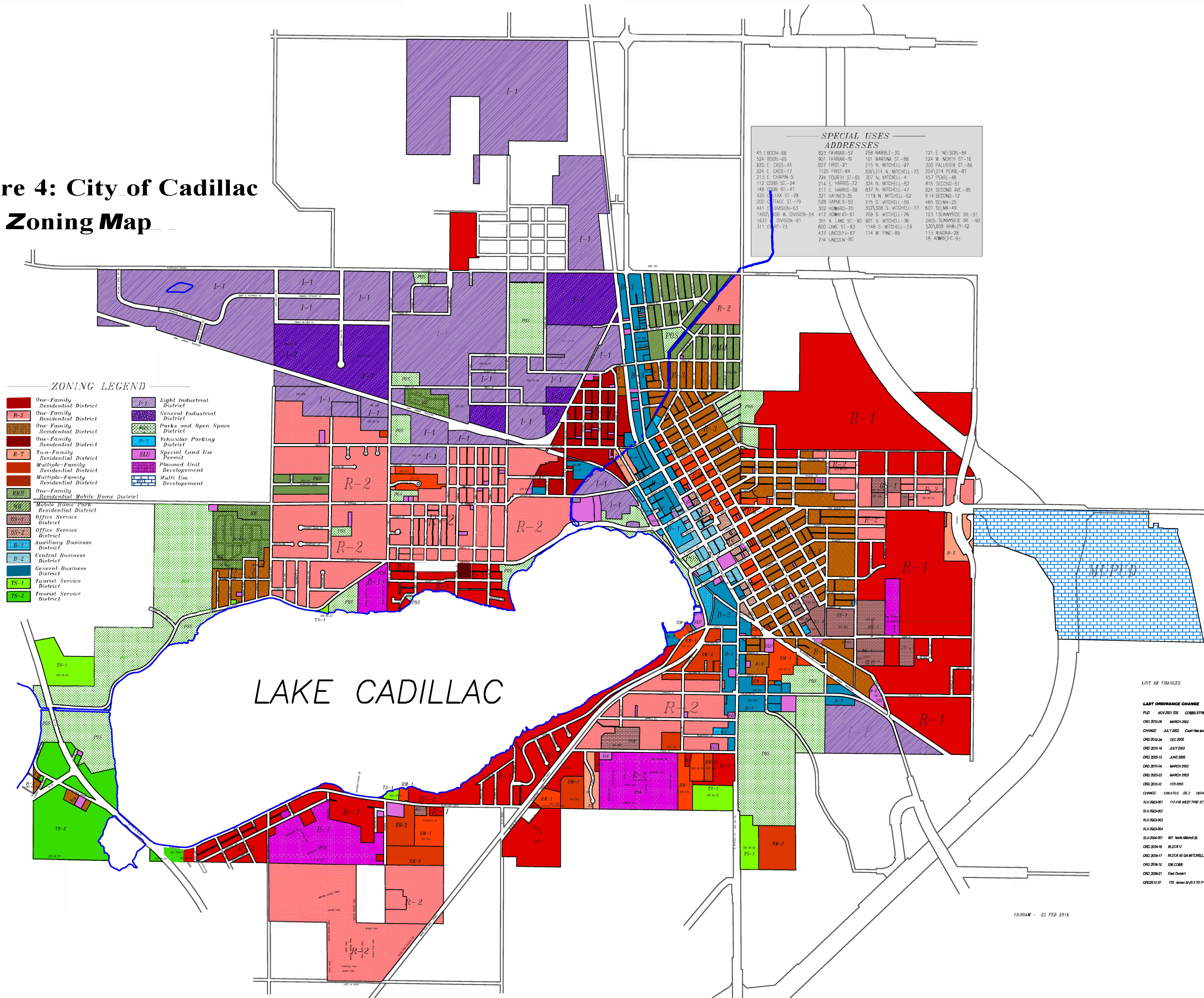
1 inch = 3,075 feet

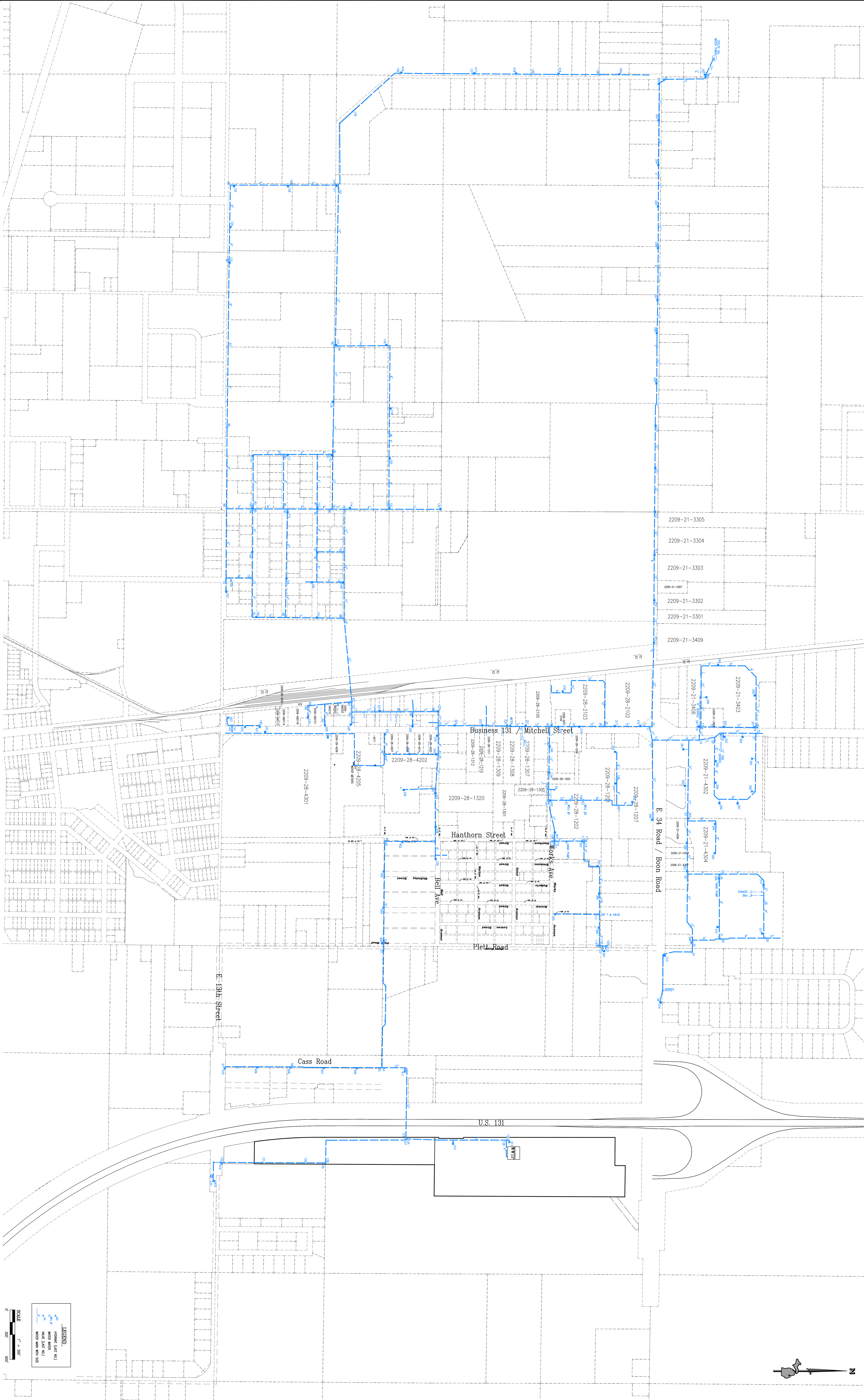
All communities in Wexford County are zoned. Cedar Creek and Haring Townships, as well as all incorporated areas, maintain their own zoning ordinance. Please contact their zoning administrator for current information.

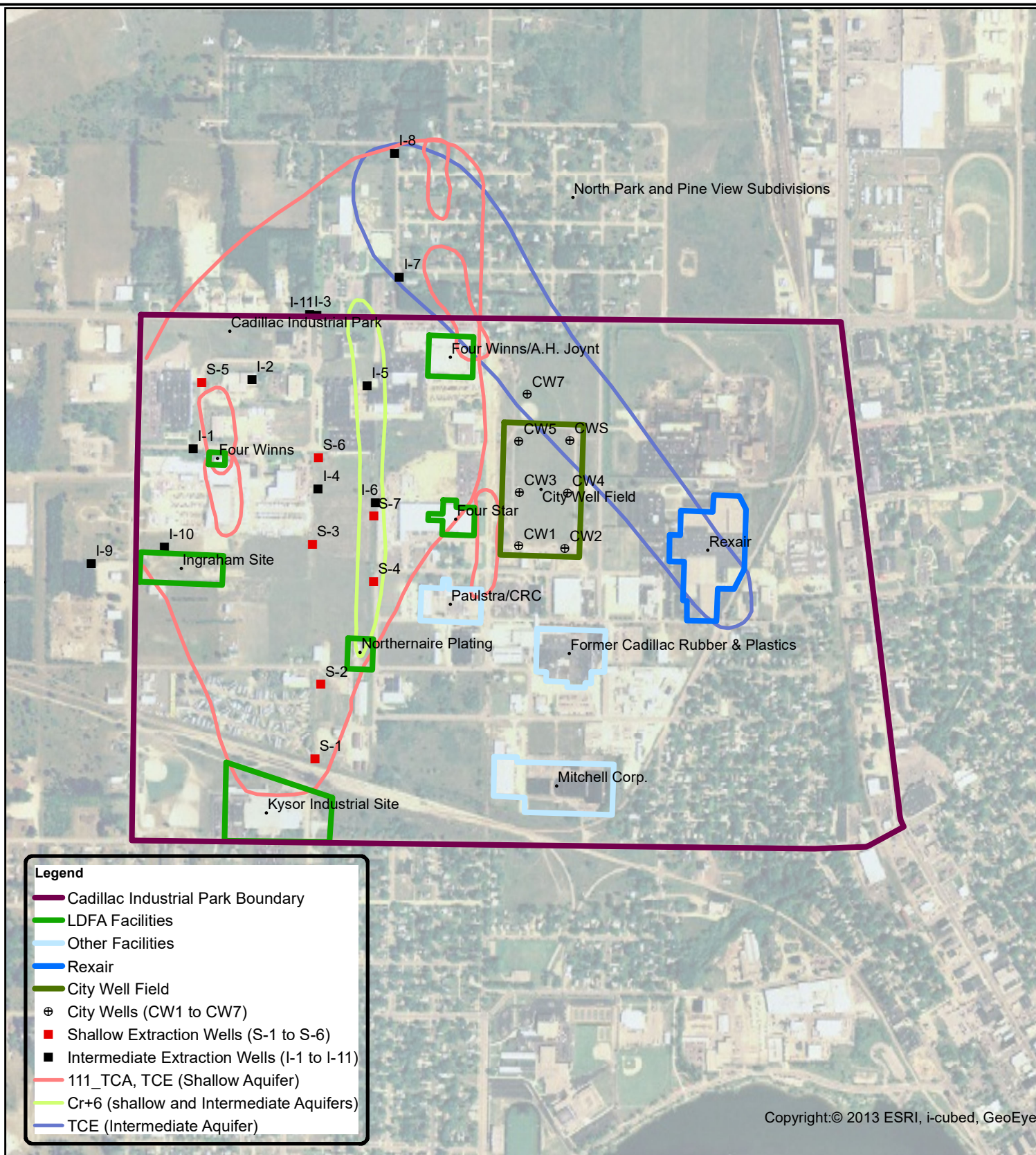
*Copyright Wexford County GIS 2017



**Figure 4: City of Cadillac
Zoning Map**







0 750 1,500 3,000 Feet

1 inch = 1,000 feet

Note: The illustrated generalized extent of groundwater contamination is based from the work of Weston (2003) and is illustrated solely for the purpose of spacial reference.



DESIGNED BY: KVM

DATE: 6/22/2020

CHECKED BY: DRB

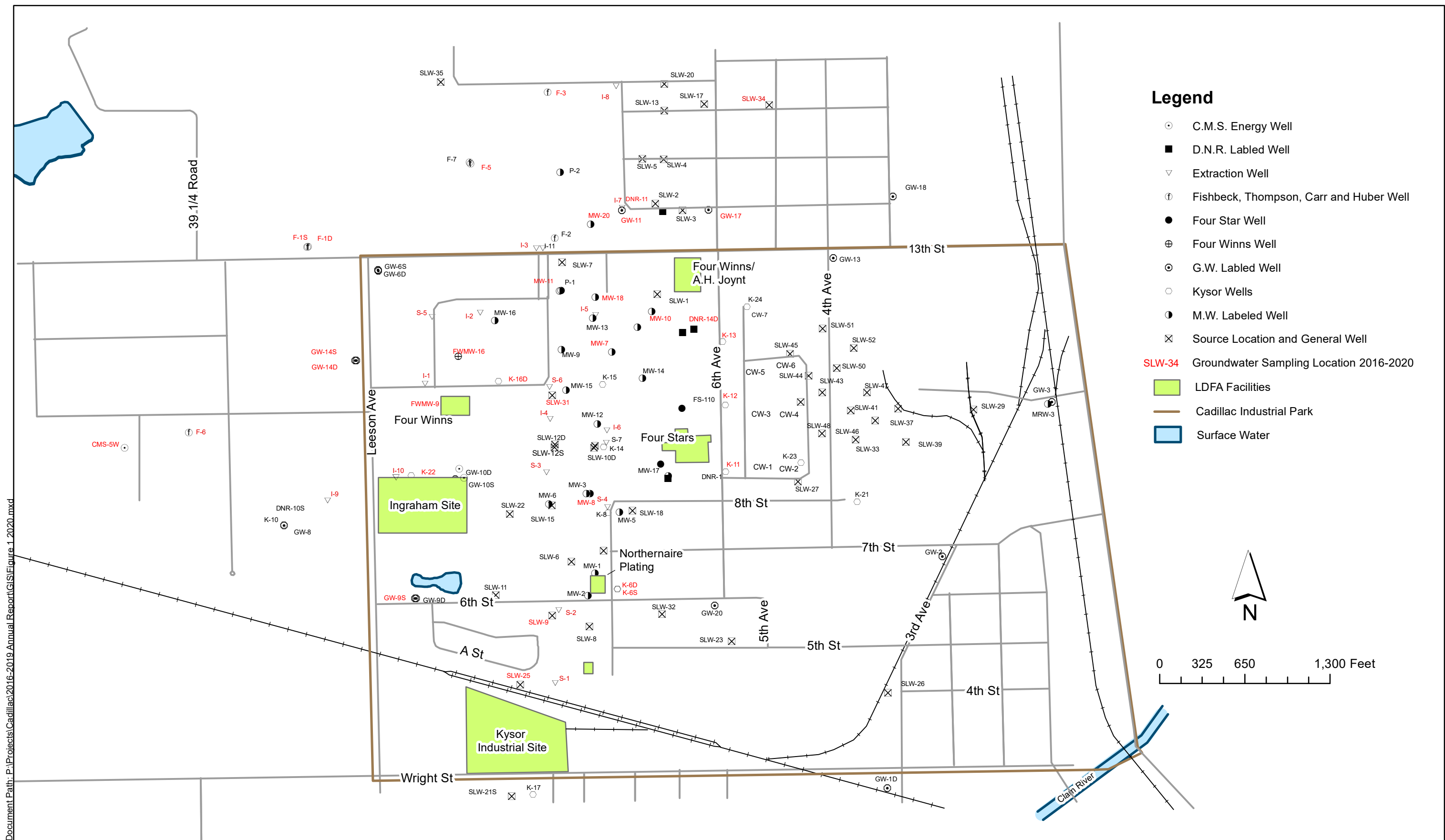
DATE: 6/26/2020

NORTHERNAIRE FINAL CLOSE OUT REPORT
LDFA
CADILLAC INDUSTRIAL PARK/CADILLAC, MI
**GENERAL EXTENT OF
GROUNDWATER IMPACTS**

FIGURE

6

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BASE MAP: CITY OF CADILLAC CADD DRAWING

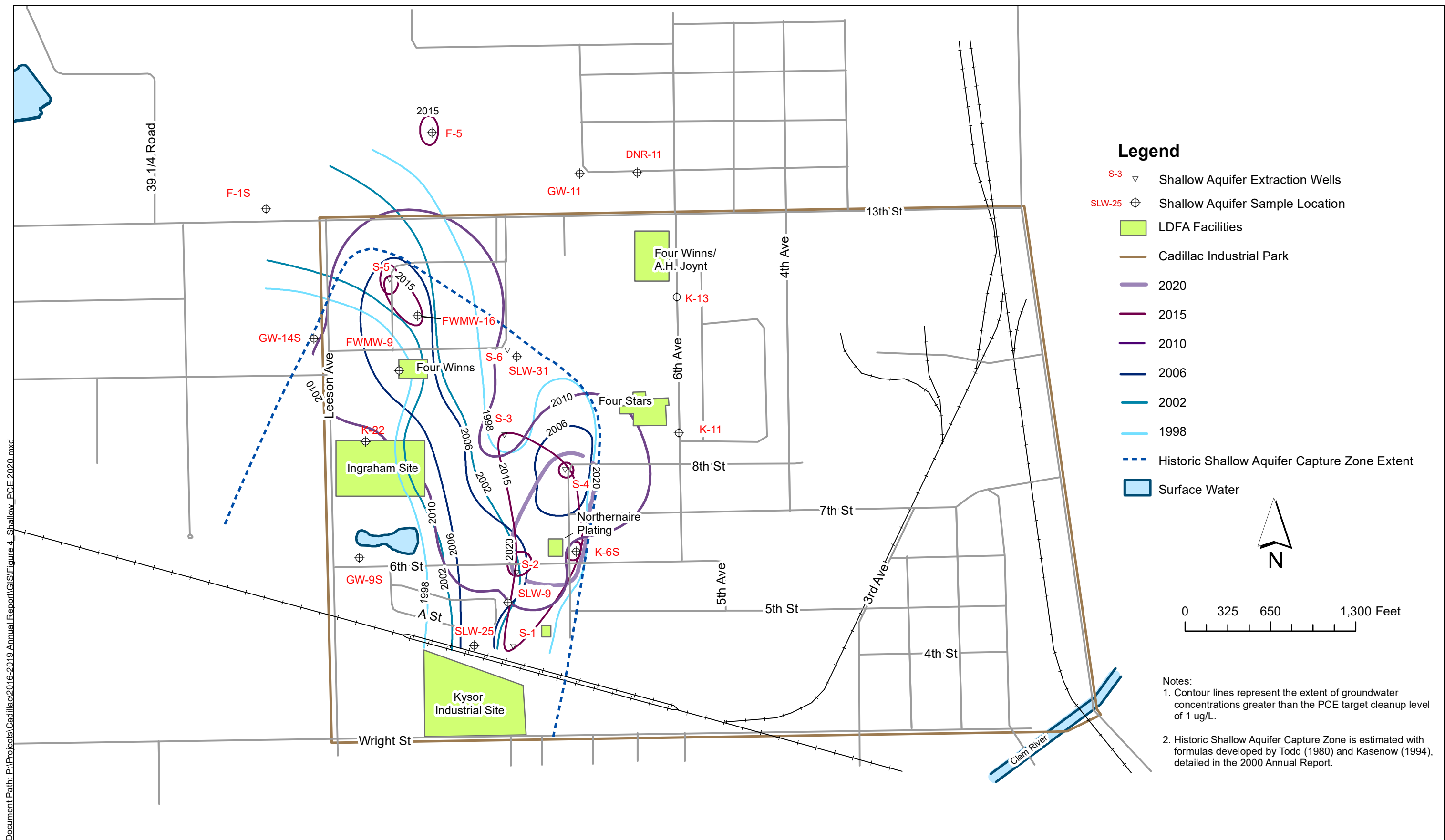
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	DATE: 7/9/2015
	REVISED BY: J. MCFETERS
	DATE: 03/18/2020

2016 - 2019 COMPENDIUM PERFORMANCE MONITORING REPORT
CADILLAC LDFA NORTHERNAIRE/KYSOR SITES
CADILLAC, MICHIGAN

GROUNDWATER SAMPLING LOCATIONS

FIGURE
7

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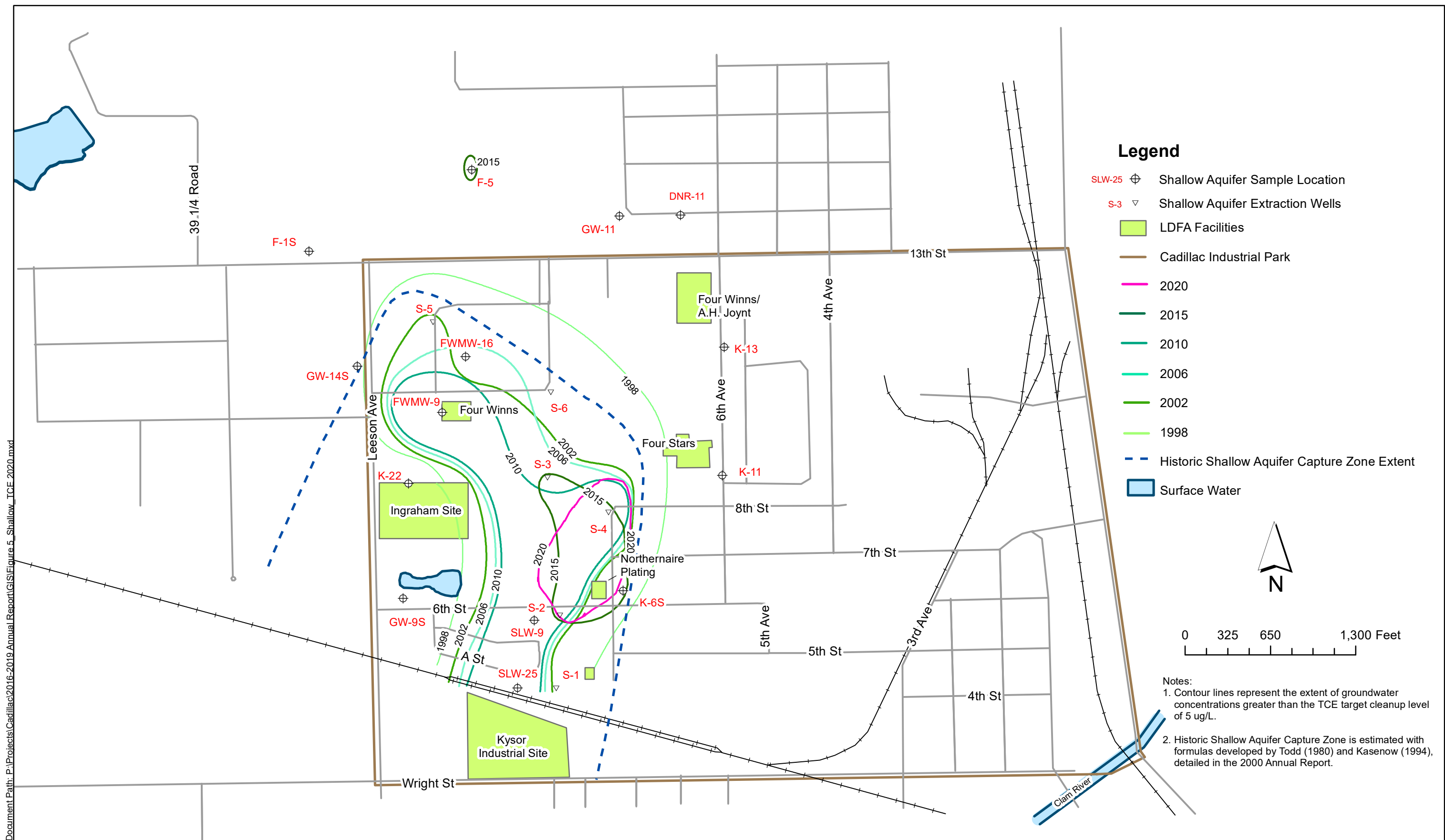
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DATE: 7/9/2015
REVISED BY: J. MCFETERS
DATE: 03/20/2020

2016 - 2019 COMPENDIUM PERFORMANCE MONITORING REPORT
CADILLAC LDFA NORTHERNAIRE/KYSOR SITES
CADILLAC, MICHIGAN

EXTENT OF TETRACHLOROETHENE IMPACTS IN THE SHALLOW AQUIFER FROM 1998-2020

FIGURE
8

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BASE MAP: CITY OF CADILLAC CADD DRAWING



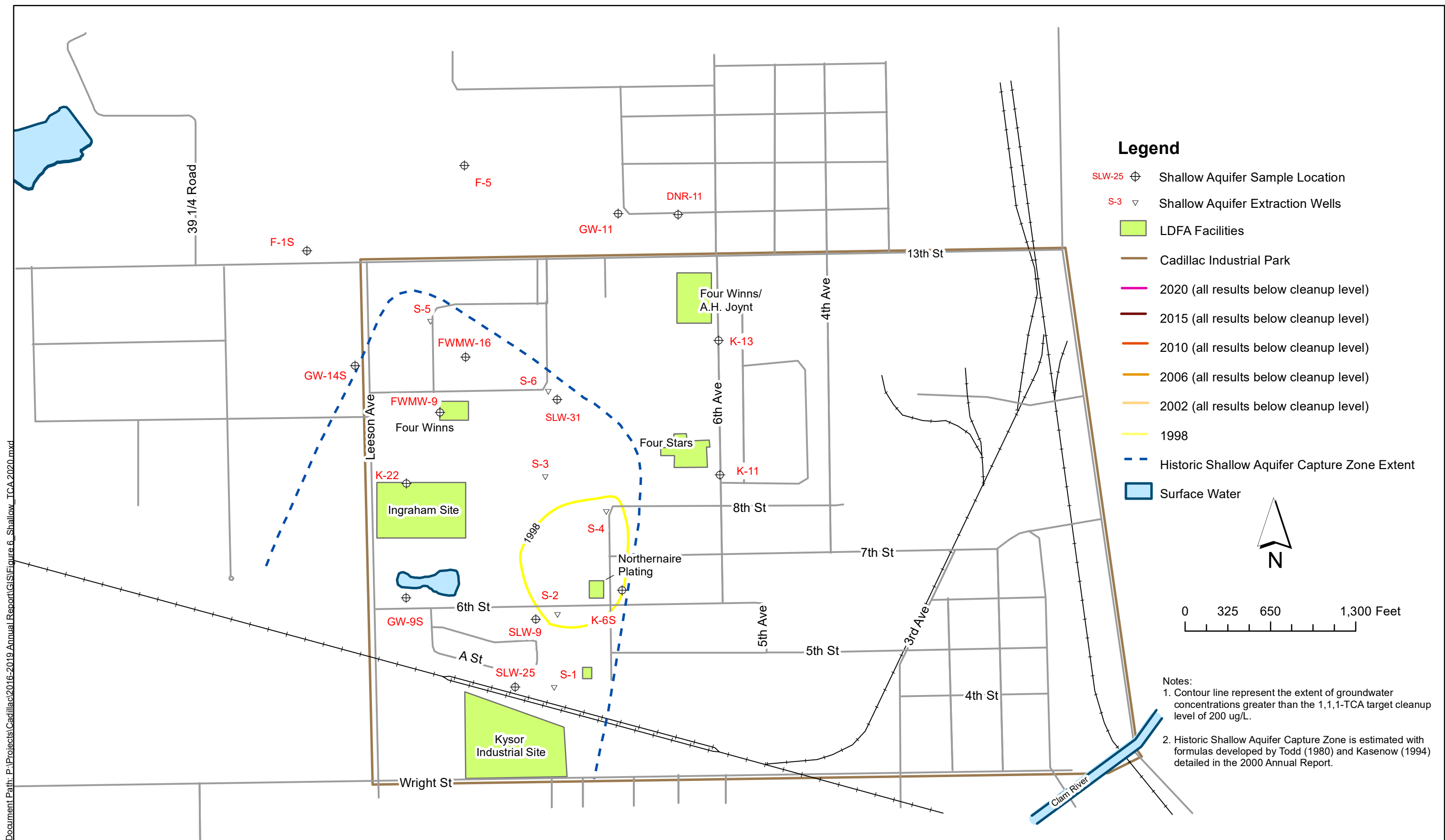
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DATE: 7/9/2015
REVISED BY: J. MCFETERS
DATE: 03/20/2020

2016 - 2019 COMPENDIUM PERFORMANCE MONITORING REPORT
CADILLAC LDFA NORTHERNAIRE/KYSOR SITES
CADILLAC, MICHIGAN

EXTENT OF TRICHLOROETHENE IMPACTS IN THE SHALLOW AQUIFER FROM 1998-2020

FIGURE
9

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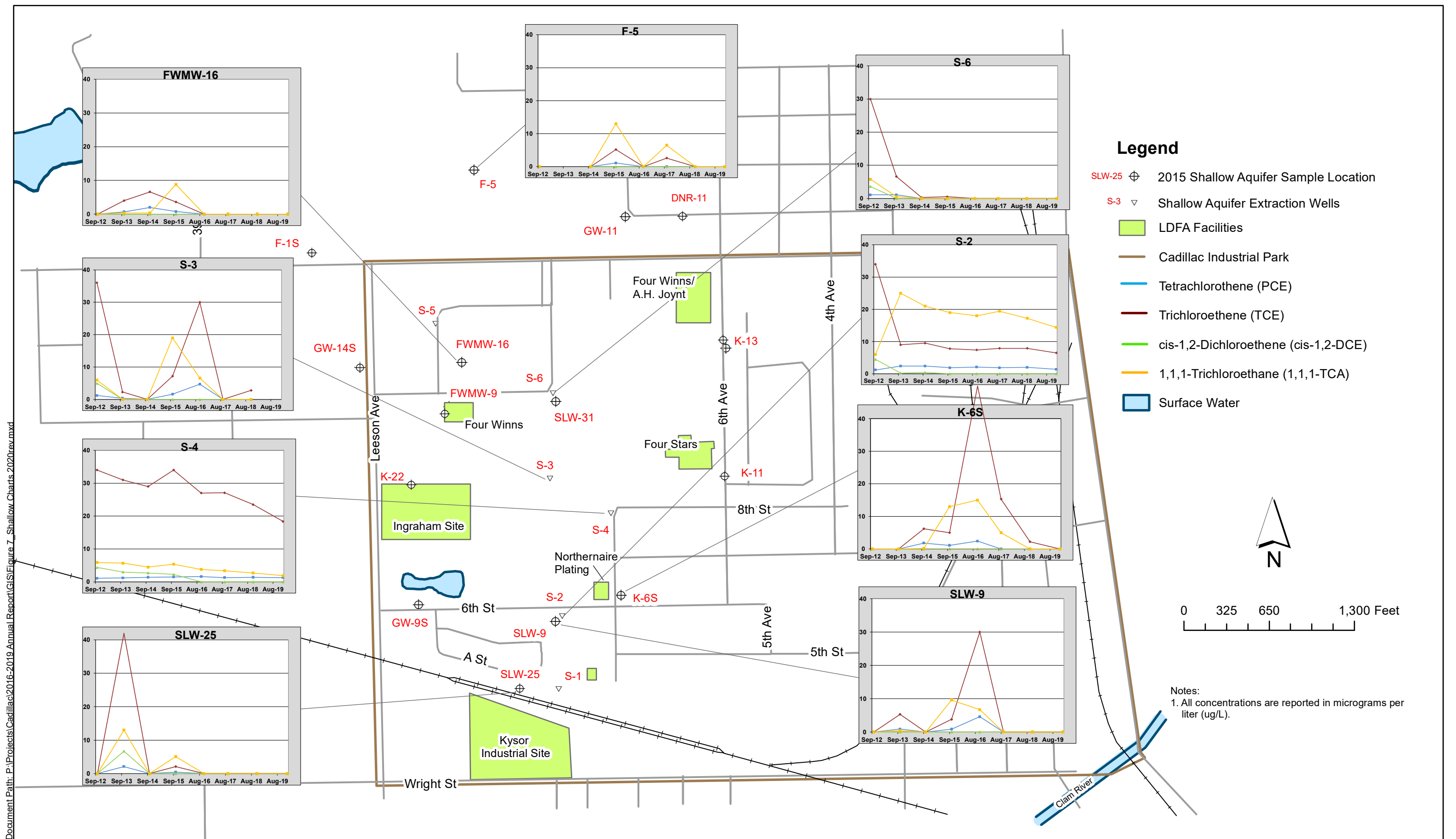
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	DATE: 7/9/2015
	REVISED BY: J. MCFETERS
	DATE: 03/20/2020

2016 - 2019 COMPENDIUM PERFORMANCE MONITORING REPORT
CADILLAC LDFA NORTHERNAIRE/KYSOR SITES
CADILLAC, MICHIGAN

EXTENT OF 1,1,1-TRICHLOROETHANE IMPACTS IN THE SHALLOW AQUIFER FROM 1998-2020

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BASE MAP: CITY OF CADILLAC CADD DRAWING



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REVISED BY: J. MCFETERS
DATE: 03/25/2020

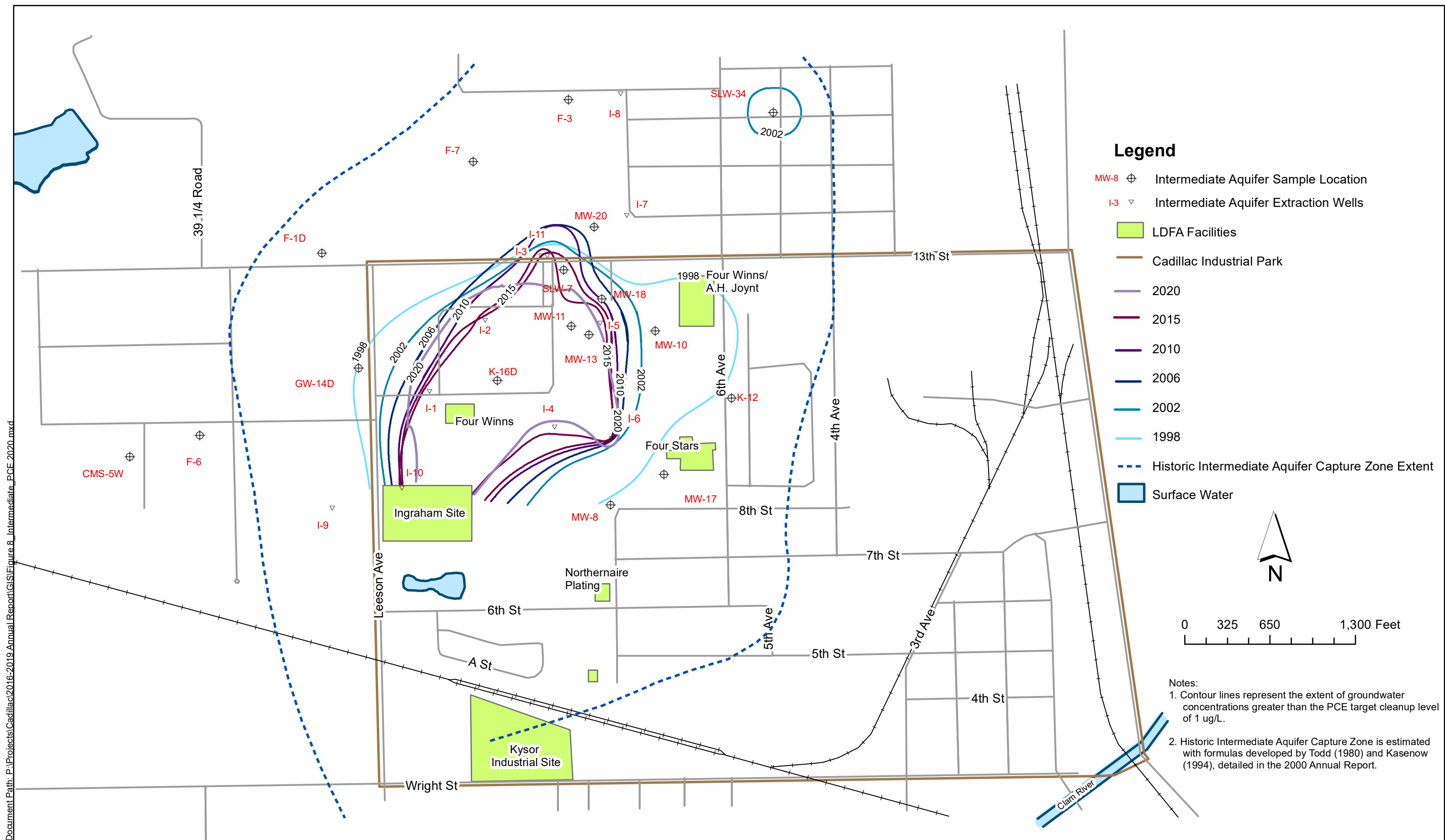
2016 - 2019 COMPENDIUM PERFORMANCE MONITORING REPORT
CADILLAC LDFA NORTHERNAIRE/KYSOR SITES
CADILLAC, MICHIGAN

SHALLOW AQUIFER: 2012-2020 CONCENTRATIONS OF VOLATILE ORGANIC COMPOUNDS OVER TIME

FIGURE

11

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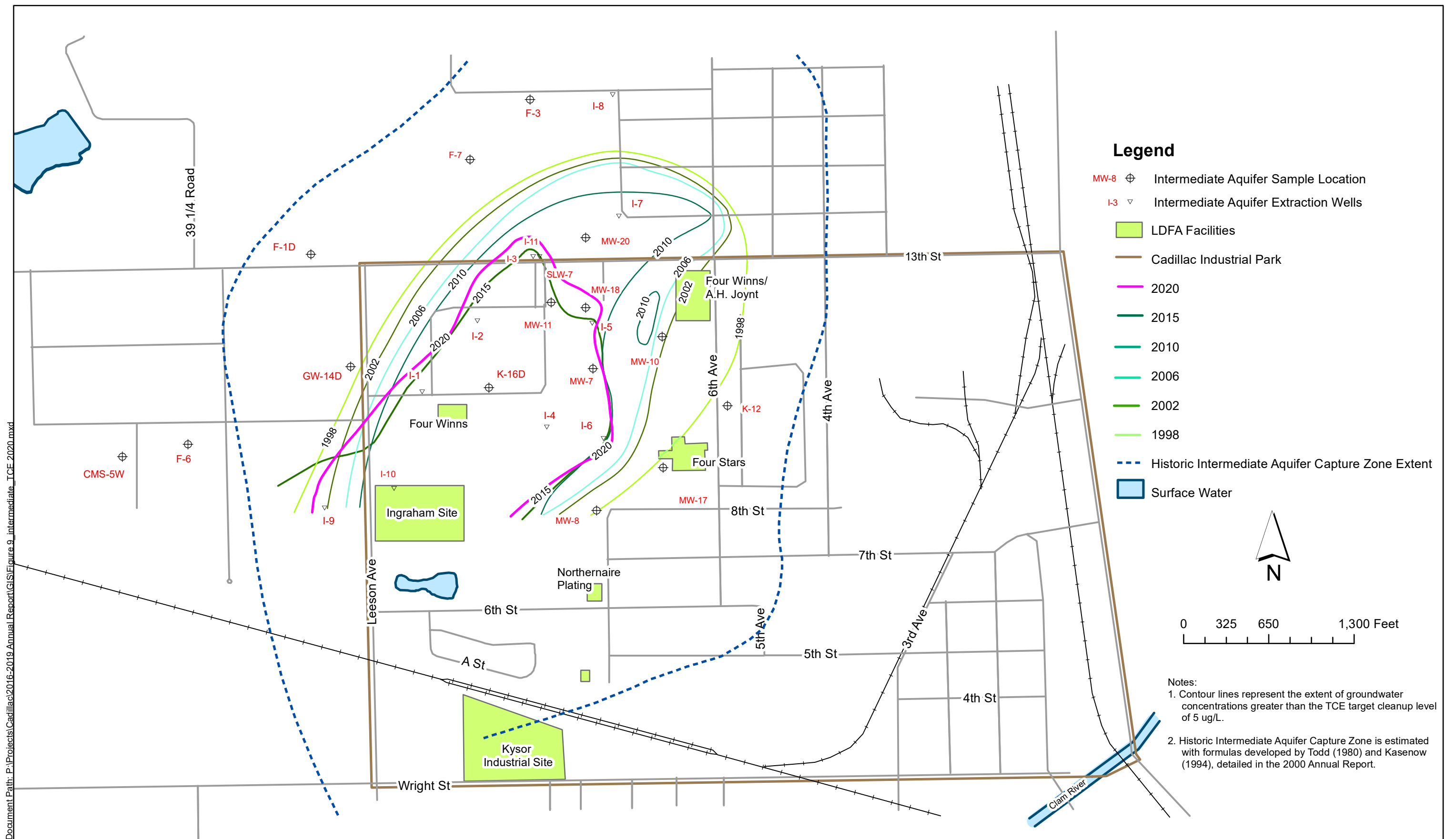
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DATE: 7/9/2015
REVISED BY: J. MCFETERS
DATE: 03/25/2020

2016 - 2019 COMPENDIUM PERFORMANCE MONITORING REPORT
CADILLAC LDFA NORTHERNAIRE/KYSOR SITES
CADILLAC, MICHIGAN

EXTENT OF TETRACHLOROETHENE IMPACTS IN THE INTERMEDIATE AQUIFER FROM 1998-2020

FIGURE
12

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	REVISED BY: J. MCFETERS
	DATE: 03/25/2020

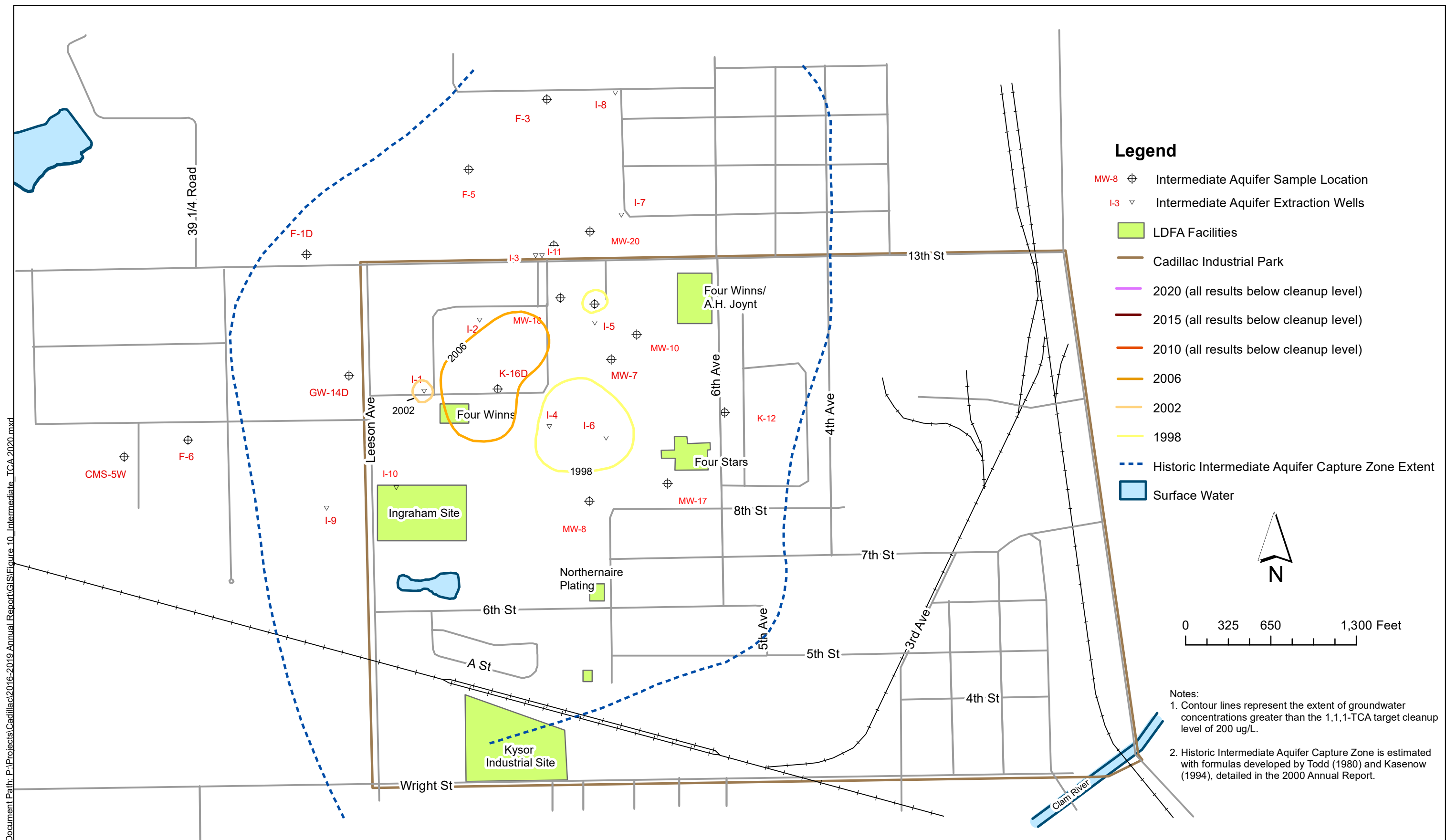
2016 - 2019 COMPENDIUM PERFORMANCE MONITORING REPORT
CADILLAC LDFA NORTHERNAIRE/KYSOR SITES
CADILLAC, MICHIGAN

EXTENT OF TRICHLOROETHENE IMPACTS IN THE INTERMEDIATE AQUIFER FROM 1998-2020

FIGURE

13

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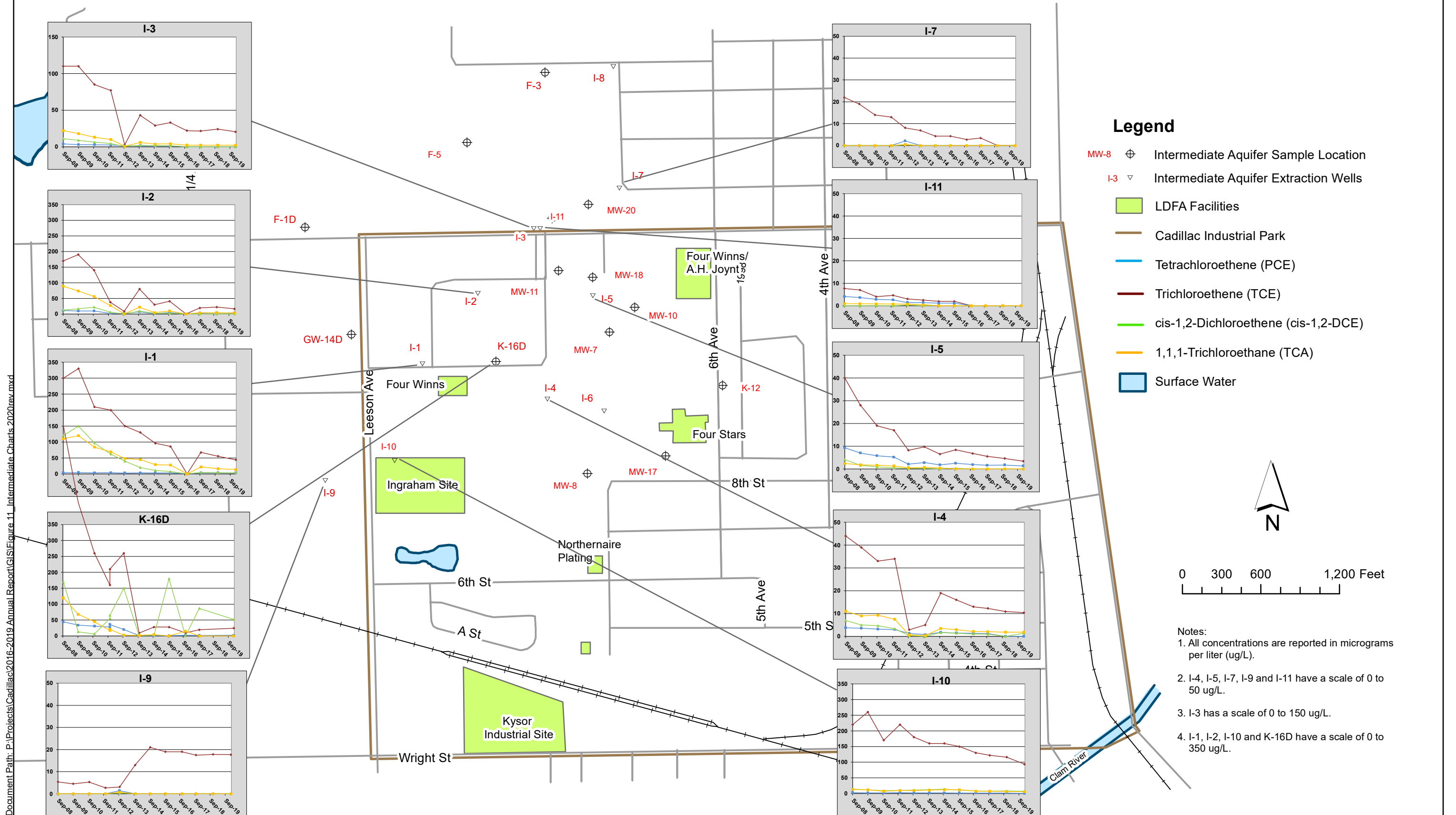
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DATE: 03/25/2020

2016 - 2019 COMPENDIUM PERFORMANCE MONITORING REPORT
CADILLAC LDFA NORTHERNAIRE/KYSOR SITES
CADILLAC, MICHIGAN

EXTENT OF 1,1,1 - TRICHLOROETHANE IMPACTS IN THE INTERMEDIATE AQUIFER FROM 1998-2020

FIGURE
14

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BASE MAP: CITY OF CADILLAC CADD DRAWING

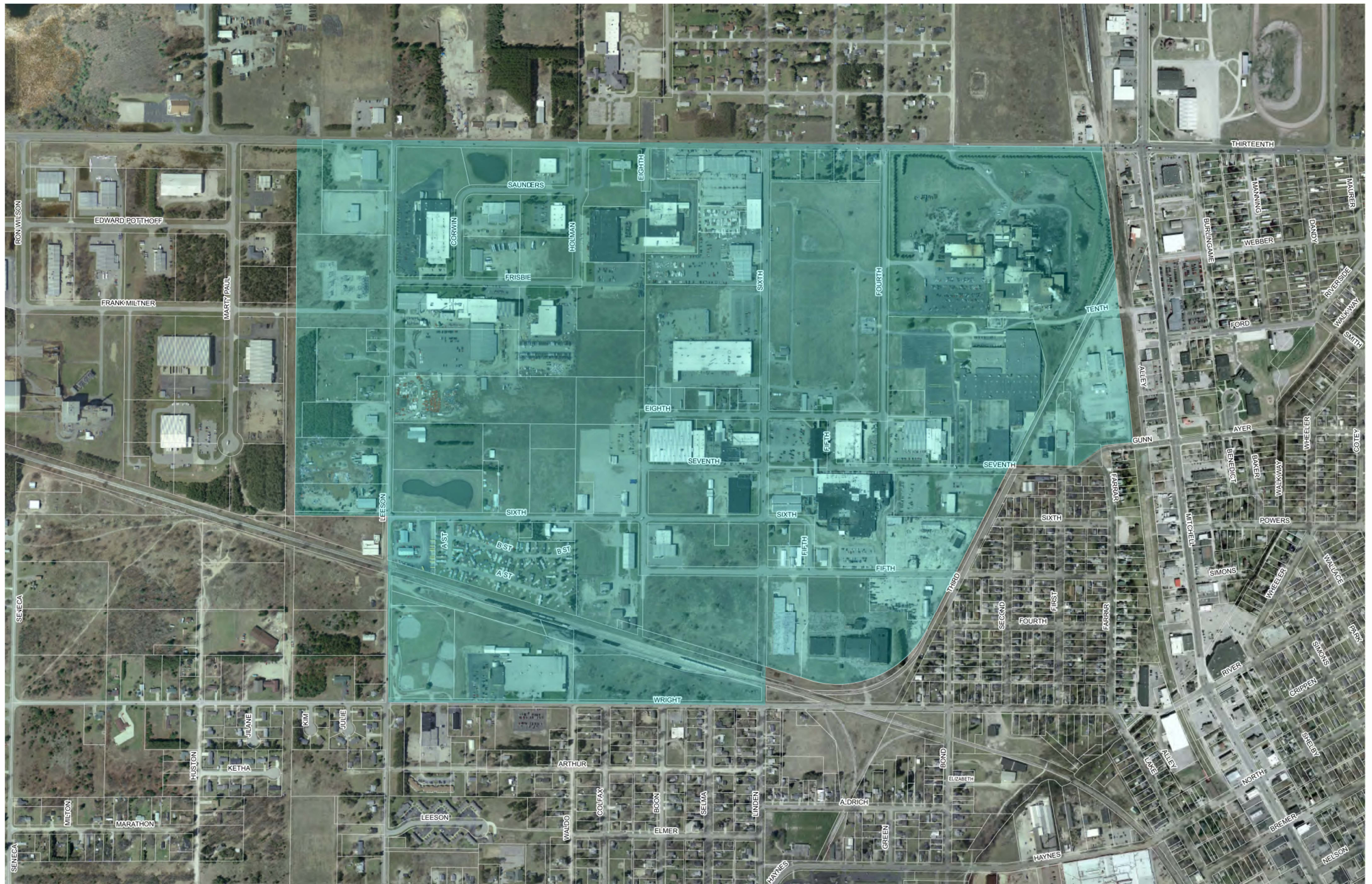


ORIGINAL: E. BAYS
DATE: 7/9/2015
REVISED BY: J. MCFETERS
DATE: 03/26/2020

2016 - 2019 COMPENDIUM PERFORMANCE MONITORING REPORT
CADILLAC LDFA NORTHERNAIRE/KYSOR SITES
CADILLAC, MICHIGAN

INTERMEDIATE AQUIFER: 2008-2020 CONCENTRATIONS OF VOLATILE ORGANIC COMPOUNDS OVER TIME

FIGURE
15

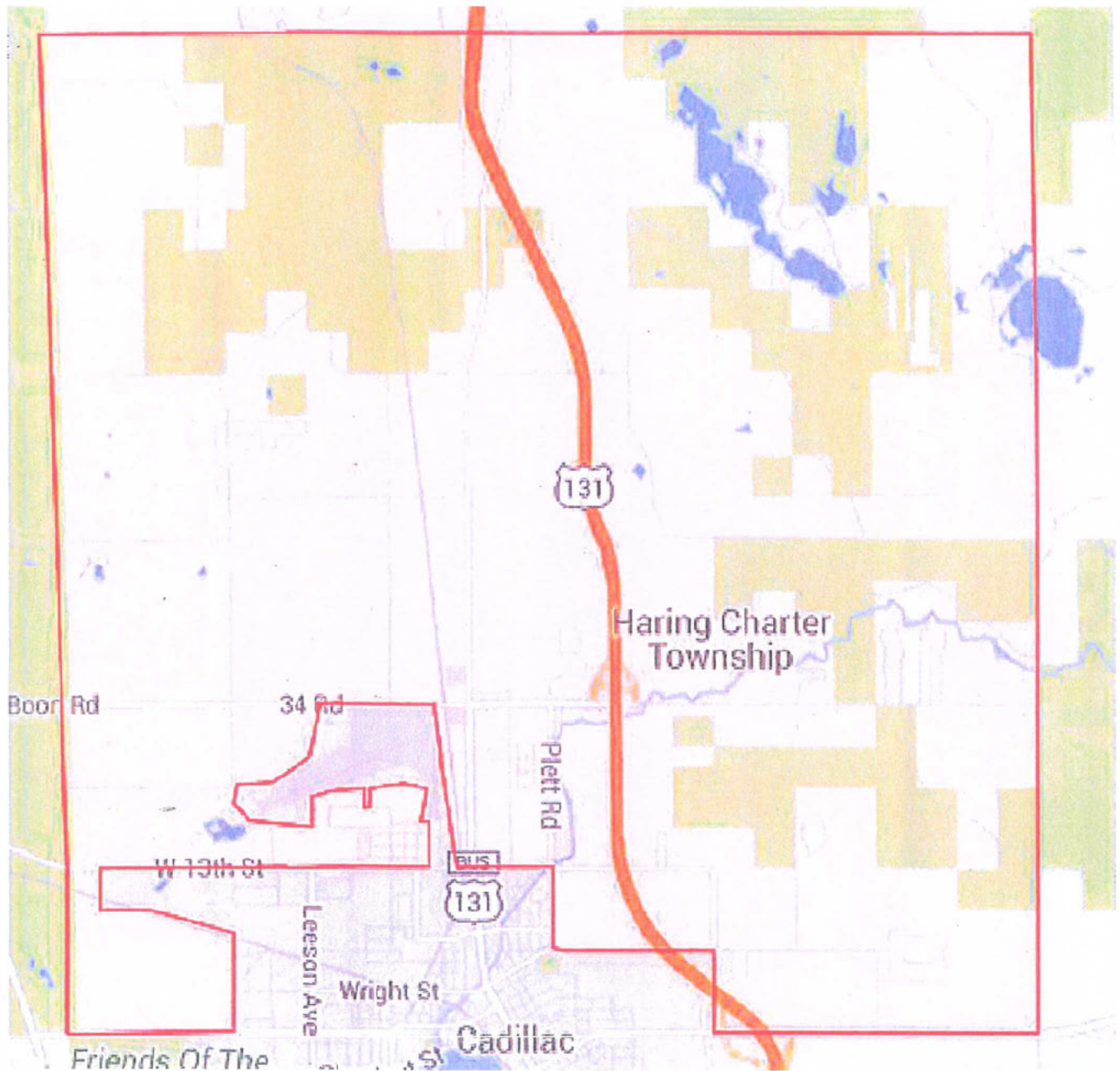


DATE: 8/16/18

FIGURE 16: CITY OF CADILLAC
EXHIBIT A

Figure 17

Haring Township Institutional Controls Map



Area Covered by ICs
(Haring Township Ordinances)

APPENDIX D – Attachments

Attachment 1 – City of Cadillac Groundwater Restriction Ordinance and Certification

Attachment 2 – Haring Township Groundwater Restriction Ordinances

Attachment 3 – Public Notice Announcing Start of Fifth Five-Year Review

Attachment 4 – Site Inspection Checklist and Photos

City Council

200 North Lake Street
Cadillac, Michigan 49601
Phone (231) 775-0181
Fax (231) 775-8755



Mayor
Carla J. Filkins

Mayor Pro-Tem
Shari Spoelman

Councilmembers
Tiyi Schippers
Robert J. Engels
Stephen King

ORDINANCE NO. 2018-10

ORDINANCE AMENDING CHAPTER 42, ARTICLE II, DIVISION 6 OF THE
CADILLAC CITY CODE REGARDING INSTITUTIONAL CONTROLS FOR KYSOR
INDUSTRIAL CORPORATION/NORTHERNAIRE PLATING COMPANY

THE CITY OF CADILLAC ORDAINS:

Section 1. Amendment of Chapter 42, Article II, Division 6

Chapter 42, Article II, Division 6 of the City Code entitled “Institutional Controls for Kysor Industrial Corporation/Northernair Plating Company” is hereby amended to incorporate additional institutional controls on the Kysor Site restricting groundwater use to prevent exposures and identify the site as Unrestricted Use/Unrestricted Exposure. The section, as amended, shall read in its entirety as follows:

Sec. 42-164. - Purpose.

- (a) The purpose of this division is for protecting the public health, welfare and the environment, and for implementing the remedial action plan at the site commonly known as the Kysor Industrial Corporation/Northernair Plating Company Superfund Site.

Sec. 42-165. – Definitions.

- (a) *Contaminated Groundwater* means groundwater in which there is present concentrations of materials that exceed drinking water criteria under the Safe Drinking Water Act, 1976 PA 399, as amended, or the residential drinking water criteria established by the MDEQ in the operational memoranda or rules promulgated pursuant to Part 201, Environmental Remediation (MCL 324.20101 *et seq.*).
- (b) *Domestic Use* means the use of water by humans for drinking, cooking, food preparation and other food-related services, cleaning, washing, bathing and similar household-type water uses in any dwelling, or in any building in which commercial/business, governmental/public or industrial activities are conducted. The term does not include

water used solely for closed-loop heat pumps, non-contact cooling, production and/or processing purposes of commercial or industrial enterprises.

- (c) *Irrigation use* means the use of water for lawn, garden or landscaping irrigation on a residential parcel of land. The term does not include water used for commercial, agricultural, or farm irrigation, except as specifically directed by the MDEQ.
- (d) *MDEQ* means the Michigan Department of Environmental Quality or its successor agency.
- (e) *Person* means any individual, partnership, corporation, association, club, joint venture, estate, trust, and any other group or combination acting as a unit, and the individuals constituting such group or unit.
- (f) *Restricted Zone* means an area or areas described within Section 42-166 of this Ordinance for which the prohibition of Wells and the use of groundwater applies and includes parcels of lands that are legally described in the map attached as **Exhibit A**.
- (g) *Well* means an opening in the surface of the earth for the purpose of removing fresh water through non-mechanical or mechanical means for any purpose other than a public emergency or conducting response actions that are consistent with the Michigan Natural Resources and Environmental Protection Act, 1994 PA 451, as amended (“NREPA”), the Comprehensive Environmental Response, Compensation and Liability Act, 42 USC § 9601 *et seq.*, the Resource Conservation and Recovery Act, 42 USC § 6901 *et seq.*, and other applicable laws.

Sec. 42-166. – Restricted Zone.

The following described area(s) in the City shall be Restricted Zone(s) under this Division. Additional Restricted Zones may be added by amending this Ordinance.

- (a) All land located in Township 22 North, Range 9 West, City of Cadillac, Wexford County, Michigan, described as follows:
 - (1) The East Quarter (E¹/₄) of the Northwest Quarter (NW¹/₄) of Section 32.
 - (2) The Northeast Quarter (NE¹/₄) of Section 32.
 - (3) The North Half (N¹/₂) of the Southeast Quarter (SE¹/₄) of Section 32.
 - (4) The Southwest Quarter (SW¹/₄) of Section 33 lying North and West of the Tuscola-Saginaw Bay Railroad.
 - (5) The Northwest Quarter (NW¹/₄) of Section 33, except the following: South of Gunn Street and Seventh Street which is East of the Ann Arbor Railroad; the property lying

East of the Pennsylvania Central Railroad; and also commencing as the Point of Beginning at the Southwest corner of Block 179 of the Improvement Board Addition; thence North to the Northwest corner of Block 188; thence East along the North line of Block 188 to the Northwest corner of Block 189; thence East along the North block line, 220 feet; thence South 71 feet; thence East 107.43 feet; thence North 71 feet; thence East 212.83 feet; thence South 16°2'30" East, 331.74 feet; thence South 3°28'30" East, 246.56 feet to the West right-of-way line of the Ann Arbor Railroad; thence Southwesterly along the West right-of-way line of the Ann Arbor Railroad to the Southeast corner of Block 177; thence West along the South line of Block 177 to the centerline of Third Avenue; thence North on the centerline of Third Avenue to the South line of Block 179 and Block 178, if extended; thence West to the Point of Beginning of the Improvement Board Addition, City of Cadillac, Wexford County, Michigan.

Hereafter referred to as the Kysor Industrial Corporation/Northern Plating Company Site (the "Site").

Sec. 42-167. – Prohibition.

Except as provided in Section 42-168, no person shall install or utilize, or allow, permit or provide for the installation or utilization of a Well in any Restricted Zone. Any existing Well at the time of the enactment of a Restricted Zone within the Restricted Zone shall be plugged/abandoned at the expense of an applicant for that particular Restricted Zone and as provided for in this Ordinance in accordance with applicable laws, regulations and ordinances, unless such existing Well falls within one of the exceptions provided in Section 42-168. Except as provided in Section 42-168, no person shall use any groundwater from a Restricted Zone.

Sec. 42-168. – Exceptions.

A person may install or utilize, or allow, permit or provide for the installation or utilization of a Well in any Restricted Zone if any of the following exceptions apply and the requirements of the exceptions are complied with. The party proposing an exception to the Well prohibition shall conduct all appropriate inquiry and prepare a due care analysis pursuant to Part 201 of NREPA:

- (a) *Proof of No Influence.* If the MDEQ determines based on information provided to it by the person seeking an exception (1) that the use of a Well in the Restricted Zone will not exacerbate existing groundwater contamination, and (2) that the water from the proposed Well will not be affected by Contaminated Groundwater, the Well may be so used. The party proposing the exception shall deliver proof/notice of those determinations to the City prior to Well operation..
- (b) *Groundwater Monitoring/Remediation.* A Well may be used for groundwater monitoring and/or remediation as part of a response activity approved by the MDEQ or the United

States Environmental Protection Agency (EPA). The party proposing the exception shall deliver proof of the MDEQ or EPA approval to the City prior to Well operation.

(c) *Construction Dewatering.* A Well may be used for construction dewatering upon approval by the MDEQ if the following conditions are satisfied:

- (1) The use of the dewatering Well will not result in unacceptable exposure to Contaminated Groundwater, possible cross-contamination between saturated zones, or exacerbation of Contaminated Groundwater as defined in Part 201 of NREPA; and
- (2) The water generated by that activity is properly handled and disposed of in compliance with all applicable laws, rules, regulations, permit and license requirements, and orders and directives of any governmental entity or agency of competent jurisdiction.

Any exposure, cross-contamination, or exacerbation caused by the use of the Well under this exception shall be the responsibility of the person operating the dewatering Well as provided in Part 201 of NREPA.

The party proposing the exception shall deliver proof/notice of MDEQ approval to the City prior to Well operation.

(d) *Processing Activities.* If the MDEQ determines that the use of a Well for non-contact heating, cooling, production, or processing involved in industrial, commercial or residential activities will not cause migration or exacerbation of Contaminated Groundwater, such use of the Well under terms and conditions specified by the MDEQ will be allowed. All information necessary for the MDEQ determination described in this subsection shall be provided by the person seeking this exception. Upon MDEQ approval, the party proposing the exception shall deliver proof/notice of those MDEQ determinations to the City prior to Well operation.

(e) *Public Emergencies.* A Well may be used in the event of a public emergency. Notice of such use shall be provided to the MDEQ within a reasonable time thereafter.

Sec. 42-169. – Enforcement.

The City Manager, or his/her designee, shall be the official having the authority to enforce this Ordinance. After the Effective Date of this Ordinance, the enforcement official shall contact all owners of Restricted Zone parcels, which appear to have Wells prohibited under this Ordinance, giving written notice of the need to cease using such Wells and of the need for (1) establishment of a Domestic Use and/or Irrigation Use water source by connecting to the City water system or (2) obtaining approval or acknowledgment of an exception under Section 42-168. The owner of a Restricted Zone parcel shall comply with the provisions of this Ordinance with regard to the provision of Domestic or Irrigation Use water within sixty (60) days from the date of such

notice. Any existing Well in violation of this Ordinance shall be plugged or abandoned in conformance with applicable legal requirements. Where, upon information available to the enforcement official, it is suspected that a Well is being used within the Restricted Zone in violation of this Ordinance, the enforcement official may inspect the Restricted Zone and serve appropriate notice and order of such violation requiring that action be taken promptly by the owner to bring the Restricted Zone in compliance. If the owner fails to act in accordance with such order, the enforcement official may seek remedies and penalties as provided in this Ordinance including, but not limited to, suspension or revocation of a previously granted exception.

Sec. 42-170. – Penalties.

Any person who violates any provision of this Ordinance shall be liable for a municipal civil infraction under Section 38-135 of the City's Code. In addition, the City may also seek an order from a court of appropriate jurisdiction requiring compliance with this Ordinance and may seek costs and attorney fees associated with such enforcement action. Any violation of this Ordinance is a public nuisance, subject to abatement, and any Well in violation of this Ordinance shall be immediately taken out of service and lawfully abandoned in compliance with applicable legal requirements. A court of competent jurisdiction may order any person violating any provision of this Ordinance to properly and lawfully remove or abandon a Well. **Sec. 42-171. – Administrative Liability.**

No officer, agent or employee of the City or member of the City Council shall render himself or herself personally liable for any damage which may occur to any person as a result of any act or decision performed in the discharge of his or her duties pursuant to this Ordinance.

Sec. 42-172. – Notice of Amendments/Repeal.

The MDEQ, an applicant, an owner, or any other interested party may request to add parcels or delete parcels from a Restricted Zone or to establish an additional Restricted Zone or to otherwise amend or repeal this Ordinance and shall provide thirty (30) days' advance notice to the MDEQ, including the reasons supporting such request. The City, upon thirty (30) days' advance notice to the MDEQ, EPA and any applicant, may also take action to amend or repeal this Ordinance as it deems appropriate. The City shall also notify the MDEQ, EPA and any applicant that this Ordinance may lapse at least thirty (30) days prior to the Ordinance being allowed to lapse.

Section 2. Validity and Severability.

Any section or subsection not expressly amended by this Ordinance shall remain in full force and effect. Should any portion of this Ordinance be found invalid for any reason, such holding shall not be construed as affecting the validity of the remaining portions of this Ordinance.

Section 3. Repealer.

All other ordinances inconsistent with the provisions of this Ordinance are hereby repealed but only to the extent necessary to give this Ordinance full force and effect.

Section 4. Effective Date

This Ordinance shall take effect twenty (20) days after its adoption.

Approved this 4th day of September, 2018.

Sandra Wasson

Sandra Wasson, Clerk

Carla G Filkins

Carla Filkins, Mayor

I, Sandra Wasson, City Clerk of the City of Cadillac, Michigan, do hereby certify that a summary of Ordinance No. 2018-10 was published in the Cadillac News on the 7th day of September, 2018.

Sandra Wasson

Sandra Wasson, City Clerk

TOWNSHIP BOARD
CHARTER TOWNSHIP OF HARING
WEXFORD COUNTY, MICHIGAN



ORDINANCE NO. 2017-107
WATER ORDINANCE

The Charter Township of Haring ordains:

1. ARTICLE I: DEFINITIONS

- 1.1. Definitions. In the interpretation of this ordinance, the following definitions shall apply unless the context clearly indicates otherwise:
- (a) "Backflow" means water of questionable quality, wastes or other contaminants entering a public water supply system due to a reversal of flow.
 - (b) "Commercial user" means a person whose premises are used to offer services and/or products such as, for example, retail and wholesale stores, gasoline stations, restaurants, schools, churches, hotels, motels, nursing homes, private clubs, theaters and governmental buildings.
 - (c) "Commodity Charge". A periodic charge levied on Users for use of the Public Water System on the basis of water consumption. The charge represents a portion of (a) that User's proportionate share of the fixed and variable Cost of Operation and Maintenance and (b) the benefit to that User derived from the use of the Public Water System. The charge may include a debt service component, as necessary, to pay all or a portion of the principal, interest and administrative costs of retiring Public Water System indebtedness.
 - (d) "Connection Fee". The charge imposed by the Township to regulate the connection of a Water Service connection, either directly or indirectly, to the Public Water System. This fee represents (a) the proportional cost attributable to each Structure in which Water Service is provided to regulate access to the Public Water System and ensures that sufficient capacity exists to accommodate the additional use without overburdening the Public Water System or adversely affecting the ability of the Township to provide service to the Public Water System's existing and future customers; and (b) the benefit to the owner of a Structure to which Water Service is provided derived from the connection to the Public Water System. See also Direct Connection and Indirect Connection.
 - (e) "Cross-connection" means a connection or arrangement of piping or appurtenances through which a backflow could occur.
 - (f) "Direct Connection". The connection of the Structure directly to the Public Water System in a manner such that the Premises served by the Water Service utilizes the existing watermain, and Water System components of the Public Water System.
 - (g) "Designated Agent" means person or persons that have been designated by the Township Board to act on behalf of the Township in enforcing and/or monitoring this Ordinance.
 - (h) "Domestic user" means a person whose premises are domiciles for single or multiple family use.

- (i) "Indirect Connection". The connection of a Structure to a water system which is installed to applicable Township specifications and with Township approval that is:
 - (1) paid for by special assessment or private funds;
 - (2) serves multiple users; and
 - (3) is connected to the Public Water System and, after construction, turned over to the Township and becomes part of the Public Water System.
 - (4) For example, if a developer constructs a water system in a plat and connects the water system to the Public Water System, the connection of each lot in the plat would be an Indirect Connection.
- (j) "Industrial user" means a person who operates a manufacturing or process facility that is engaged in producing a product.
- (k) "Local distribution lines" means those pipes that serve only the abutting property within only one local service area.
- (l) "Potable water" means water intended for human consumption or prolonged bodily contact that is free from impurities in amounts sufficient to cause disease or harmful physiological effects. Its bacteriological and chemical quality shall conform to the applicable requirements of the Federal Drinking Water Standards and to the regulations of District Health Department #10 and the Michigan Department of Environmental Quality.
- (m) "Premises" means the lands included within the boundaries of a single description as set forth, from time to time, on the general tax rolls of the Township as a single taxable parcel of property, including all structures located thereon.
- (n) "Public Water System or Water System". All Township or other publicly owned facilities and all subsequent additions and expansions, including wells, pumps, water treatment facilities, transmission and distribution mains, hydrants, storage tanks, meters, and all other facilities used or useful in the pumping, treatment, and distribution of public water to properties in the Township.
- (o) "Readiness to Serve Charge". A periodic charge levied on a User based upon the size of the User's public water meter. The charge represents a portion of (a) that User's proportionate share of the fixed and variable cost of operation and maintenance of the Public Water System and (b) the benefit to that User derived from the availability of the Public Water System and the peak usage that User may demand from the Public Water System. The charge may include a debt service component, as necessary, to pay all or a portion of the principal, interest and administrative costs of retiring Public Water System indebtedness.
- (p) "Secondary water supply" means a water supply system maintained in addition to a public water supply, including but not limited to water systems from ground or surface sources not meeting the requirements of Act No. 98 of the Public Acts of 1913, as amended, being Sections 325.201 to 325.214 of the Compiled Laws of 1948, or water from a public water supply which in any way has been treated, processed or exposed to any possible contaminant or stored in other than an approved storage facility.

- (q) "Service Connection". The portion of the Public Water System which extends either to or onto the parcel of land adjacent to the path of the Public Water System, and includes the tee, valve, connector pipes, shut off valve, meter, and appurtenances, but not including the water piping on private property between the shut off valve and the Structure served.
- (r) "Structure". A building in which toilet, kitchen, laundry, bathing, or other facilities which need or are required to have Potable water for use for domestic, commercial, industrial, or other purposes.
- (s) "Submerged inlet" means a water pipe or extension thereto from a public water supply terminating in a tank, vessel, fixture or appliance which may contain water of questionable quality, waste or other contaminants and which is unprotected against backflow.
- (t) "System" or "Water distribution system" means the Township water supply and distribution system to water service customers within the Township.
- (u) "Township" means the Charter Township of Haring, Wexford County, Michigan.
- (v) "Unit or Residential Equivalent Unit (REU)". A factor established for each type of User as shown on Appendix A, which is based upon an average daily Water usage of 200 gallons for one Unit and which represents the quantity of Water ordinarily required from the occupancy of a freestanding single-family residential dwelling by a single family of ordinary size and the benefit derived from the provision thereof. A listing of the relative relationships between the various Users of the Public Water System is hereby determined by the Township and set forth in Appendix A. The assignment of Unit(s) to a particular User shall be determined from time to time by the Township, based upon available information and investigation of the use to which the User's property is put. The assignment of Unit(s) for any use not enumerated in Appendix A shall, in the sole discretion of the Township, be based upon the most similar use enumerated in Appendix A.
- (w) "User" means the owner, lessee or occupant of any premises connected to or served by the System.
- (x) "Water connection" means that part of the water distribution system connecting the watermain at the curb stop with the premises served.
- (y) "Watermain" means the primary and intermediate transmission and local distribution lines of the water distribution system.

2. ARTICLE II: CONNECTIONS

2.1. Water Connection Required.

- (a) Connection Required. As a matter of public health, the owners of all improved premises in the Township that are used for human occupancy, employment, recreation, or other purposes, which require or need the use of potable water and which abut any right-of-way, easement, highway, street, alley, or public way in which there is located, or in the future may be located, an available watermain or local distribution line of the System, are hereby required, at the owners sole expense, to directly connect all buildings and structures on the premises used for said purpose(s) to the water distribution system, provided that said watermain or local distribution line is within 200 feet from the nearest point of a premises using or requiring potable water.

- (1) A separate and independent Service Connection shall be provided for every building. Where, however, one building stands at the rear of another on an interior lot and no private water is available or can be constructed to the rear through an adjoining alley, court, yard, or driveway, the Service Connection from the front building may be extended to the rear building and the whole considered as one Building Service upon application to and approval by the Township. Other exceptions may be allowed only by special permission granted by the Township Board acting as the Board of Appeals in accordance with Article X of this Ordinance.
 - (2) Leased spaces within a common Structure may each have an independent Service Connection only if the provisions of section 4.3.a are met.
 - (3) Existing Structures that are connected to the Public Water System at the time of adoption of this Ordinance will not be required to modify an existing Service Connection to comply with section 2.1.a.1 unless the Structure undergoes a change in use, ownership, or as may be directed by the Township as required to protect public health.
- (b) Connection Deadline. When connection to an available watermain or local distribution line is declared a necessity by the Township for the public health and welfare, or as required under this Ordinance, all such connections required hereunder shall be completed no later than 180 days after the last to occur of (i) the date the watermain or local distribution line becomes available to the premises, or (ii) the modification of a building so as to become a building using or requiring potable water. Newly constructed structures required to connect shall be connected prior to occupancy thereof. Notwithstanding the preceding, if the Township Board or Health Department requires completion of a connection within a shorter period of time for reasons of public health, such connection shall be so completed. Persons who fail to complete a required connection when required shall be liable for a penalty equal in amount to the Availability Charge and Commodity Charge that, based upon similarly situated Customers, the System would have accrued and been payable, effective upon the expiration of the connection period, had the connection been made as required.
- (c) Enforcement in the Event of Failure to Connect to System. In the event a required connection is not made within the time provided by Subsection (b), the Township shall require the connection to be made immediately after notice given by first class or certified mail or by posting on the property. The notice shall give the approximate location of the available watermain or local distribution line and shall advise the owner of the affected property of the requirement and enforcement provisions provided by Township ordinance. In the event the required connection is not made within 90 days after the date of mailing or posting of the written notice, the Township may bring an action in court of competent jurisdiction for a mandatory injunction or court order to compel the property owner to immediately connect the affected property.
- (d) Private Wells Prohibited. Except as otherwise provided by Section 2.1(e), private water wells are prohibited on premises connected to the System. Pre-existing private water wells located on premises that are to be subsequently connected to the System shall be abandoned prior to or upon physical connection of the premises to the System. Wells shall be abandoned in accordance with applicable state law and District Health Department #10 regulations. Residential premises serviced by a permitted private well shall be allowed to maintain their well and postpone connecting to the System until such time as their existing well fails. Once the private well fails the premises shall be required to connect to the System and abandon their existing well.

- (e) Exceptions. Notwithstanding Section 2.1(d), private water wells may be maintained on premises that are connected to the System where (i) the premises maintains a closed geothermal heating system and the private well is an integral part of said system, or (ii) the premises is required by applicable federal or state law, regulation or order to monitor ground water on the premises, and the well is used for groundwater monitoring purposes only. Any such private well must be physically and completely separated from all plumbing used for potable water. All piping for a private water well is subject to the cross connection provisions of this Ordinance.

2.2. Service Connections:

- (a) Permit Application. A connection to the Public Water System shall be made only by an authorized contractor or plumber upon written authorization and a connection permit issued by the Township. Prior to said connection, the property owner or his agent shall submit a permit application to the Township. This permit application shall be on a form furnished by the Township, an example of which is included in Appendix B, and shall be accompanied by payment of the applicable Connection Fee determined in accordance with Section 2.2.i, any civil penalty which has accrued, and the Inspection and Administration Fee, the plans and specifications of all plumbing construction within the Premises (when requested), and all other information required by the Township.
- (b) Length of permits. Any water connection permit issued by the Township shall be valid for a period of three (3) years from the date of issuance. Following the expiration of the three (3) year period, a water connection permit may be continued to be held for as long as the holder of the permit pays a periodic ready to serve charge for all permitted but not yet active water use equivalents. The amount and frequency of the payment of such charges shall be established from time to time by resolution of the Township. The failure of a permit holder to make a ready to serve charge payment within 30 days of its due date will result in the revocation of the permit.
- (c) Inspection. A connection permittee shall notify the Inspector 48 hours in advance of when the Water Connection and Service Connection are ready for inspection. The excavation shall be left open until inspection is complete. If the Inspector determines that the Water Connection and Service Connection have been constructed and installed in accordance with the requirements of the permit and this Ordinance, a water connection approval shall be issued and the Water Connection shall then be connected with the Public Water System under the observation of the Inspector.
- (d) Plan review/inspection. If the Township determines that the standard application/inspection fees will not cover the actual costs for review of a water plan or inspection of a water connection, or that it is necessary or advisable to have any portion of the plan reviewed or inspected by other engineers or other consultants, then the applicant shall be required to pay the actual costs of this additional review or inspection. Such actual costs shall be billed to and paid by the applicant, and no occupancy permit shall be issued or maintained for the subject premises until such actual costs have been paid in full.
- (e) Escrow deposit. The Township may require an applicant to deposit an amount equal to the estimated actual costs associated with the review of an application or an inspection. This amount shall be held in escrow in the applicant's name and may be used solely to pay the actual costs associated with the subject application or inspection. Any unexpended funds held in escrow shall be returned to the applicant, without interest. Any actual costs in excess of the

amount held in escrow shall be billed to the applicant and no occupancy permit shall be issued or maintained for the subject premises until such actual costs have been paid in full.

- (f) Installation costs; Indemnification. All costs and expenses incidental to the installation of the Water Connection from the Building and the connection of the Water Connection to the Public Water System shall be borne by the owner of the property being connected. Any Person is authorized to install a Water Service Connection, subject to applicable building and plumbing codes and inspection in accordance with Section 2.2.c. Only an authorized contractor or plumber approved by the Township in accordance with Section 2.2.f.1 may connect the Water Connection to the Public Water System, subject to the requirements of this Ordinance, including inspection in accordance with Section 2.2.c. No such work shall be commenced before such owner obtains any necessary permission to work in the public right of way from the County Road Commission or MDOT, as the case may be. Said owner shall indemnify the Township from all loss or damage that may directly or indirectly be caused by the installation and connection of the Water Connection and the installation and connection of the Water Connection to the Public Water System.
- (1) Contractor Requirements. Any Person desiring to construct a Service Connection or connect a Water Connection to a Service Connection or uncover, make any connection with or opening into, use, alter or disturb any Public Water or appurtenances thereof, must be approved by the Township. Approval may include placing a deposit with the Township, a cash bond or irrevocable letter of credit in the sum of \$5,000, conditioned that he will faithfully perform all work with due care and skill, and in accordance with the laws, rules codes, and regulations established under the authority of the Township pertaining to water services and plumbing. This bond shall state that the Person will indemnify and save harmless the Township and the owner of the Premises against all damages, costs, expenses, outlays and claims of every nature and kind arising out of mistakes or negligence on his part in connection with the installation and/or connection as prescribed in this Ordinance. Such bond shall remain in force for the duration of the work, except that, upon completion, the bond shall remain in force as to all penalties, claims and demands that may have accrued thereunder prior to such expiration. The approval process shall also provide to the Township, evidence of public liability insurance insuring the interests of the Township, the property owner, and all Persons, for all damages caused by accidents attributable to the work, with limits of \$100,000 for one (1) Person, \$300,000 for bodily injuries per accident, and \$100,000 for property damages.
- (g) Change in use of premises. In the event there is an increase or change in the use of any particular multi-family, commercial, or industrial user's premises being served by the Public Water System, or a change in use from a single family residential use to another type of use, additional connection charges consistent with the new or increased use shall be assessed and shall be payable in the same manner as a water use charge when billed. This charge shall be based upon the difference of the then applicable connection charges for the new or increased use and the then applicable connection charges for the immediately preceding use.
- (h) No refunds. There shall be no refund of any water connection charges paid to the Township under this ordinance or any predecessor thereof.
- (i) Connection Fee. For a direct connection to the water system, a user shall pay a direct connection fee. For purposes of this section, a "direct connection" is a connection of the premises to a system watermain. For an indirect connection to the water system, a user shall

pay an indirect connection fee. For purposes of this section, an "indirect connection" is the connection of a premise to a watermain or local distribution line installed and paid for by the user. The direct connection fee and indirect connection fees shall be established in accordance with this section and as determined from time to time by resolution of the Township Board. The Connection Fee shall be determined based on the Residential Equivalent Unit table provided in Appendix "A" and the applicable direct or indirect connection charge per REU.

- (j) Installment Payment of Connection Fee. The property owner may elect to pay the Connection Fee either by a single payment or in installments over a period not to exceed five (5) years. The property owner electing to pay by the installment method shall in writing agree to pay the Connection Fee in five (5) or fewer equal annual installments of principal, plus interest of ten percent (10%) per annum. Any annual installment, including interest due, not paid when due, from and after the date when due, shall be subject to a one-time four percent (4%) penalty charge, plus interest at the rate of seventy-five hundredths of one percent (0.75%) for each month or portion thereof payment is past due. In addition, interest at the rate of ten percent (10.0%) per annum shall continue to accrue on the entire amount of unpaid principal including any late installment subject to the penalties. The first installment shall be due and payable prior to the issuance of a connection permit.
 - (k) Meters. Water meters shall be installed by the Township, unless otherwise provided by resolution of the Township Board. The meter installation fee shall be established in accordance with this section and as determined from time to time by resolution of the Township Board. All meters and water connections shall be the property of the Township.
 - (l) Water Service. Water service will not commence until payment in full for the installation has been made to the Township. Construction of a water service line shall be done as expeditiously as possible after written notice to proceed, but the time for construction shall be at the convenience of the Township.
- 2.3. Institution of or Restarting Water Service. Written notice given not less than 48 hours in advance shall be made to the Township by the property owner and/or occupant of the premises when water service is desired to be activated. It shall be unlawful for any person to connect to or use water supplied by the Township without first giving notice as provided herein. The Township reserves the right to require that an amount of money equal to an anticipated bill for three months of water service be placed on deposit with the Township for the purpose of establishing or maintaining any customer's credit. No person, other than an authorized employee of the Township, shall turn on or off any water service, except that a licensed plumber may, with the prior express approval of the Township, turn on water service for testing his work (after which it must be immediately turned off) or upon receiving a written order from the Township; provided, that upon written permit from the Township, water may be turned on for construction purposes upon payment to the Township of the charges applicable thereto.
- 2.4. Meters. The Township reserves the right to determine the size and type of meter used. The Township reserves the right to require the installation of remote meter reading equipment. The cost of that equipment and its installation shall be charged to the user at the prevailing rates and actual cost of material and labor.
- 2.5. Access to Meters. The Township shall have the right to shut off the supply of water to any premises where the Township is not able to obtain access to the meter. Any qualified employee of the Township shall, at all reasonable hours, have the right to enter the premises where such meters are installed for the purpose of reading, testing, removing or inspecting the same and no person shall hinder, obstruct,

or interfere with such employee in the lawful discharge of his duties in relation to the care and maintenance of such water meter.

- 2.6. Curb Box. No person shall remove the cover from any curb box or place any dirt, stone or other obstruction in it or tamper with any meter or valve or commit any act tending to obstruct the use thereof.
- 2.7. Injury to Facilities. No person, except an employee of the Township in the performance of his duties, shall break, damage, destroy, uncover, deface or tamper with any structure, appurtenance or equipment which is a part of the water distribution system.
- 2.8. Reimbursement for Damage. Any damage which a meter may sustain resulting from carelessness of the owner, agent, or tenant or from neglect of either of them to properly secure and protect the meter, as well as any damage which may be wrought by frost, hot water, or steam backing from a boiler, shall be paid by the owner of the property to the Township on presentation of a bill therefor; and in cases where the bill is not paid, the water may be shut off and shall not be turned on until all charges have been paid to the Township.
- 2.9. Responsibility for Damages. The Township will not be responsible for any damages because of failures of or within the System, or actions by the Township to correct such failures.
- 2.10. Water Leakage. It shall be the responsibility of the owner to repair any leaks between the meter and the curb valve within 72 hours of notice by the Township or an additional fee shall be charged for water loss. Water leakage within the structure that is metered shall be billed and paid unless the leakage is due to a sudden, one time event and is reported to the Township within thirty (30) days.
- 2.11. Hydrant Use. No person, except an employee of the Township in the performance of his duties, shall open or use any fire hydrant, except in case of emergency, without first securing a written permit from the Township and paying such charges as may be prescribed by the Township.
- 2.12. Unlawful Connections. No person shall make a connection on a service line between the water meter and the street mains, or install a by-pass around the meter.
- 2.13. Cross-Connections Control. The Township shall eliminate and prevent all cross-connections pursuant to the plan submitted to and approved by the Department of Public Health, pursuant to the requirements of Sections 325.11401 through 325.11407 of the 1979 Michigan Administrative Code, as the same may be amended from time to time.
 - (a) Cross-Connections Prohibited:
 - (1) A cross-connection shall not be made between the water distribution system and a secondary water supply.
 - (2) A cross-connection shall not be made by submerged inlet.
 - (3) A cross-connection shall not be made between the water distribution system and piping which may contain sanitary waste or a chemical contaminant.
 - (4) A cross-connection shall not be made between the water distribution system and piping immersed in a tank or vessel which may contain a contaminant.
 - (b) Inspections. It shall be the duty of the Township to cause inspections to be made of all properties served by the public water supply where cross connections with the public water supply is deemed possible. The frequency of inspections and re-inspections based on potential health hazards involved shall be as established by the Township.

- (c) Right of Entry. The representative of the Township shall have the right to enter at any reasonable time any property served by a connection to the water distribution system for the purpose of inspecting the piping system or systems thereof for cross-connections. On request, the owner, lessees or occupants of any property so served shall furnish to the Township any pertinent information regarding the piping system or systems on such property. The refusal of such information or refusal of access, when reasonably requested, shall be deemed evidence of the presence of cross connections.
- 2.14. Protection of Water Supply. The potable water supply made available on the premises served by the System shall be protected from possible contamination as specified by this ordinance and by the State of Michigan's and any other applicable plumbing codes. Any water outlet which could be used for potable or domestic purposes and which is not supplied by the water distribution system must be labeled in a conspicuous manner as water unsafe for drinking.
- 2.15. Discontinuance of Service. In the event a water customer desires water service to be discontinued, said customer shall so request in writing not less than 48 hours prior to time of such discontinuance of service is desired. The Township will not recognize the discontinuance of water service unless the provisions of the foregoing sentence have been fully met. Under any other circumstances, discontinuance of service will be charged to the customer at the actual cost of labor, materials and equipment. The Township may refuse or discontinue water service for any violation of any rule, regulation, or condition of service.
- 2.16. Shut off of Water. The Township reserves the right at all times (and will endeavor to give due notice) to shut off the water at the watermains or local distribution lines or to require reduced use or no use of the System for the purpose of making repairs or extensions or for other purposes. All persons having equipment on their premises and depending on water from the water-mains or local distribution lines are hereby cautioned against danger which might arise from emergency shutting off of water. In the event of such emergency, the Designated Agent may designate in any notice the extent of any regulation, limitation or prohibition and the date and time on which it shall take effect.
- 2.17. Additional Regulations. The Township Board may make and issue additional rules and regulations concerning the water distribution system, connections thereto, meter installations and maintenance, connection and meter installation fees, hydrants and watermains and the appurtenances thereto, not inconsistent herewith. Such rules and regulations shall be effective upon approval by the Township Board.
- 2.18. Other Laws. If any other ordinances of the County or Township or any statutes of the State of Michigan shall impose greater restrictions than herein set forth, then such ordinances or statutes shall control.
- 3. ARTICLE III: CONTROLLED USE
 - 3.1. Limitation of Water Use. The Township Board, may regulate, limit or prohibit the use of water for any purpose. Such regulations shall restrict less essential water uses to the extent deemed necessary to assure an adequate supply for essential domestic and commercial needs and for fire fighting. No such regulation, limitation or prohibition shall be effective until twenty-four (24) hours after the publication thereof in a newspaper of general circulation in the Township, except in an emergency as may be deemed by the Designated Agent. In the case of such emergency the regulation, limitation or prohibition shall be and take effect as indicated by the Township Board and notice shall be given on a local radio station. Any person violating such rule or regulation shall, upon conviction thereof, be punished as prescribed in this ordinance.

4. ARTICLE IV: WATER RATES

4.1. Basis of Charges. Except as otherwise provided in this ordinance, all water service shall be charged for on the basis of water consumed as determined by the meter installed in the premises of water customers by the Township. No free water service shall be furnished to any person.

4.2. Water Rates and Charges.

- (a) Water rates and charges shall be levied monthly against each premises having any water connection to the water distribution system.
- (b) Said water rates and charges shall be based on the quantity of water used on or in the property, as measured by the water meter therein used, except as otherwise provided. If a meter cannot be read, an estimated charge will be made and adjustment, if necessary, will be made when the meter reading can be obtained. Rates and charges for users inside and outside the Township shall be in amounts to be established from time to time by the Township Board.
- (c) Any charges remaining unpaid shall be charged against the real property on which the service was rendered, and may be spread on the next regular Township ad valorem property tax roll after the date on which such charge shall become due and payable, and shall become a lien of the same character and effect as the lien created by State of Michigan and County taxes, until paid,

4.3. Billing and Penalties and Remedies for Nonpayment or Late Payment.

- (a) The owner of the premises serviced and the occupants thereof shall be jointly and severally liable for the water service provided to said premises.

A lien shall not attach for Water Rates and Charges to a Premises which is subject to a legally executed lease or land contract that expressly provides that the tenant (and not the landlord) of the Premises or a dwelling unit thereon shall be liable for payment of Water Rates and Charges, effective for services which accrue after the date an affidavit is filed by the landlord with the Township. This affidavit shall include the names and addresses of the parties, the expiration date of the lease and an agreement by the landlord to give the Township thirty (30) days written notice of any cancellation, change in or termination of the lease. The filing of the affidavit by the landlord shall be accompanied by a true copy of the lease and a security deposit in the amount equal to the Debt Service Charge and the User Charge for the preceding twelve (12) monthly billing periods or four (4) quarterly billing periods. Upon the failure of the tenant to pay the Water Rates and Charges when due, the security deposit shall be applied by the Township against the unpaid balance, including time price differential, interest and penalties. The tenant shall immediately make sufficient payment to the Township to cover the amount of the security deposit so advanced. Upon the failure of the tenant to do so within ten (10) days of said advance, the penalties, rights and remedies set forth in ARTICLE 6 shall be immediately applicable with respect to the unpaid Water Rates and Charges, including time price differential, interest and penalties. The security deposit shall be held by the Township without interest and shall be returned to the landlord upon proof of termination of the lease. In the case of land contract purchasers, the deposit shall be returned less any balance due when the service is discontinued or the purchaser obtains a deed for the premises.

- (b) Bills for rates and charges as herein established shall be mailed to users monthly. All bills shall be payable on the 20th day of the month following the period of service and shall payable to the Township. If any bill shall not be paid by the 25th of the month in which it is due, a late charge of ten percent (10%) of the amount billed shall be applied to the current amount past due and collected

therewith. Late charges will not be compounded.

(c) Water service may be discontinued to any premises to enforce the payment of rates and charges after the user has been given the opportunity for a hearing before the Township Supervisor or his designated representative to show cause why service should not be discontinued. Services so discontinued shall not be restored until such time as all the rates, charges, and penalties are paid or satisfactory arrangements made for the payments thereof.

(d) Delinquent bills may be collected by any method authorized by the law including, without limitation, making such amounts due a lien on the premises served.

4.4. Meter Failure. If any meter shall fail to register properly, the Township shall estimate the consumption on the basis of former consumption and bill accordingly.

(a) Inaccurate Meters. A user may require that the meter be tested. If the meter is found accurate, a charge as set from time to time by the Township Board will be made. If the meter is found defective, it shall be repaired or an accurate meter installed and no charge shall be made.

(b) Accuracy Required. A meter shall be considered accurate if, when tested, it registers not to exceed two percent (2%) more to two percent (2%) less than the actual quantity of water passing through it. If a meter registers in excess of two percent (2%) more than the actual quantity of water passing through it, it shall be considered "fast" to that extent. If a meter registers in excess of two percent (2%) less than the actual quantity of water passing through it, it shall be considered "slow" to that extent.

(c) Bill Adjustment. If a meter has been tested at the request of a user and shall have been determined to register "fast," the Township shall credit the consumer with a sum equal to the percent "fast" multiplied by the amount of all bills incurred by said consumer, within the three (3) months prior to the test, and if a meter so tested is determined to register "slow," the Township may collect from the consumer a sum equal to the percent "slow" multiplied by the amount of the bills incurred by the consumer for the prior three (3) months. When the Township, on its own initiative, makes a test of a water meter, it shall be done without cost to the consumer, other than his paying the amount due the Township for water used by him as above provided, if the meter is found to be "slow."

5. ARTICLE V: WATER DISTRIBUTION LINES

5.1. Purpose of Rates and Charges. Charges for the installation of local water distribution lines of the water distribution system are hereby established for the purpose of recovering the cost of construction, reconstruction, maintenance, operation and replacement of said local water distribution lines. Such charges shall be made in accordance with the provisions hereinafter set forth.

5.2. Local Distribution Line Installation Charges:

(a) Payment of Local Distribution Line Installation Charges. Whenever a watermain used for local distribution is constructed, except if pursuant to a special assessment project or if funded entirely by State or federal grants, the property owners whose property is served by the line and on which property the line is used by a commercial user, a domestic user or an industrial user shall pay their share of the cost of construction and installation as herein established at the time of construction. Any property owner requesting service who has not previously paid his share of the cost of constructing the watermain used for local distribution shall pay those costs as

established by the then effective resolution of the Township Board setting the local water distribution installation charges.

- (b) Computation of Local Distribution Charges and Annual Review. The charge shall be known as the "front foot rate" and be determined by multiplying the rate per foot by the number of feet the property owner has fronting on the watermain used for local distribution. The front foot rate shall be as set from time to time by resolution of the Township Board. Following an annual review of the charges, the Township Board may change the charges to reflect the current cost of construction and installation of local distribution lines.
 - (c) Assessment of Charges. Upon completion of construction or at the time connection is requested (if this cost has not been paid at the time of construction), the Township shall certify the cost to be charged to the property owner. The Township shall bill the owner of the premises affected, advising him that the amount so billed is to be paid prior to connection to the local distribution line serving the property of said property owner.
 - (d) Deferral of Assessments. If the Township Board so decides, such installations and connections may be made at the expense of the System when the owner of the premises signs a written agreement agreeing to pay the expense of such installation and connections upon terms agreeable to the Township Board and that the unpaid balance shall constitute a lien upon his property of the same character and subject to the same methods of collection as prescribed for special assessments.
- 5.3. Technical Standards. Materials and methods of construction of the Public Water System shall conform to the most current version of the "Water System Guidelines for Haring Township", attached as Appendix B which may be revised from time to time by a Resolution of the Board.

6. ARTICLE VI: ENFORCEMENT

- 6.1. Penalties for Violation of Ordinance. Whoever violates or fails to comply with any provision of this ordinance shall be fined up to Five Hundred Dollars (\$500.00), imprisoned not more than ninety (90) days, or both. A separate offense shall be deemed committed each day during or which a violation or non-compliance occurs or continues. The Township Supervisor or the County Sheriff or his deputies may issue appearance citations for violations of this ordinance.
- 6.2. Civil Action. The Township may institute any action at law or equity to compel compliance with this ordinance or to collect amounts due under this ordinance. If such action is instituted the Township shall recover the costs and expenses incurred to bring and maintain the action including, without limitation, actual reasonable attorneys' fees.
- 6.3. Lien. All rates, fees and charges billed or due here-under including those due pursuant to section 6.2 hereof shall to the extent permitted by law be liens upon the premises served from the delivery of such service which shall be enforceable in the same manner as ad valorem property tax liens.

7. ARTICLE VII: REPEAL CLAUSE

- 7.1. All ordinances or parts of ordinances in conflict herewith are repealed to the extent necessary to give this ordinance full force and effect.

8. ARTICLE VIII: SEVERABILITY

- 8.1. If any section, clause, sentence or provision of this Ordinance is determined to be invalid, said invalidity shall not affect the validity of any other part of this Ordinance which can be given effect without such invalid part or parts.

9. ARTICLE IX: PUBLICATION; EFFECTIVE DATE

- 9.1. Publication. A true copy or a summary of this Ordinance shall be published in The Cadillac News, a newspaper of general circulation within the boundaries of the Township qualified under state law to publish legal notices within thirty (30) days after the adoption of the Ordinance by the Township. This Ordinance shall be recorded in the minutes of the Township Board of the meeting at which this Ordinance was adopted and, in addition, shall be recorded in the Ordinance Book of the Township.
- 9.2. Effective Date. This Ordinance shall be in full force and effect thirty (30) days after its publication as provided by law.

10. ARTICLE X: APPEALS

- 10.1. Informal Hearing. In order that the provisions of this Ordinance may be reasonably applied and substantial justice done in instances where this Ordinance is misapplied or unnecessary financial hardship would result from carrying out the strict letter of this Ordinance, an informal hearing before the Township Utilities Committee may be requested in writing by any Person deeming itself aggrieved by a citation, order, charge, fee, surcharge, penalty or action within ninety (90) days after the date thereof, stating the reasons therefore with supporting documents and data. The informal hearing shall be scheduled at the earliest practicable date, but not later than thirty (30) days after receipt of the request, unless extended by mutual written agreement. The hearing shall be conducted on an informal basis at the Township offices or at such place as designated by the Township Utilities Committee. The Township Utilities Committee may grant the appeal, reject the appeal or schedule a second informal hearing not more than thirty (30) days after the initial hearing to allow time for study or to gather additional information. The Township Utilities Committee shall issue a written statement of its decision within fifteen (15) business days after the informal hearing.
- 10.2. Board of Appeals. In order that the provisions of this Ordinance may be reasonably applied and substantial justice done in instances where this Ordinance is misapplied or unnecessary hardship would result from carrying out the strict letter of this Ordinance, the Township Board shall serve as a Board of Appeals. The duty of such Board shall be to consider appeals from the decision of the Township Utilities Committee and to determine, in particular cases, whether this Ordinance has been misapplied or any deviation from strict enforcement will violate the intent of the Ordinance or jeopardize the public health, safety or welfare. In all appeals, the appellant shall have the burden of proof.
- 10.3. Appeals from Informal Hearing. Appeals from the written decisions of the Township Utilities Committee may be made to the Township Board, acting as a Board of Appeals, within thirty (30) days from the date of the written decision of the Township Utilities Committee. Such appeal may be taken by any Person aggrieved. The appellant shall file a Notice of Appeal with the Township Utilities Committee and with the Board of Appeals, specifying the ground therefore. Prior to a hearing, the Township Utilities Committee shall transmit to the Board of Appeals a summary report of all previous action taken. The Board of Appeals may, at its discretion, call upon the Township Utilities Committee to explain the action. The final disposition of the appeal shall be in the form of a resolution either reserving, modifying, or affirming, in whole or in part, the appealed decision or determination. In order to find for the appellant, a majority of the Board of Appeals must concur. The Board of Appeals shall fix a reasonable time for the hearing of the appeal, give due notice thereof to interested parties, and decide the same within a reasonable time. Within the limits of its jurisdiction, the same Board of Appeals may reserve or affirm, in whole or in part, or may make such order, requirements, decision or determination as, in its opinion, ought to be made in the case under consideration, and to that end have

all the powers of the official from whom said appeal is taken. The decision of said Board of Appeals shall be final.

The Board of Appeals shall meet at such times as the Board of Appeals may determine. There shall be a fixed place of meeting and all meetings shall be open to the public in accordance with applicable laws. The Board of Appeals shall adopt its own rules or procedure and keep a record of its proceedings, showing findings of fact, the action of the Board of Appeals, and the vote of each member upon each question considered. The presence of four (4) members shall be necessary to constitute a quorum.

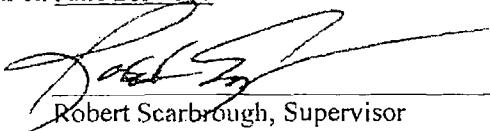
Board member Whetstone, seconded by Board member Olson, moved the adoption of the "Water Ordinance 2017-107"

YEAS: Board member(s) Soule, Scarbrough, McCain, Whetstone, Baldwin, Olson, Dewey

NAYS: Board member(s) None

ABSENT: Board member(s) None

Passed and adopted by the Township Board of the Charter Township of Haring, County of Wexford, Michigan on June 28, 2017


Robert Scarbrough, Supervisor
Charter Township of Haring

Attest:



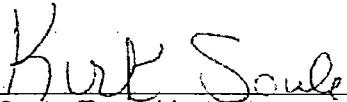
Kirk Soule, Clerk

Charter Township of Haring

CLERK'S CERTIFICATE

I hereby certify that the foregoing is a true and complete copy of an Ordinance adopted by the Township Board of the Charter Township of Haring, County of Wexford, State of Michigan, at a meeting held on the 28th day of June, 2017, and that public notice of said meeting was given pursuant to Act No. 267, Public Acts of Michigan, 1976, including in the case of a special or rescheduled meeting, notice by posting at least eighteen (18) hours prior to the time set for said meeting.

I further certify that said Ordinance has been recorded in the Ordinance Book of the Township and such recording has been authenticated by the signatures of the Supervisor and the Township Clerk.


Kirk Soule, Township Clerk

Attachment 3

Local COVID cases held steady on Friday

BY KAREN HOPPER USHER
CADILLAC NEWS

CADILLAC — After a small uptick in COVID-19 cases on Thursday, local counties were again holding steady on Friday.

There were no new cases Friday in Wexford, Missaukee, Osceola or Lake Counties.

In the District Health Department No. 10 region, Wexford had 15 cases, Missaukee had 18 and Lake had six. Additionally, probable cases were seven in Wexford, six in Missaukee and one in Lake county.

Many local COVID-19 cases are recoveries, with

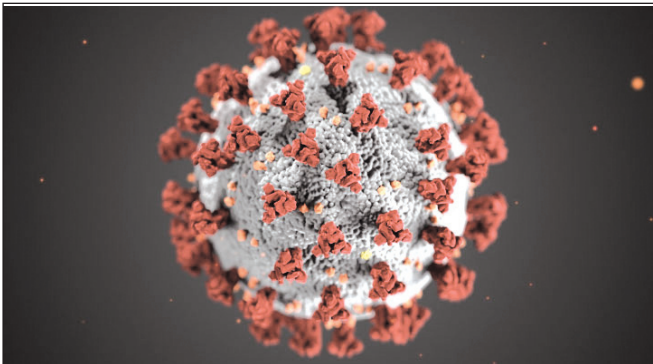
nine in Wexford County considered recovered, 15 in Missaukee and three in Lake County.

In the Central Michigan District Health Department jurisdiction, Osceola County has had 19 cases with 8 recoveries.

In Manistee County, which has had 13 cases overall, there was a new case on Friday.

District Health Department No. 10 announced plans to hold a drive-through testing clinic in Manistee next week.

The free testing will be available at the Manistee High School Parking Lot, 525 Twelfth Street, Manist-



FILE PHOTO

There were no new COVID-19 cases in Wexford, Missaukee Osceola or Lake counties. Manistee County added one on Friday.

ee, Thursday June 25 from 3 p.m. to 7 p.m. and Friday, June 26 from 10 a.m. to 2 p.m.

State numbers on Friday reached 60,829 with 5,823

deaths. Of the 60,00-plus cases, 212 were new since Thursday. There were five new deaths statewide.

kusher@cadillacnews.com | 775-NEWS (6397)

CAPS BOE set to adopt next budget for 2020-2021

BY RICK CHARMOLI
CADILLAC NEWS

CADILLAC — By Monday, Cadillac Area Public Schools will have its 2020-2021 budget in place.

While the board is scheduled to vote in the new budget at a special meeting on Monday, CAPS Superintendent Jennifer Brown said that doesn't mean the budget won't potentially be changed multiple times after it is adopted. She also said the board is scheduled to address the current year's budget which ends on June 30.

"We are assuming a \$650 proration for this year and holding flat the next budget year," Brown said.

"We know we will have to amend next year's budget and we anticipate multiple amendments to next year's budget when information becomes available from the state."

The final numbers

When it comes to balancing the current year's budget, Brown said the board will likely utilize savings the district experienced during the recent COVID-19 shutdown and the use of roughly \$800,000 from fund balance.

If the state does not prorate the current year's budget, CAPS won't likely know that until the fall after the August revenue estimating conference, according to Brown.

If that happens, Brown said the district will use those savings to help offset the cut that will come during the next fiscal year that is slated to start July 1.

"We just don't know if it is coming this year, next year or both," she said.

During its June 8 meeting, the board discussed the current year's budget and the budget for the next fiscal year.

Part of the discussion revolved around concerns the current year's budget could be prorated as well as a cut in funding for the next fiscal to make up the lost revenues in the state's school aid fund.

Brown told the board on June 8 the longer proration is not discussed the better, but there is potential for a \$650 proration this year to help cover the deficit and another \$550 cut per pupil for the next fiscal year.

Although it was discussed that the district was not going to make any cuts until it had firm numbers from the state, the board is scheduled Monday to discuss and act on layoffs of five professional staff.



RICK CHARMOLI | CADILLAC NEWS

By Monday, Cadillac Area Public Schools will have its 2020-2021 budget in place, but it likely won't be the last time they look at the budget in the next few months.

Brown said these staffing reductions are not connected to the potential budget cuts from the state or on the employees' performance in their duties.

Instead, she said it has everything to do with the consolidation and transition the district is undergoing at the elementary level, i.e. the closing of Kenwood Elementary.

"This is not due to the budget cuts and was planned to support student and program needs. We are closing an elementary building so we will need to be far more efficient with staffing," she said.

The potential layoffs include an administrator, an administrative assistant and three teachers. Although the district is looking at reducing its staffing, Brown said it is not reducing its programming.

The CAPS board is scheduled to meet at 7 p.m. Monday via Zoom.

For Monday's meeting, there are two ways the public can attend: by computer or by smartphone. A password also is needed to enter the Zoom meeting and can be obtained by calling (231) 876-5000. The Meeting ID also can be obtained when calling Central Office.

rcharmoli@cadillacnews.com | 775-NEWS (6397)

Wall Street dips as virus fears drown out economy hopes

NEW YORK (AP) — Wall Street careened through all the forces that have pushed and pulled it through the week, at first rising on Friday amid hope for the economy and then falling on worries about worsening coronavirus levels in some states, all before ending with modest losses.

The S&P 500 dropped 0.6%, a relatively small

move to cap its fourth weekly gain in the last five. But the market swung between a gain of 1.3% and a decline of 1%, another example of how uncertainty is the dominant force over Wall Street as investors weigh budding improvements in the economy against worsening infection levels in the South and West.

The Dow Jones Industrial Average dropped 208.64 points, or 0.8%, to 25,871.46 after earlier swinging from a gain of 371 points to a loss of 320 points. The Nasdaq composite inched up by 3.07 points, or less than 0.1%, to 9,946.12. The S&P 500 fell 17.60 points to 3,097.74.

Also exacerbating volatility was Friday's simultaneous expiration of contracts

for stock options and futures, an occasional occurrence that can drive bouts of buying and selling and is known as "quadruple witching day."

Early in the day, U.S. stocks appeared set to follow European and Asian markets higher and follow through on Wall Street's momentum from earlier in the week.

Business Brief

Forest Area Federal Credit Union Awards \$8,000 in scholarships to area high school seniors

Forest Area Federal Credit Union's Board of Directors has approved the awarding of \$500 scholarships to 16 high school seniors attending Fife Lake, Kalkaska, Kingsley and Manton, Lake City and Mesick High Schools.

Four seniors from Manton High School, one senior from Mesick High School and one from Lake City High School will be receiving their scholarships through various methods over the next month.

The Forest Area Federal Credit Union Scholarship program is open to area high school seniors who are

members of Forest Area Federal Credit Union and maintain a GPA of at least 3.0. Preference, when needed, is given to those going into a business-related field. The scholarship program was designed to give one student in each of the school districts a helping hand with their college expenses. Due to the overwhelming support of our members and our communities, Forest Area Federal Credit Union was able to expand the reach of this program once again this year and touch the lives of many of the youth in our area with this scholarship award.

Livestock auctions

Clare County Livestock Auction, LLC

Monday, June 22 2020 at 1 p.m.

June 15 results:

Choice colored steers, \$95 to \$104; choice colored heifers, \$90 to \$102; good colored steers and heifers, \$80 to \$100; choice Holstein steers, \$75 to \$95; heavy bulls, \$54 to \$106.50; heifer cows, \$60 to \$75; commercial/utility cows, \$45 to \$65; thin-lean cows, \$20 to \$50; cow/calf pairs and bred females, \$675 to \$825; colored feeder steers 500 pounds and under, \$80 to \$127; colored feeder steers 500 pounds and over, \$70 to \$125; colored feeder heifers 500 pounds and under, \$70 to \$137; colored feeder heifers 500 pounds and over, \$90 to \$135; Holstein feeder steers 500 pounds and under, \$75 to \$130; Holstein feeder steers 500 pounds and over, \$75 to \$95; colored

feeder bulls 800 pounds and under, \$70 to \$90; Holstein feeder bulls 800 pounds and under, \$50 to \$80; heifer calves, \$10 to \$100; Beef calves 300 pounds and under, \$125 to \$185; calves 1, \$100 to \$140; calves 2, \$50 to \$100; calves 3, \$10 to \$50; lambs, \$125 to \$195; ewes, \$40 to \$80; rams, \$65 to \$75; goats, \$80 to \$230; butcher hogs (200-299 pounds), \$40 to \$80; fleshy sows, \$30 to \$40; boars and stags, \$5 to \$40; feeder pigs, \$50 (per head); (322) 1st cutting hay (per bale), \$2.75 to \$4; (0) 2nd and 3rd cutting hay (per bale), \$3.50 to \$5.25; (0) straw (per bale), \$2.75 to \$3.75; (0) round bales, \$30.

Cattle Producers lets get BQA Certified online at www.bqa.org, select the FEEDYARD option.

Fall 2020 Feeder sale, 1 p.m., September 3, October 1, November 5 and December 3.

Public Notices



MCBAIN RURAL AGRICULTURAL SCHOOL NOTICE OF A PUBLIC HEARING ON TRUTH-IN-TAXATION/ PROPOSED 2020-2021 BUDGET

PLEASE TAKE NOTICE that on June 24, 2020, at 5:30 o'clock p.m., at McBain Public Schools' board room, McBain, Michigan, the Board of Education of McBain Rural Agricultural School will hold a public hearing to consider the District's proposed 2020-2021 budget.

The Board may not adopt its proposed 2020-2021 budget until after a public hearing. A copy of the proposed 2020-2021 budget, including the proposed property tax millage rate, is available for public inspection at Central Office.

The property tax millage rate proposed to be levied to support the proposed budget will be a subject of this hearing.

This notice is given by order of the Board of Education.

Karen Abrahamson, Secretary
June 19, 20, 22, 23, 24

LOST / REWARD: A set of size 6 white gold wedding rings in the Cadillac/Lake City area, week of September 13th, if found please call 000-0000 to verify and collect \$200 reward.

found

"Thank you so much."

"With the ad placed in your paper, I got my rings back.

I thought they were gone forever until the phone rang last week.

A lady had found them.

She checked the lost and found section of your paper and called me.

I just want to say thank you from the depth of my soul for your help."

Suz Knapp

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Classifieds Work!

So many people think they don't need to wear their hearing aids at home alone...

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1-800-286-1378



Here is why they should be.

Even when we are home and socially isolating from those geographically around us, it is important for us to keep our brain active and processing sound. In a study that tracked 639 adults for nearly 12 years, researchers at Johns Hopkins found that a mild hearing loss doubled dementia risk, and those with severe hearing losses were five times more likely to develop dementia when hearing loss was left untreated.

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CADILLAC FAMILY PHARMACY
A BUSINESS BUILT ON CUSTOMER SERVICE



EPA Begins Review of the Kysor Industrial Corp. Superfund site Cadillac, Wexford County, Michigan

The U. S. Environmental Protection Agency is conducting a five-year review of the Kysor Industrial Corp. Superfund site at 1100 Wright St. in Cadillac, Mich. The Superfund law requires regular checkups of sites that have been cleaned up – with waste managed on-site – to make sure the cleanup continues to protect people and the environment. This is the fifth five-year review of this site.

The cleanup of the Kysor Industrial site included installing a soil vapor extraction system to clean contaminated soil behind the Kysor plant, pumping and treating contaminated groundwater, and restricting access to the site as well as the use of groundwater wells. The soil vapor extraction system was turned off in 2009. EPA will evaluate if the pumping and treating of contaminated groundwater is still effective. EPA will also check to ensure that access restrictions protect the neighboring subdivision, including any private wells; that the current local ordinances in place restrict groundwater use and prevent new well installation; and that there are no activities that would compromise the cleanup.

More information is available at the Cadillac-Wexford Library, 411 S. Lake St., and at www.epa.gov/superfund/kysor-industrial. The review should be completed by this September.

The five-year review is an opportunity for you to tell EPA about site conditions and any concerns you have. Contact:

Sheila Sullivan
Remedial Project Manager
312-886-5251
sullivan.sheila@epa.gov

Heriberto León
Community Involvement Coordinator
312-886-6163
leon.heriberto@epa.gov

You may also call EPA toll-free at 800-621-8431, 9:30 a.m. to 5:30 p.m., weekdays.

**Kysor Industrial Corporation Site
Cadillac, Michigan**

Five Year Review Site Inspection

July 29, 2020

**Site Inspection Checklist
Site Inspection Photos**

I. SITE INFORMATION	
Site name: Kysor Industrial Corp.	Date of inspection: 7/29/20
Location and Region: Cadillac, Michigan; R5	EPA ID: MID043681840
Agency, office, or company leading the FYR: Michigan EGLE for EPA	Weather/temperature: 75 F, partly sunny
Remedy Includes: (Check all that apply) ☐ Landfill cover/containment ☒ Access controls ☒ Institutional controls ☒ Groundwater pump and treatment ☐ Surface water collection and treatment ☒ Other _____ ☐ Monitored natural attenuation ☐ Groundwater containment ☐ Vertical barrier walls	
Attachments: ☐ Inspection team roster attached ☐ Site map attached	

II. INTERVIEWS (Check all that apply)
1. O&M site manager: Jeffrey Dietlin, Director, Cadillac Utilities, July 2020 Interviewed ☐ at site ☐ at office ☒ by phone Phone No. (231) 779-7346 David Stearns, Assistant to Cadillac Utilities Director, Phone No. (231) 775-0181, ext. 125 Interviewed ☒ at site ☐ at office ☐ by phone Problems, suggestions; ☒ Report attached None reported
2. O&M staff Derek Goodrich, O&M Site Manager, LDFA Plant, July 2020 Douglas Langworthy, Asst. Water Resources Supv. Phone No.: (231) 775-2841 Interviewed: ☒ at site ☐ at office ☐ by phone Problems, suggestions; ☒ Report attached – No problems reported.

3.	Local regulatory authorities and response agencies (i.e., State and Tribal offices, emergency response office, police department, office of public health or environmental health, zoning office, recorder of deeds, or other city and county offices, etc.) Fill in all that apply. Agency: Michigan Department of EGLE Contact: Nick Swiger, Site Manager, Phone no. (231) 876-4458 Problems; suggestions; ☒ Report attached—No problems reported. Agency: Haring Township Contact: Robert Scarborough, Supervisor, Haring Twp., Phone No. (231) 779-9465 Problems; suggestions; ☒ Report attached—No problems reported. Agency: Wexford County Health Dept. District #10 Contact: Thomas Reichard, Environmental Health Director, Phone No. (231) 775-9942 Problems; suggestions; ☒ Report attached –No problems reported. Agency: Haring Township Utilities Department, Contractor (Infrastructure Alternatives, Inc. Contact: Bob Polanic, O&M Manager, Phone no. (231) 577-8793 Problems; suggestions; ☒ Report attached: --No problems reported.												
4.	Other interviews (optional) ☒ Report attached. See the Five-Year Review Report RPM conducted interviews with the following representatives: Jeffrey Dietlin, Director of Utilities, City of Cadillac; 2) Robert Scarborough, Supervisor, Haring Township and Building Official, Wexford County; 3) Bob Polanic, Infrastructure Alternatives, Inc. O&M contractor, Haring Township utilities; 4) Nicholas Swiger, Project Manager, EGLE; 5) Tom Reichard, Environmental Health Director, Wexford County Health Department, District #10.												
III. ON-SITE DOCUMENTS & RECORDS VERIFIED (Check all that apply)													
1.	O&M Documents <table style="width: 100%; border: none;"> <tr> <td style="width: 40%;">☒ O&M manual</td> <td style="width: 20%;">☒ Readily available</td> <td style="width: 20%;">☐ Up to date</td> <td style="width: 20%;">☐ N/A</td> </tr> <tr> <td>☒ As-built drawings</td> <td>☐ Readily available</td> <td>☒ Up to date</td> <td>☐ N/A</td> </tr> <tr> <td>☒ Maintenance logs</td> <td>☒ Readily available</td> <td>☒ Up to date</td> <td>☐ N/A</td> </tr> </table> Remarks _____	☒ O&M manual	☒ Readily available	☐ Up to date	☐ N/A	☒ As-built drawings	☐ Readily available	☒ Up to date	☐ N/A	☒ Maintenance logs	☒ Readily available	☒ Up to date	☐ N/A
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☒ As-built drawings	☐ Readily available	☒ Up to date	☐ N/A										
☒ Maintenance logs	☒ Readily available	☒ Up to date	☐ N/A										
2.	Site-Specific Health and Safety Plan ☒ Readily available ☐ Up to date ☐ N/A ☐ Contingency plan/emergency response plan ☐ Readily available ☐ Up to date ☐ N/A Remarks _____												
3.	O&M and OSHA Training Records ☒ Readily available ☐ Up to date ☐ N/A Remarks _____												

4.	Permits and Service Agreements ☒ Air discharge permit ☐ Readily available ☒ Up to date ☐ N/A ☒ Effluent discharge ☒ Readily available ☐ Up to date ☐ N/A ☒ Waste disposal, POTW ☐ Readily available ☒ Up to date ☐ N/A ☐ Other permits _____ ☐ Readily available ☒ Up to date ☐ N/A Remarks _____ _____
5.	Gas Generation Records ☐ Readily available ☐ Up to date ☒ N/A Remarks _____ _____
6.	Settlement Monument Records ☐ Readily available ☐ Up to date ☒ N/A Remarks _____ _____
7.	Groundwater Monitoring Records ☐ Readily available ☐ Up to date ☒ N/A Remarks: RPM Sheila Sullivan received these. _____
8.	Leachate Extraction Records ☐ Readily available ☐ Up to date ☒ N/A Remarks _____ _____
9.	Discharge Compliance Records ☐ Air ☐ Readily available ☐ Up to date ☐ N/A ☒ Water (effluent) ☒ Readily available ☐ Up to date ☐ N/A Remarks _____ _____
10.	Daily Access/Security Logs ☐ Readily available ☐ Up to date ☒ N/A Remarks _____ _____

IV. O&M COSTS	
1.	O&M Organization ☐ State in-house ☐ Contractor for State ☐ PRP in-house ☐ Contractor for PRP ☐ Federal Facility in-house ☐ Contractor for Federal Facility ☒ Other: City of Cadillac and Contractor

2.	O&M Cost Records ☒ Readily available ☒ Up to date ☒ Funding mechanism/agreement in place Original O&M cost estimate _____ ☐ Breakdown attached Total annual cost by year for review period if available <table style="width: 100%; border: none;"> <tr> <td style="width: 20%;">From _____</td> <td style="width: 20%;">To _____</td> <td style="width: 20%;"></td> <td style="width: 20%;"></td> <td style="width: 20%;"></td> <td style="width: 10%;"></td> </tr> <tr> <td style="text-align: center;">Date</td> <td style="text-align: center;">Date</td> <td style="text-align: center;">Total cost</td> <td colspan="3"></td> </tr> <tr> <td colspan="6">☐ Breakdown attached</td> </tr> </table> <table style="width: 100%; border: none;"> <tr> <td style="width: 20%;">From _____</td> <td style="width: 20%;">To _____</td> <td style="width: 20%;"></td> <td style="width: 20%;"></td> <td style="width: 20%;"></td> <td style="width: 10%;"></td> </tr> <tr> <td style="text-align: center;">Date</td> <td style="text-align: center;">Date</td> <td style="text-align: center;">Total cost</td> <td colspan="3"></td> </tr> <tr> <td colspan="6">☐ Breakdown attached</td> </tr> </table> <table style="width: 100%; border: none;"> <tr> <td style="width: 20%;">From _____</td> <td style="width: 20%;">To _____</td> <td style="width: 20%;"></td> <td style="width: 20%;"></td> <td style="width: 20%;"></td> <td style="width: 10%;"></td> </tr> <tr> <td style="text-align: center;">Date</td> <td style="text-align: center;">Date</td> <td style="text-align: center;">Total cost</td> <td colspan="3"></td> </tr> <tr> <td colspan="6">☐ Breakdown attached</td> </tr> </table> <table style="width: 100%; border: none;"> <tr> <td style="width: 20%;">From _____</td> <td style="width: 20%;">To _____</td> <td style="width: 20%;"></td> <td style="width: 20%;"></td> <td style="width: 20%;"></td> <td style="width: 10%;"></td> </tr> <tr> <td style="text-align: center;">Date</td> <td style="text-align: center;">Date</td> <td style="text-align: center;">Total cost</td> <td colspan="3"></td> </tr> <tr> <td colspan="6">☐ Breakdown attached</td> </tr> </table> <table style="width: 100%; border: none;"> <tr> <td style="width: 20%;">From _____</td> <td style="width: 20%;">To _____</td> <td style="width: 20%;"></td> <td style="width: 20%;"></td> <td style="width: 20%;"></td> <td style="width: 10%;"></td> </tr> <tr> <td style="text-align: center;">Date</td> <td style="text-align: center;">Date</td> <td style="text-align: center;">Total cost</td> <td colspan="3"></td> </tr> <tr> <td colspan="6">☐ Breakdown attached</td> </tr> </table>	From _____	To _____					Date	Date	Total cost				☐ Breakdown attached						From _____	To _____					Date	Date	Total cost				☐ Breakdown attached						From _____	To _____					Date	Date	Total cost				☐ Breakdown attached						From _____	To _____					Date	Date	Total cost				☐ Breakdown attached						From _____	To _____					Date	Date	Total cost				☐ Breakdown attached					
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3.	Unanticipated or Unusually High O&M Costs During Review Period Describe costs and reasons: No unusual cost issues. 																																																																																										

V. ACCESS AND INSTITUTIONAL CONTROLS ☒ Applicable ☐ N/A	
A. Fencing	
1.	Fencing damaged ☐ Location shown on site map ☒ Gates secured ☐ N/A Remarks _____
B. Other Access Restrictions	
1.	Signs and other security measures ☐ Location shown on site map ☐ N/A Remarks: Signs posted on fencing and wellheads
C. Institutional Controls (ICs)	

1.	Implementation and enforcement Site conditions imply ICs not properly implemented ☐ Yes ☒ No ☐ N/A Site conditions imply ICs not being fully enforced ☐ Yes ☒ No ☐ N/A Type of monitoring (e.g., self-reporting, drive by) _____ Frequency _____ Responsible party/agency: City of Cadillac, Haring Township Contact _____					
	<table style="width: 100%; border: none;"> <tr> <td style="width: 33%; border-bottom: 1px solid black;">Name</td> <td style="width: 33%; border-bottom: 1px solid black;">Title</td> <td style="width: 33%; border-bottom: 1px solid black;">Date</td> <td style="width: 33%; border-bottom: 1px solid black;">Phone no.</td> </tr> </table>	Name	Title	Date	Phone no.	
Name	Title	Date	Phone no.			
	Reporting is up-to-date ☐ Yes ☐ No ☒ N/A Reports are verified by the lead agency ☐ Yes ☐ No ☒ N/A Specific requirements in deed or decision documents have been met ☒ Yes ☐ No ☐ N/A Violations have been reported ☐ Yes ☐ No ☐ N/A Other problems or suggestions: ☐ Report attached					
	See main FYR report document for details.. _____					
2.	Adequacy ☒ ICs are adequate ☐ ICs are inadequate ☐ N/A Remarks: ICs are adequate for the City of Cadillac, but Haring Township ICs should be strengthened. _____					
D. General						
1.	Vandalism/trespassing ☐ Location shown on site map ☒ No vandalism evident Remarks _____ _____					
2.	Land use changes on site ☒ N/A Remarks: No significant changes _____					
3.	Land use changes off site ☒ N/A Remarks: No significant changes _____					
VI. GENERAL SITE CONDITIONS						
A. Roads ☒ Applicable ☐ N/A						
1.	Roads damaged ☐ Location shown on site map ☒ Roads adequate ☐ N/A Remarks _____ _____					
B. Other Site Conditions						
Remarks _____ _____ _____						

VII. LANDFILL COVERS ☐ Applicable ☒ N/A			
A. Landfill Surface			
1.	Settlement (Low spots) Areal extent _____ Remarks _____	☐ Location shown on site map Depth _____	☐ Settlement not evident
2.	Cracks Lengths _____ Widths _____ Depths _____ Remarks _____	☐ Location shown on site map	☐ Cracking not evident
3.	Erosion Areal extent _____ Remarks _____	☐ Location shown on site map Depth _____	☐ Erosion not evident
4.	Holes Areal extent _____ Remarks _____	☐ Location shown on site map Depth _____	☐ Holes not evident
5.	Vegetative Cover ☐ Grass ☐ Cover properly established ☐ No signs of stress ☐ Trees/Shrubs (indicate size and locations on a diagram) Remarks _____		
6.	Alternative Cover (armored rock, concrete, etc.) ☐ N/A Remarks _____		
7.	Bulges Areal extent _____ Remarks _____	☐ Location shown on site map Height _____	☐ Bulges not evident
8.	Wet Areas/Water Damage ☐ Wet areas ☐ Ponding ☐ Seeps ☐ Soft subgrade Remarks _____ ☐ Wet areas/water damage not evident ☐ Location shown on site map ☐ Location shown on site map ☐ Location shown on site map ☐ Location shown on site map Areal extent _____ Areal extent _____ Areal extent _____ Areal extent _____		
9.	Slope Instability ☐ Slides ☐ Location shown on site map ☐ No evidence of slope instability Areal extent _____ Remarks _____		
B. Benches ☐ Applicable ☐ N/A (Horizontally constructed mounds of earth placed across a steep landfill side slope to interrupt the slope in order to slow down the velocity of surface runoff and intercept and convey the runoff to a lined channel.)			

1.	Flows Bypass Bench Remarks_____	☐ Location shown on site map	☐ N/A or okay
2.	Bench Breached Remarks_____	☐ Location shown on site map	☐ N/A or okay
3.	Bench Overtopped Remarks_____	☐ Location shown on site map	☐ N/A or okay
C. Letdown Channels ☐ Applicable ☐ N/A (Channel lined with erosion control mats, riprap, grout bags, or gabions that descend down the steep side slope of the cover and will allow the runoff water collected by the benches to move off of the landfill cover without creating erosion gullies.)			
1.	Settlement Areal extent_____ Depth_____	☐ Location shown on site map	☐ No evidence of settlement
2.	Material Degradation Material type_____ Areal extent_____	☐ Location shown on site map	☐ No evidence of degradation
3.	Erosion Areal extent_____ Depth_____	☐ Location shown on site map	☐ No evidence of erosion
4.	Undercutting Areal extent_____ Depth_____	☐ Location shown on site map	☐ No evidence of undercutting
5.	Obstructions Type_____ ☐ No obstructions ☐ Location shown on site map Areal extent_____		
6.	Excessive Vegetative Growth Type_____		
	☐ No evidence of excessive growth ☐ Vegetation in channels does not obstruct flow ☐ Location shown on site map Areal extent_____		
D. Cover Penetrations ☐ Applicable ☐ N/A			
1.	Gas Vents ☐ Active ☐ Passive ☐ Properly secured/locked ☐ Functioning ☐ Routinely sampled ☐ Good condition ☐ Evidence of leakage at penetration ☐ Needs Maintenance ☐ N/A Remarks_____		

2.	Gas Monitoring Probes ☐ Properly secured/locked ☐ Functioning ☐ Routinely sampled ☐ Good condition ☐ Evidence of leakage at penetration ☐ Needs Maintenance ☐ N/A Remarks _____ _____
3.	Monitoring Wells (within surface area of landfill) ☐ Properly secured/locked ☐ Functioning ☐ Routinely sampled ☐ Good condition ☐ Evidence of leakage at penetration ☐ Needs Maintenance ☐ N/A Remarks _____ _____
4.	Leachate Extraction Wells ☐ Properly secured/locked ☐ Functioning ☐ Routinely sampled ☐ Good condition ☐ Evidence of leakage at penetration ☐ Needs Maintenance ☐ N/A Remarks _____ _____
5.	Settlement Monuments ☐ Located ☐ Routinely surveyed ☐ N/A Remarks _____ _____
E. Gas Collection and Treatment ☐ Applicable ☐ N/A	
1.	Gas Treatment Facilities ☐ Flaring ☐ Thermal destruction ☐ Collection for reuse ☐ Good condition ☐ Needs Maintenance Remarks _____ _____
2.	Gas Collection Wells, Manifolds and Piping ☐ Good condition ☐ Needs Maintenance Remarks _____ _____
3.	Gas Monitoring Facilities (e.g., gas monitoring of adjacent homes or buildings) ☐ Good condition ☐ Needs Maintenance ☐ N/A Remarks _____ _____
F. Cover Drainage Layer ☐ Applicable ☐ N/A	
1.	Outlet Pipes Inspected ☐ Functioning ☐ N/A Remarks _____ _____
2.	Outlet Rock Inspected ☐ Functioning ☐ N/A Remarks _____ _____
G. Detention/Sedimentation Ponds ☐ Applicable ☐ N/A	
1.	Siltation Areal extent _____ Depth _____ ☐ N/A ☐ Siltation not evident Remarks _____ _____

2.	Erosion Areal extent _____ Depth _____ ☐ Erosion not evident Remarks _____ _____
3.	Outlet Works ☐ Functioning ☐ N/A Remarks _____ _____
4.	Dam ☐ Functioning ☐ N/A Remarks _____ _____
H. Retaining Walls ☐ Applicable ☐ N/A	
1.	Deformations ☐ Location shown on site map ☐ Deformation not evident Horizontal displacement _____ Vertical displacement _____ Rotational displacement _____ Remarks _____ _____
2.	Degradation ☐ Location shown on site map ☐ Degradation not evident Remarks _____ _____
I. Perimeter Ditches/Off-Site Discharge ☐ Applicable ☐ N/A	
1.	Siltation ☐ Location shown on site map ☐ Siltation not evident Areal extent _____ Depth _____ Remarks _____ _____
2.	Vegetative Growth ☐ Location shown on site map ☐ N/A ☐ Vegetation does not impede flow Areal extent _____ Type _____ Remarks _____ _____
3.	Erosion ☐ Location shown on site map ☐ Erosion not evident Areal extent _____ Depth _____ Remarks _____ _____
4.	Discharge Structure ☐ Functioning ☐ N/A Remarks _____ _____
VIII. VERTICAL BARRIER WALLS ☐ Applicable ☒ N/A	
1.	Settlement ☐ Location shown on site map ☐ Settlement not evident Areal extent _____ Depth _____ Remarks _____ _____
2.	Performance Monitoring Type of monitoring _____ ☐ Performance not monitored Frequency _____ ☐ Evidence of breaching Head differential _____ Remarks _____ _____

IX. GROUNDWATER/SURFACE WATER REMEDIES ☒ Applicable ☐ N/A		
A. Groundwater Extraction Wells, Pumps, and Pipelines ☒ Applicable ☐ N/A		
1.	Pumps, Wellhead Plumbing, and Electrical ☒ Good condition ☒ All required wells properly operating ☐ Needs Maintenance ☐ N/A Remarks: Wells are thoroughly checked about 3-4 times/year, which entails opening the vaults, checking the pumps and electrical panels. 	
2.	Extraction System Pipelines, Valves, Valve Boxes, and Other Appurtenances ☒ Good condition ☐ Needs Maintenance Remarks: see Item #1 above. 	
3.	Spare Parts and Equipment ☒ Readily available ☐ Good condition ☐ Requires upgrade ☐ Needs to be provided Remarks: _____ 	
B. Surface Water Collection Structures, Pumps, and Pipelines ☒ Applicable ☐ N/A		
1.	Collection Structures, Pumps, and Electrical ☒ Good condition ☐ Needs Maintenance Remarks: Checked regularly. 	
2.	Surface Water Collection System Pipelines, Valves, Valve Boxes, and Other Appurtenances ☐ Good condition ☐ Needs Maintenance Remarks: _____ 	
3.	Spare Parts and Equipment ☐ Readily available ☐ Good condition ☐ Requires upgrade ☐ Needs to be provided Remarks: _____ 	
C. Treatment System ☒ Applicable ☐ N/A		
1.	Treatment Train (Check components that apply) ☐ Metals removal ☐ Oil/water separation ☐ Bioremediation ☒ Air stripping ☐ Carbon adsorbers ☐ Filters _____ ☐ Additive (e.g., chelation agent, flocculent) _____ ☐ Others _____ ☒ Good condition ☐ Needs Maintenance ☐ Sampling ports properly marked and functional ☐ Sampling/maintenance log displayed and up to date ☐ Equipment properly identified ☐ Quantity of groundwater treated annually _____ ☐ Quantity of surface water treated annually _____ Remarks: Treatment system inspected daily. 	

2.	Electrical Enclosures and Panels (properly rated and functional) ☐ N/A ☒ Good condition ☐ Needs Maintenance Remarks _____ _____
3.	Tanks, Vaults, Storage Vessels ☐ N/A ☒ Good condition ☐ Proper secondary containment ☐ Needs Maintenance Remarks: Stripping Towers cleaned out every 3 years on average, but are monitored for algae growth and cleaned as needed. _____
4.	Discharge Structure and Appurtenances ☐ N/A ☒ Good condition ☐ Needs Maintenance Remarks _____ _____
5.	Treatment Building(s) ☐ N/A ☒ Good condition (esp. roof and doorways) ☐ Needs repair ☐ Chemicals and equipment properly stored Remarks _____ _____
6.	Monitoring Wells (pump and treatment remedy) ☒ Properly secured/locked ☒ Functioning ☒ Routinely sampled ☒ Good condition ☐ All required wells located ☐ Needs Maintenance ☐ N/A Remarks: Monitoring wells are thoroughly inspected 3-4 times/week. _____
D. Monitoring Data	
1.	Monitoring Data ☐ Is routinely submitted on time ☒ Is of acceptable quality
2.	Monitoring data suggests: ☒ Groundwater plume is effectively contained ☐ Contaminant concentrations are declining
E. Monitored Natural Attenuation	
1.	Monitoring Wells (natural attenuation remedy) ☐ Properly secured/locked ☐ Functioning ☐ Routinely sampled ☐ Good condition ☐ All required wells located ☐ Needs Maintenance ☐ N/A Remarks _____ _____
X. OTHER REMEDIES	
If there are remedies applied at the site which are not covered above, attach an inspection sheet describing the physical nature and condition of any facility associated with the remedy. An example would be soil vapor extraction. None	

XI. OVERALL OBSERVATIONS	
A.	Implementation of the Remedy
	Describe issues and observations relating to whether the remedy is effective and functioning as designed. Begin with a brief statement of what the remedy is to accomplish (i.e., to contain contaminant plume, minimize infiltration and gas emission, etc.). System is operating as designed. Please see the text of the FYR Report for details.
B.	Adequacy of O&M
	Describe issues and observations related to the implementation and scope of O&M procedures. In particular, discuss their relationship to the current and long-term protectiveness of the remedy. No issues noted. Please refer to the main FYR document for O&M details.
C.	Early Indicators of Potential Remedy Problems
	Describe issues and observations such as unexpected changes in the cost or scope of O&M or a high frequency of unscheduled repairs that suggest that the protectiveness of the remedy may be compromised in the future. No indicators of potential problems noted
D.	Opportunities for Optimization
	Describe possible opportunities for optimization in monitoring tasks or the operation of the remedy. Please refer to the main FYR document for discussion of optimization.

**Kysor Industrial Corporation Site, Cadillac Michigan
Five-Year Review Site Inspection**

July 29, 2020

Inspection Photos

Kysor Industrial Corporation Site Five-Year Review Inspection

July 29, 2020

Photo 1: View of the two-stage air stripping system that removes VOCs from groundwater. The two air stripping towers are in the center. The LDFA treatment plant is visible behind the towers.



Photo 2 – View of east air stripping tower. The piping connections to the LDFA groundwater treatment plant are visible.



Kysor Industrial Corporation Site Five-Year Review Inspection

July 29, 2020

Photo 3: Inside the LDFA treatment plant. View of one of the pumps (left back-ground) and air line (center) which feed air into the air stripping towers.



Photo 4: Inside the LDFA treatment plant. Two groundwater extraction well pumps are visible on the right side of photo in the background.



Kysor Industrial Corporation Site Five-Year Review Inspection

July 29, 2020

Photo 5: Inside the LDFA treatment plant. Two carbon tanks in the center are used to treat chromium from the Northern Site. The green pipe is a dedicated chromium line from extraction well I-5. The effluent merges with the influent VOC-contaminated groundwater before air stripping and the combined effluent is discharged.



Photo 6: Inside the LDFA treatment plant. Panel showing flow controls



Kysor Industrial Corporation Site Five-Year Review Inspection

July 29, 2020

Photo 7: Meter showing total flow through system.



Photo 8: Inside LDFA treatment plant. The alarm panel (top) indicates no alarm.



Kysor Industrial Corporation Site Five-Year Review Inspection

July 29, 2020

Photo 9: Looking northeast from the LDFA treatment plant. The Northernair Plating facility is the red building in the lower right-hand corner of the photo. The facility is now defunct. The Four Star plant is visible in the background. (File photo).



Photo 10: Looking southward from the LDFA stripping tower platform. The Kysor facility is the blue building south of the railroad tracks. The plant is now owned by Borg-Warner. (File photo).



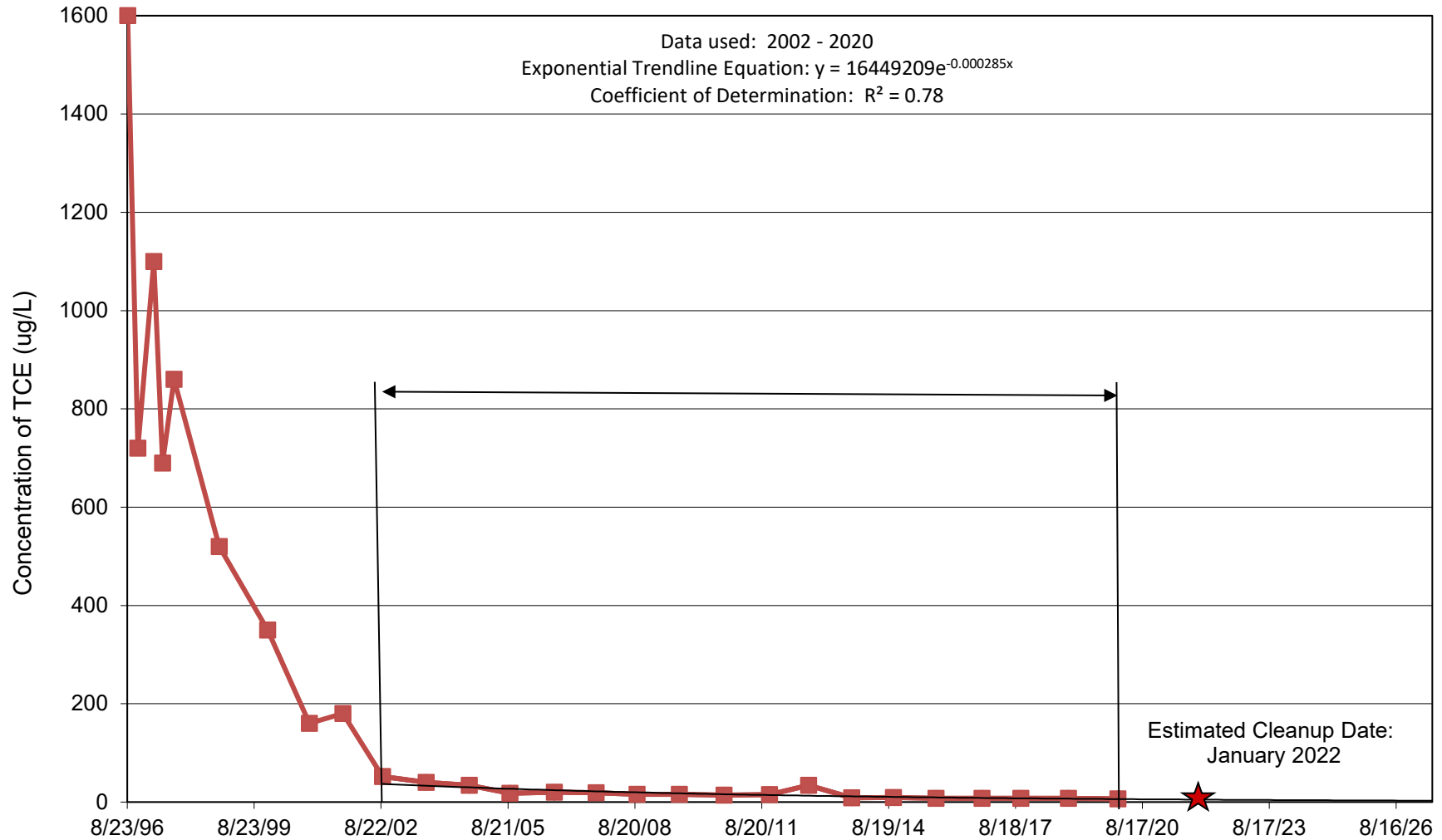
APPENDIX E – GRAPHS

Trend Analysis for Shallow Wells

Trend Analysis for Intermediate Wells

Appendix E
TCE Trends in the Shallow Aquifer
2016 - 2019 Compendium Performance Monitoring Report
Cadillac LDFA Northernnaire/Kysor Sites
Cadillac, Michigan

S-2

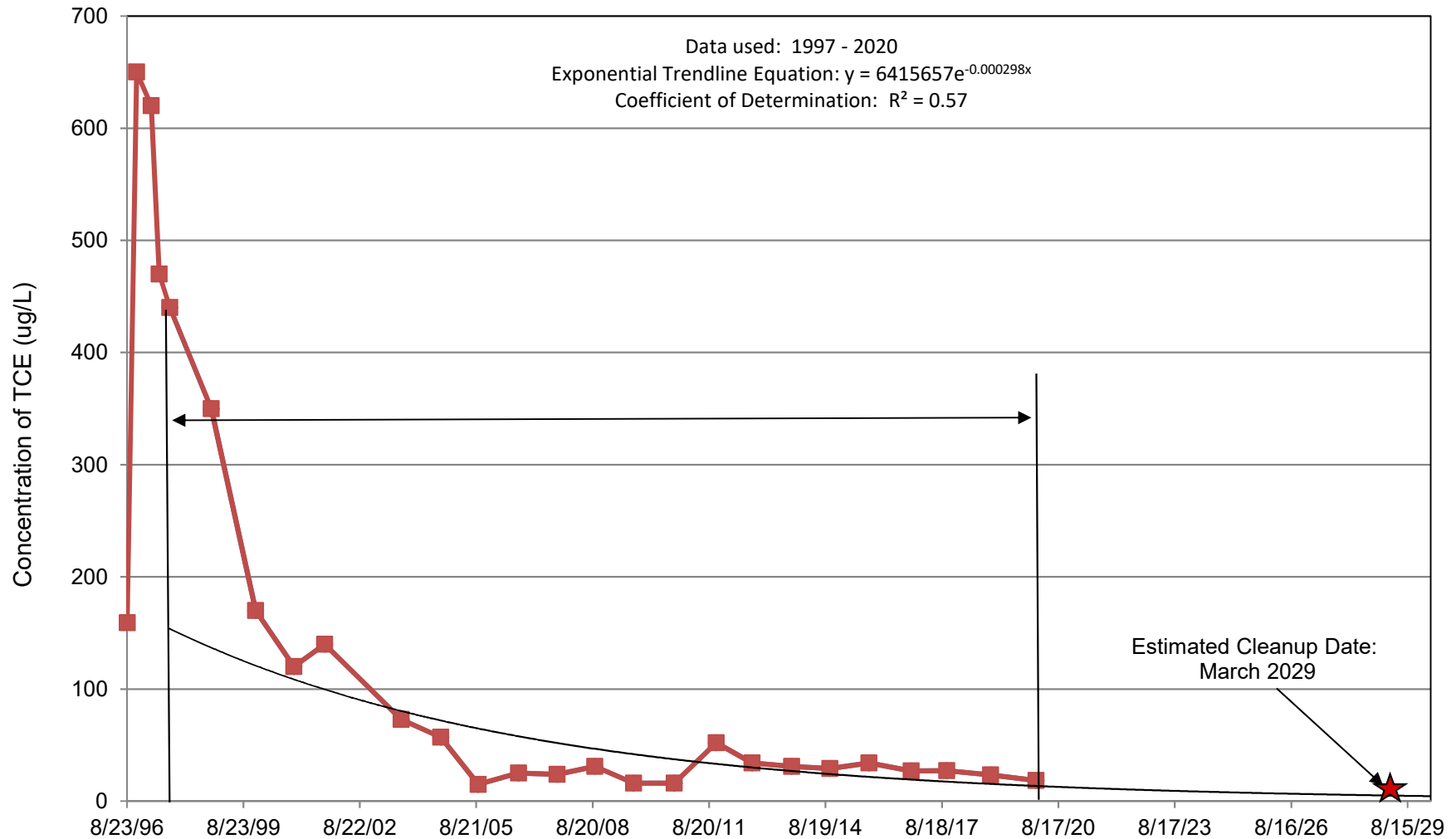


Notes:

1. S-2 monitoring data spans from 1996 to 2020.
2. The trendline above is calibrated to achieve similarity between estimated concentrations from 2002 to 2020 when compared to actual concentrations over the same period. This was achieved by excluding data prior to 2002 from the trendline equation.

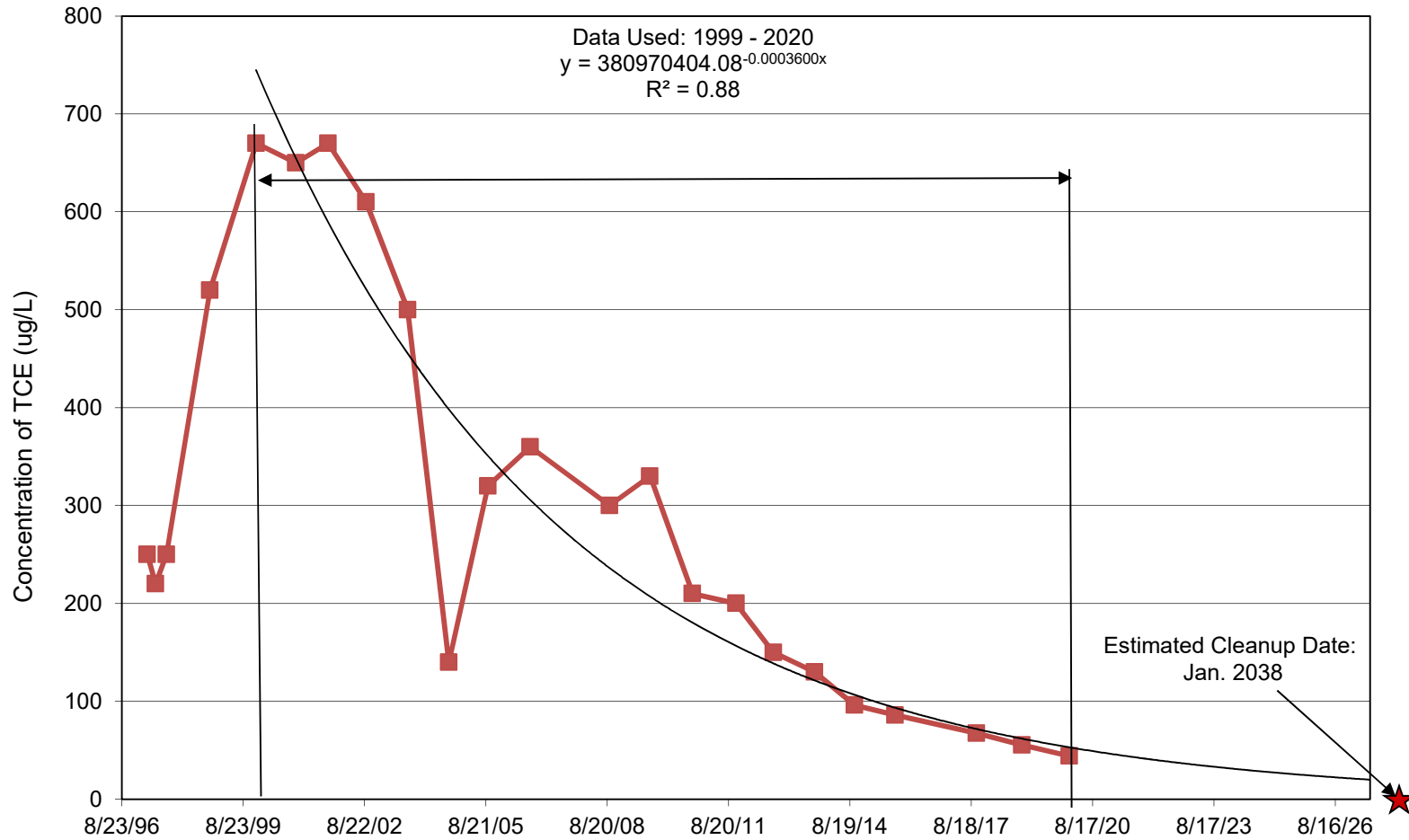
Appendix E
TCE Trends in the Shallow Aquifer
2016 - 2019 Compendium Performance Monitoring Report
Cadillac LDFA Northernnaire/Kysor Sites
Cadillac, Michigan

S-4



Appendix E
TCE Trends – Intermediate Aquifer
2016 - 2019 Compendium Performance Monitoring Report
Cadillac LDFA Northnaire/Kysor Sites
Cadillac, Michigan

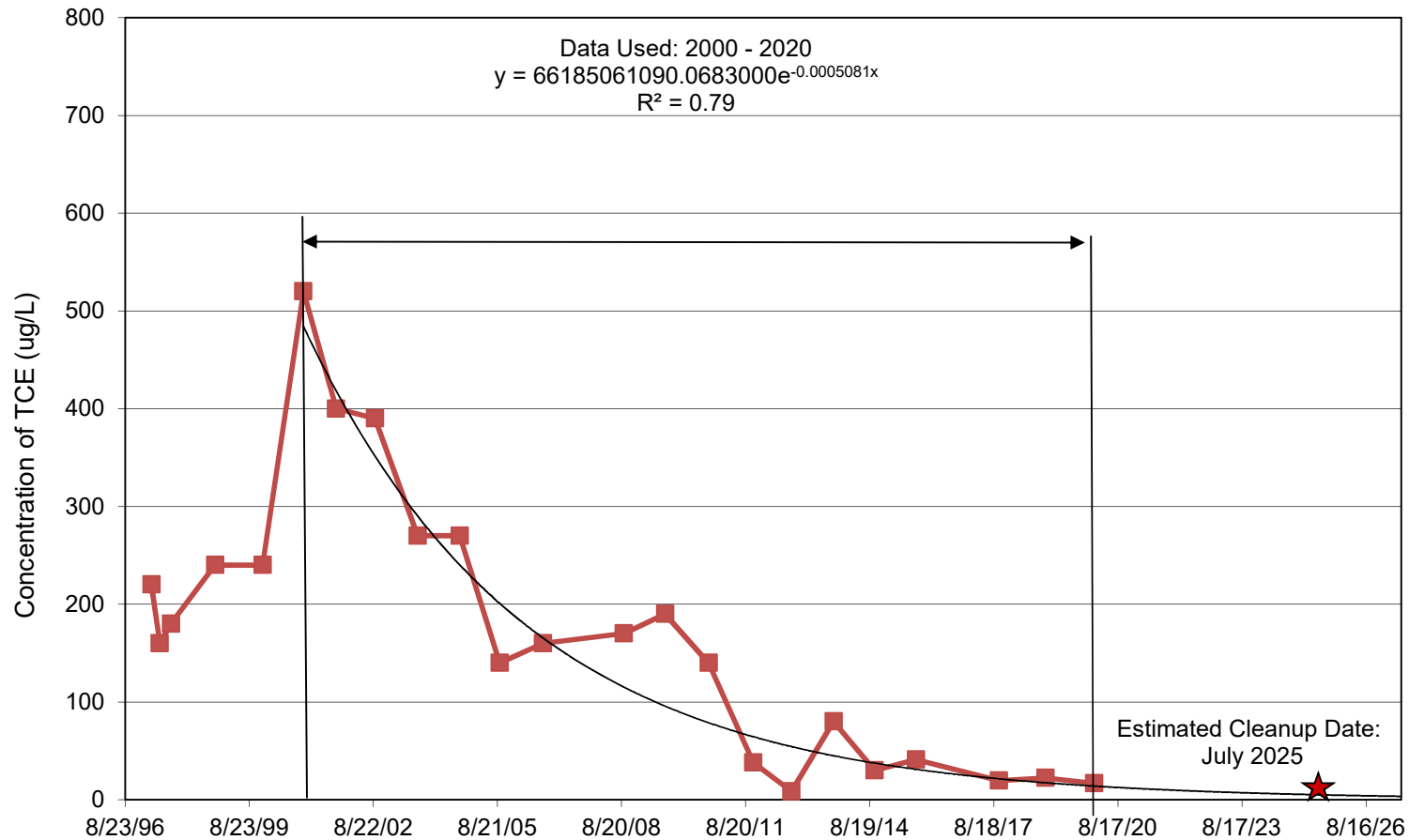
I-1



- Notes:
1. I-1 monitoring data spans from 1997 to 2020.
 2. The trendline above is calibrated to achieve similarity between concentrations from 2000 to 2020. This was achieved by excluding data prior to 12/99 from the trendline equation.

Appendix E
TCE Trends – Intermediate Aquifer
2016 - 2019 Compendium Performance Monitoring Report
Cadillac LDFA Northnaire/Kysor Sites
Cadillac, Michigan

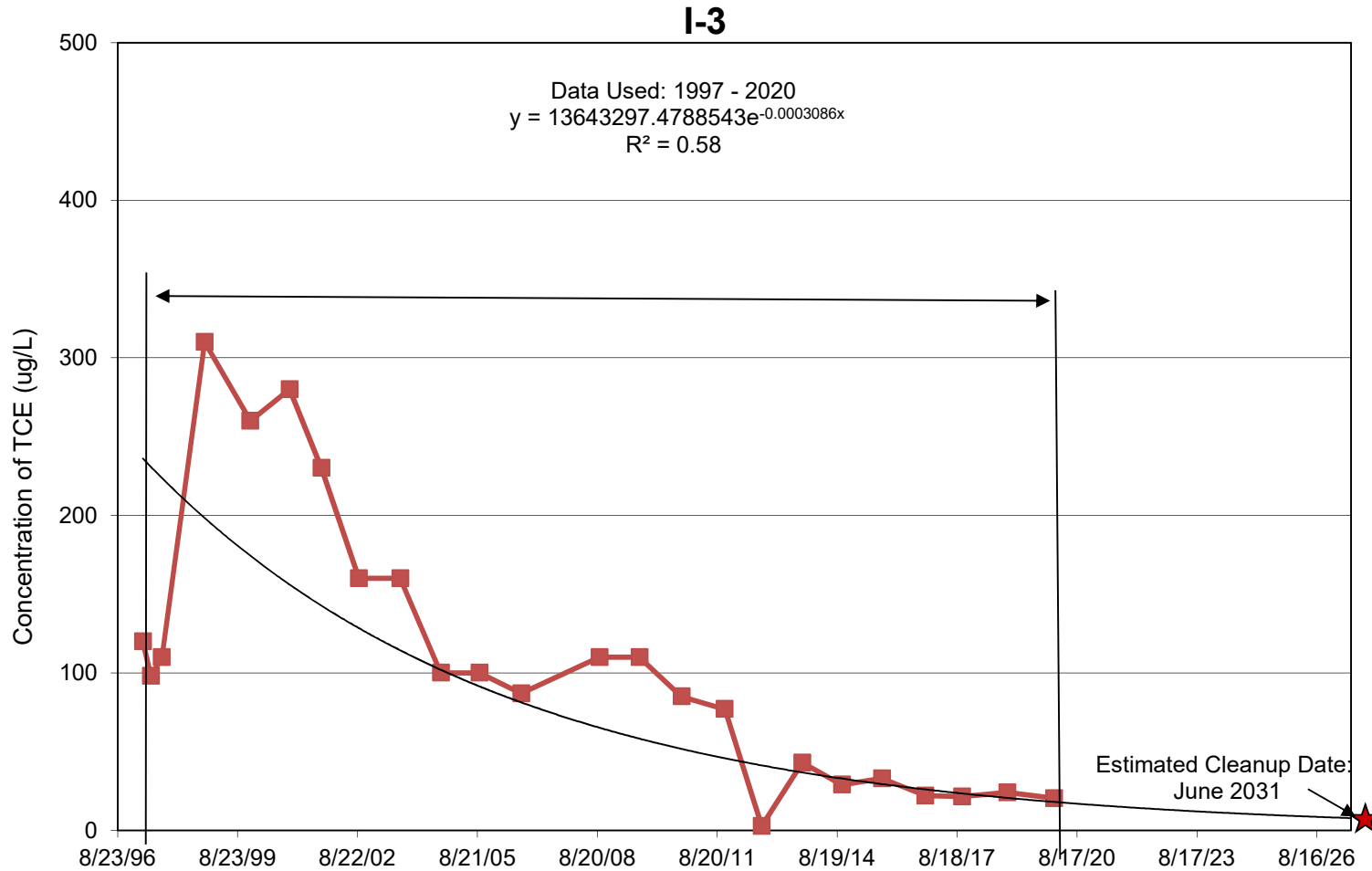
I-2



Notes:

1. I-2 monitoring data spans from 1997 to 2020.
2. The trendline above is calibrated to achieve similarity between concentrations from 2000 to 2020. This was achieved by excluding data prior to 2000 from the trendline equation.

Appendix E
TCE Trends – Intermediate Aquifer
2016 - 2019 Compendium Performance Monitoring Report
Cadillac LDFA Northnaire/Kysor Sites
Cadillac, Michigan

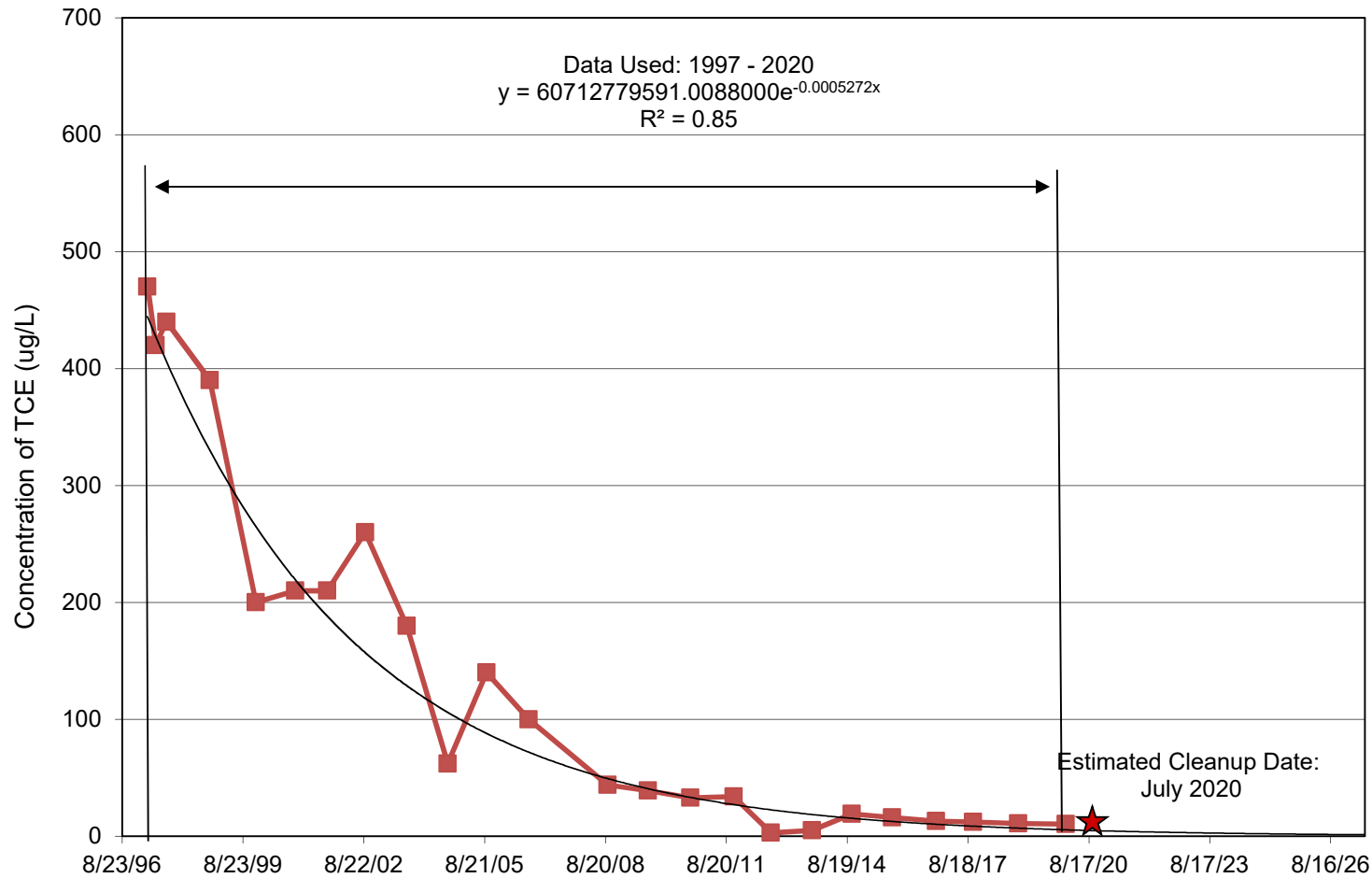


Notes:

1. I-3 monitoring data spans from 1997 to 2020.
2. An exponential trendline for the entire data set yielded similar estimated concentrations from 2004 to 2020 when compared to actual concentrations over the same period.

Appendix E
TCE Trends – Intermediate Aquifer
2016 - 2019 Compendium Performance Monitoring Report
Cadillac LDFA Northnaire/Kysor Sites
Cadillac, Michigan

I-4

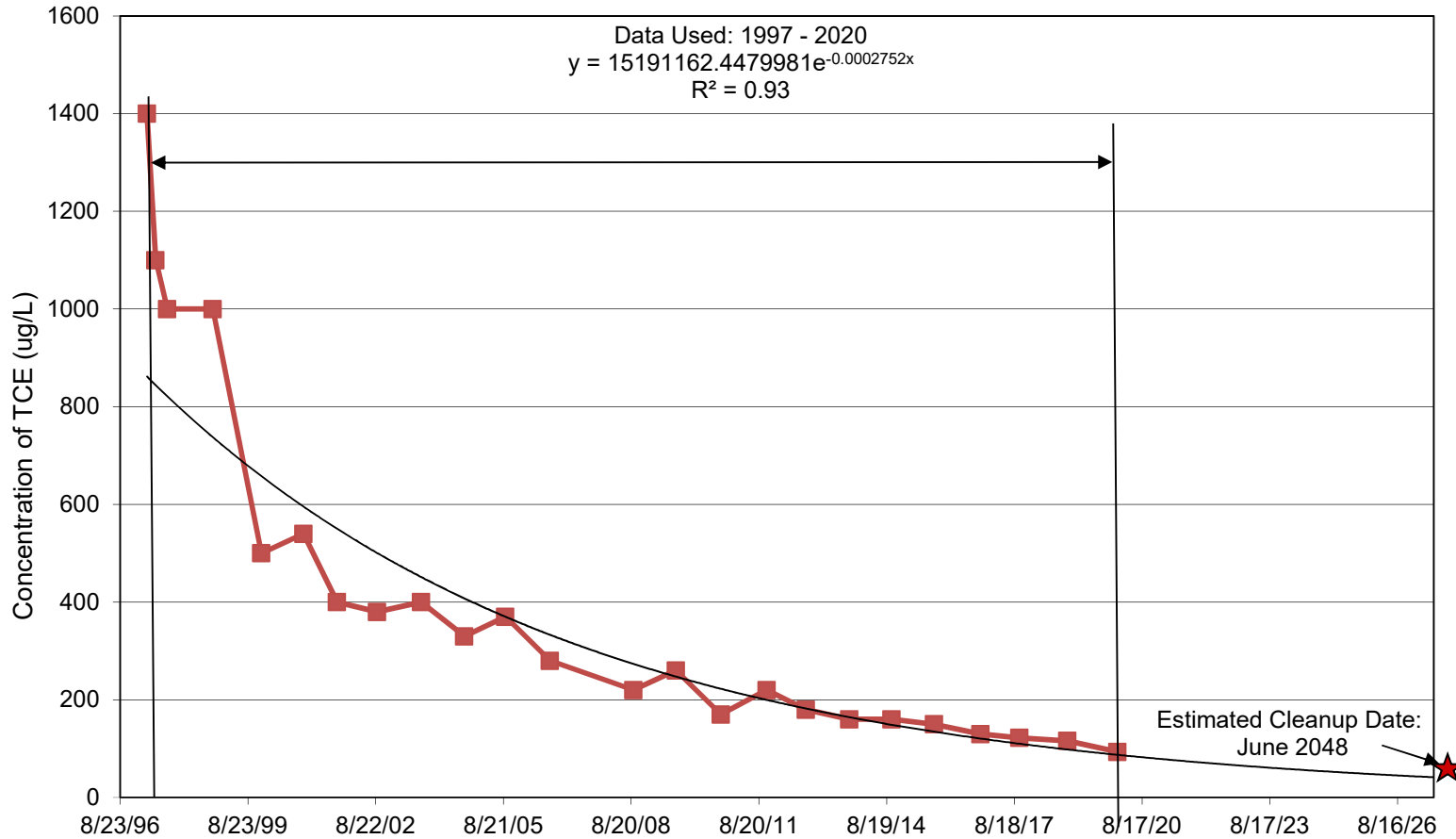


Notes:

1. I-4 monitoring data spans from 1997 to 2020.
2. An exponential trendline for the entire data set yielded similar estimated concentrations from 2008 to 2020 when compared to actual concentrations over the same period.

Appendix E
TCE Trends – Intermediate Aquifer
2016 - 2019 Compendium Performance Monitoring Report
Cadillac LDFA Northnaire/Kysor Sites
Cadillac, Michigan

I-10



Notes:

1. I-10 monitoring data spans from 1997 to 2015.
2. An exponential trendline for the entire data set yielded similar estimated concentrations from 2008 to 2015 when compared to actual concentrations over the same period.